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Invasion analysis for structured populations: analytical principles and numerical workflows

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Abstract

All real populations show some form of structure—differences among individuals, such as life stage or physiological condition, that shape a species' ability to persist. Yet it remains challenging to incorporate population structure into models of species interactions. We provide a general framework for studying competition in structured populations by quantifying species' invasion growth rates. The crux of our approach is separation of timescales: the initial dynamics of a structured population depend on how exactly it is introduced, but these transient differences do not affect long-term invasion success. Depending on the research question, the relevant long-term growth rate when rare may equivalently be derived using analytical separation of timescales, matrix population theory, or numerical simulations. Using case studies spanning developmental stage, environmental state, and infection status, we illustrate how our methodology derives simple, ecologically meaningful invasion conditions for a variety of systems. Finally, we extend the approach to fluctuating resident dynamics, showing how to use simulations or a general mathematical framework known as Floquet theory to correctly quantify invasion growth. By unifying the quantitative methods of population biology and community ecology, our framework will allow ecologists to study species persistence and coexistence across a greater variety of demographically complex systems.

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1 Introduction

All living things change over time: whether through their own growth and development, or through interactions with their biotic and abiotic environment, individuals differ in ways that affect their success. Yet community ecology typically treats entire species as its fundamental unit. In other words, classic models of ecological communities represent each species' population using a single variable (Nakazawa, 2015). This is in contrast to the field of population ecology, which has long recognized the importance of population structure—the distribution of individuals among different states such as sex, age, and size—for understanding population dynamics (e.g., De Roos & Persson, 2013). Such structure is not limited to individual development, as individuals may also differ in their physiological conditions (e.g., dormancy, infection status; Letten *et al.*, 2024), spatial location (Nakazawa & Huang, 2016), or environmental state (Ke & Levine, 2021). Classic population-level theory highlights how differences in stage structure affect populations' ability to persist (Caswell, 2001). Meanwhile, a growing perspective emphasizes that stage structure can also determine the outcome of species competition (Werner *et al.*, 2025). Accordingly, stage structure plays an important yet understudied role in determining whether species can coexist—a long-standing question in community ecology.

Invasion analysis, the foundational tool for studying species' persistence within a community, nonetheless remains challenging in structured populations. Ecologists quantify a species' invasion growth rate (IGR) as its growth rate when it is at low density within the environment created by its competitors. Species with positive invasion growth rates are predicted to persist because they recover from low abundance and avoid competitive exclusion. Beyond predicting coexistence based on mutual invasibility (Barabás *et al.*, 2018; Spaak & De Laender, 2020), invasion analysis is the first step for partitioning the mechanisms driving coexistence (Ellner *et al.*, 2019; Shoemaker *et al.*, 2020) and provides a fundamental way to quantify fitness in evolutionary biology (Yamamichi *et al.*, 2022). However, the invasion growth rate is hard to define in structured populations: whether a rare population initially grows or shrinks depends strongly on which stages are dominant (Iles *et al.*, 2016). Formal theory has established that the persistence of structured populations is determined by precise long-term growth metrics (e.g., their dominant Lyapunov exponents; Benaïm & Schreiber, 2009; Hofbauer & Schreiber, 2010; Roth & Schreiber, 2014). Nonetheless, these quantities are rarely derived in more system-specific ecological models. Instead, studies of coexistence in structured populations have more commonly relied on numerical simulations (Werner *et al.*, 2025) or strong simplifying assumptions (DeMalach *et al.*, 2021; Song *et al.*, 2021), indicating that general analytical workflows for calculating invasion growth rates remain poorly established (but see Klausmeier, 2008). Taken together, these studies highlight the need

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for a generally applicable approach for invasion analysis in structured populations.

Structured populations may often present an additional challenge: complex dynamics such as population cycles or chaos may emerge due to delayed density dependence, reflecting stage-specific feedbacks in reproduction, maturation, and mortality (De Roos & Persson, 2013; De Roos *et al.*, 2008; Gurney & Nisbet, 1985). These population fluctuations introduce two additional complications to invasion analysis. An invading species' growth rate varies over time, while its initial success depends on the particular state of residents' population dynamics it experiences at introduction. With the loss of a fixed resident equilibrium, analytical treatments quickly become mathematically cumbersome or intractable (Ellner *et al.*, 2016; Klausmeier, 2008). In these cases, invasion analysis requires concrete numerical simulations to estimate asymptotic growth over the resident cycle while simultaneously accounting for the invader's own internal population structure. Yet, to date, numerical approaches for structured invasion analysis remain scattered across specific models (Letten *et al.*, 2024; Werner *et al.*, 2025), with few overarching principles for estimating and validating invasion growth rates in time-varying structured populations.

Here, we develop a general framework for invasion analysis in structured populations, enabling researchers to study coexistence in diverse new systems and research questions. After briefly reviewing invasion analysis in classical models (Box 1), we use a simple age-structured competition model to illustrate how population structure complicates invasion analysis (section 2). Then we show how to address this challenge using two approaches. First, we demonstrate that the invader's population structure has much faster dynamics than population growth, allowing us to calculate invasion growth rates using separation of timescales (section 3). Second, we show that invasion analysis in structured populations is mathematically closely related to matrix population models, and thus, that linear algebra–based tools for population ecology provide a succinct and general method to quantify invasion (section 4). We illustrate the generality of our framework through case studies in which population structure represents not only developmental stages, but also diverse factors such as disease status or spatially structured habitats (section 5). Finally, for systems that are not analytically tractable, we provide reproducible numerical recipes guided by the analytical insights for rigorously studying invasion dynamics in arbitrarily complicated systems (section 6). By making the connection between invasion analysis and structured-population theory explicit, we provide a coherent and accessible toolkit for studying species persistence and coexistence across a wider range of ecological systems.

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Box 1. Review: Invasion analysis in unstructured populations

In this Box, we briefly summarize how classic invasion analysis can be used to study species coexistence in models without population structure. Consider a model of competition between n different species' populations N_i, where $i = 1, \ldots, n$. In continuous time, we can model the dynamics of species i according to

$$\frac{dN_i}{dt} = f_i\left(N_1, \ldots, N_n\right) \cdot N_i, \tag{B1.1}$$

where f_i is the per-capita growth rate as a function of species abundances; for instance, $f_i\left(N_1, \ldots, N_n\right) = r_i - \sum_j \alpha_{ij} N_j$ for the Lotka–Volterra model. We are interested in determining whether a species of interest will persist in the community after starting at some positive abundance. This species, which we will label 1, is known as the *invader*, while all other species are known as *residents*.

Resident equilibrium The first step is to consider the equilibrium reached by the residents when the invader is absent (i.e., $N_1 = 0$). To do so, we find the abundances $N_2^*, \ldots, N_n^*$ such that all residents experience zero growth, giving the system of equations

$$f_i\left(0, N_2^*, \ldots, N_n^*\right) = 0, \quad \text{for } i = 2, \ldots, n. \tag{B1.2}$$

For instance, in a two-species Lotka–Volterra model, the resident equilibrium abundance of species 2 is obtained by solving $f_2(0, N_2^*) = r_2 - \alpha_{22} N_2^* = 0$, yielding $N_2^* = r_2/\alpha_{22}$.

Invasion growth rate Assuming the residents reach a stable and well-defined equilibrium, we can then evaluate the per-capita growth rate of species 1, the invader, when introduced at extremely low abundance ($N_1 \approx 0$) into the resident equilibrium. Crucially, under this assumption, known as the *invasion scenario*, the invader is too rare to affect the residents' abundances or its own growth. Thus, at the beginning of invasion, it grows exponentially with a constant per-capita rate known as the *invasion growth rate*, given by

$$\mathrm{IGR}_1 = f_1\left(0, N_2^*, \ldots, N_n^*\right). \tag{B1.3}$$

The invader persists when its invasion growth rate is positive, i.e., $\mathrm{IGR}_1 > 0$. Stated conceptually, by ensuring that the species recovers from low abundance, this condition ensures that the species will not be excluded from the community. In our two-species Lotka–Volterra example, the invasion growth rate of species 1

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Box 1 (continued)

is $\mathrm{IGR}_1 = r_1 - r_2\alpha_{12}/\alpha_{22}$, and it persists if and only if $(r_1/r_2) \cdot (\alpha_{22}/\alpha_{12}) > 1$.

Coexistence conditions By repeating the invasion analysis for all species, we can predict when the entire community can robustly coexist. Specifically, coexistence requires *mutual invasibility*: each species i must have positive invasion growth. In other words, all species persist because they can recover from low abundance.

2 Problem: Accounting for population structure in invasion analysis

Invasion analysis is an important technique routinely covered in theoretical ecology textbooks and is relatively straightforward in simple theoretical models (Box 1). However, applying it in more realistic settings is complicated by the problem of *population structure*. Rather than considering a species' total population N_i, we might consider a series of discrete states $N_i^{(1)}, N_i^{(2)}, \ldots$, with superscripts representing our state(s) of interest (e.g., life stages, habitat, or disease status). In this section, we introduce how such models account for population structure, and illustrate why this complicates invasion analysis.

Simple stage-structured competition model As a concrete example of our general framework (Appendix S1), we begin with one of the simplest models where population structure matters. In particular, we consider the case where our invader of interest (species 1) has two life stages (juveniles $N_1^{(1)}$ and adults $N_1^{(2)}$), but its competitor (species 2) does not (represented simply by N_2; Fig. 1a). Dynamics obey the following system of ordinary differential equations:

$$\frac{dN_1^{(1)}}{dt} = \left(r_1 - \alpha_{11}N_1^{(2)} - \alpha_{12}N_2\right) \cdot N_1^{(1)} + f_1 \cdot N_1^{(2)}, \tag{1a}$$

$$\frac{dN_1^{(2)}}{dt} = g_1 \cdot N_1^{(1)} - m_1 \cdot N_1^{(2)}, \tag{1b}$$

$$\frac{dN_2}{dt} = \left(r_2 - \alpha_{21}N_1^{(2)} - \alpha_{22}N_2\right) \cdot N_2. \tag{1c}$$

Here, N_2 follows the standard Lotka–Volterra dynamics considered in Box 1, but species 1 experiences additional complexity due to transitions between its two states (Fig. 1a, solid arrows): juveniles $N_1^{(1)}$ replenish their own population at a rate r_1 and mature into adults $N_1^{(2)}$ at a rate g_1, while adults produce new juveniles at a rate f_1 and die at a rate m_1. Note that the juvenile-to-juvenile replenishment rate r_1 reflects the balance of juvenile reproduction minus mortality and maturation, and thus may be positive or

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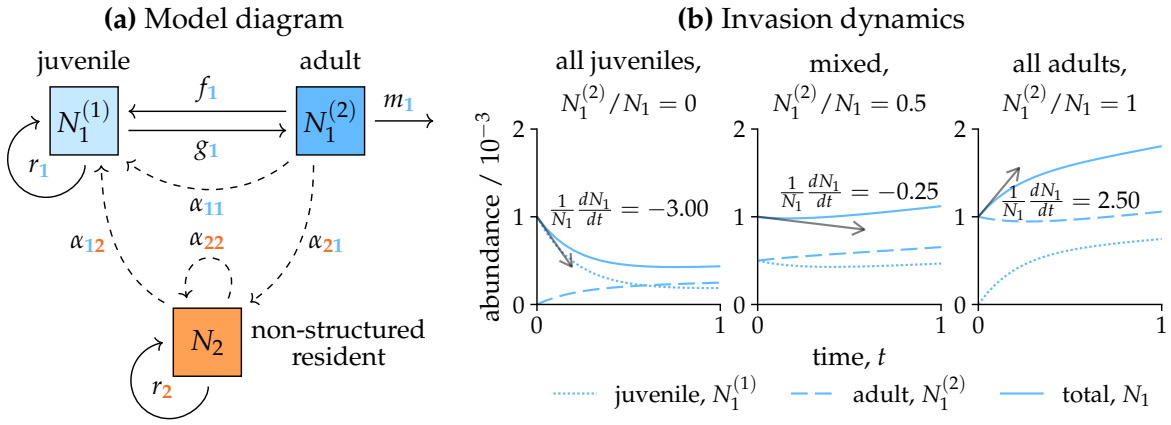


Figure 1: Simple model with stage structure. (a) Model diagram of a stage-structured species 1 competing against species 2 with no stage structure. Species 1 has two stages, juveniles $N_1^{(1)}$ and adults $N_1^{(2)}$. Solid arrows indicate population transitions: juveniles mature into adults at a rate g_1 and replenish their own population at a rate r_1, whereas adults produce new juveniles at a rate f_1 and die at a rate m_1. Dashed arrows indicate competition: juvenile replenishment rate r_1 is linearly reduced by competitors of species j according to the competition coefficient α_{ij}; we assume that all competitive effects due to species 1 are solely imposed by adults. **(b)** Instantaneous invasion dynamics under different initial population structures: 0% adults (left), 50% adults (middle), and 100% adults (right). In all cases, an initial total population of $N_1^{(1)} + N_1^{(2)} = 10^{-3}$ invades a resident population of species 2 at its equilibrium (not shown). We plot the abundance (y-axis) over time of species 1 (juveniles $N_1^{(1)}$: dotted; adults $N_1^{(2)}$: dashed; total N_1: solid). The initial rate of change of the total population (gray tangent arrow) differs between panels. Parameter values: $r_1 = -1$, $f_1 = 3$, $g_1 = 1$, $m_1 = 0.5$, $r_2 = 1$, $\alpha_{11} = 5$, $\alpha_{12} = 3$, $\alpha_{21} = 0.5$, $\alpha_{22} = 1$.

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negative. The species also experience competition: competitor density negatively affects each species i's rate r_i, representing the juvenile-to-juvenile transition for species 1 and general population growth for species 2. We assume this effect is mediated by a linear competition coefficient α_{ij}, where j is the competitor species, and that all competitive effects due to species 1 are solely mediated by adults $N_1^{(2)}$ (Fig. 1a, dashed lines). For the general analysis, we assume all variables are positive except r_1, which is negative due to mortality and maturation (specific parameter values are given in Fig. 1).

Invasion dynamics depend on population structure To study whether species 1 can persist when facing species 2, we consider its invasion scenario: species 2 is at its equilibrium abundance $N_2^* = r_2/\alpha_{22}$, as calculated in Box 1, while species 1 is introduced at low total abundance (i.e., $N_1^{(1)} + N_1^{(2)} \approx 0$). A naive approach might be to simply introduce one of the stages of species 1 at low abundance and calculate the instantaneous per-capita growth rate. For instance, introducing juveniles alone, we would find the per-capita growth rate of the population $(dN_1/dt)/N_1$ approaches $r_1 - \alpha_{12}N_2^* + g_1$ when the initial abundance is near zero. For adults, we would instead find that $(dN_1/dt)/N_1 = f_1 - m_1$. If we interpreted these outcomes as evidence of invasion success, our conclusion would change depending on the invader's initial stage distribution (Fig. 1b): introducing only juveniles would suggest that species 1 cannot persist, whereas introducing only adults would suggest successful invasion. How can ecologists obtain consistent, and more importantly, biologically meaningful predictions about invasion success in structured populations?

3 Method 1: Taking advantage of fast population structure dynamics

Fortunately, there is a conceptually simple solution: during invasion, the transient dynamics of population structure can be separated from the long-term growth of invader abundance. Under the invasion scenario, the invader remains sufficiently rare that it has negligible effects on the resident community and its own density-dependent demographic rates. Thus, the invader experiences an effectively fixed resident environment, and the effects of its initial stage distribution disappear before the overall population grows to appreciable abundance. We illustrate this by simulating the invasion of species 1, with various initial stage distributions, into a resident population of species 2 in the example model (dark to light blue in Fig. 2). The proportion of adults converges within a few time units regardless of the initial stage distribution (left of the x-axis break; Fig. 2a). Accordingly, there is a well-defined invasion growth rate that predicts invasion success: as shown in Fig. 2b, species 1's per-capita growth rate

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remains stable (top panel) while the population remains rare (bottom panel). Thus, invasion success should be evaluated using the per-capita growth rate after population structure has settled, rather than the instantaneous growth rate associated with any particular initial stage distribution. In this section, we use this insight to derive analytical expressions for invasion growth rates in structured populations using the technique of timescale separation (Otto & Day, 2011). We present two variants of this approach: a direct method applicable to any model, and a mathematical transformation technique which greatly simplifies calculations in many cases (see Appendix S2 for detailed derivations).

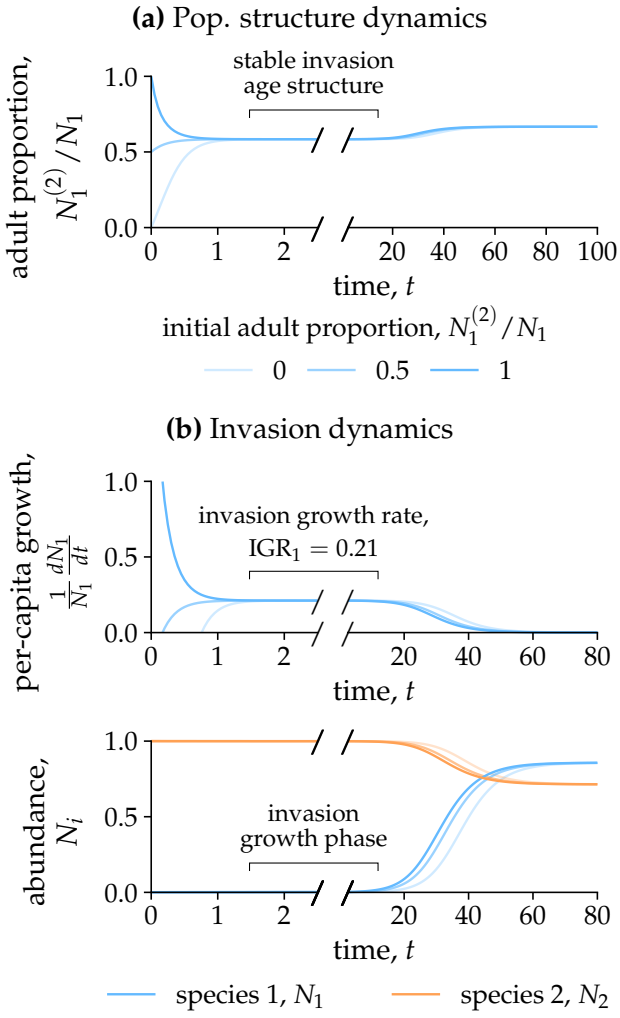


Figure 2: Fast and slow dynamics under invasion. We plot the temporal trajectory of **(a)** population structure (the proportion of adults within the entire population) and **(b)** per-capita growth rate for the invader population (top) and the total species abundance (bottom). Parameters are as in Figure 1 with an initial population of $N_1^{(1)} + N_1^{(2)} = 10^{-3}$. Different choices of initial population structure are shown as differently colored lines. Note that the resolution differs before (i.e., during the initial invasion) and after (i.e., subsequently approaching equilibrium) the break in the axis (after $t = 2$). The period when the invader reaches a stable population structure while remaining sufficiently rare gives the relevant invasion growth rate (IGR = 0.21; eq. 4).

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Method 1.1: Direct analytical timescale separation Under timescale separation, we can consider population structure as a fast variable, and overall population growth as a slow variable. To do so, we transform the model into two new variables representing each process: let Q represent population structure and N represent overall population size. There are many ways to capture these two aspects, but a mathematically simple choice is simply to use the juvenile-to-adult ratio and the adult abundance: $Q = N_1^{(1)}/N_1^{(2)}$ and $N = N_1^{(2)}$ (see Appendix S2 for other choices). Using the chain rule, we can rewrite the dynamics of species 1 in terms of these new variables:

$$\frac{dQ}{dt} = -g_1 Q^2 + (r_1 + m_1 - \alpha_{12} N_2^*)\, Q + f_1 \tag{2a}$$

$$\frac{dN}{dt} = g_1 QN - m_1 N, \tag{2b}$$

where we have dropped the $\alpha_{11} N_1^{(2)}$ term because it is near zero during invasion. Next, to perform timescale separation, we assume that Q rapidly approaches a steady state known as the *quasi-equilibrium* Q^* for a given near-constant value of the slow variable N. Setting $dQ/dt = 0$ in eq. 2a, we can solve a quadratic equation to find that the sole positive root is

$$Q^* = \frac{A + \sqrt{A^2 + g_1 f_1}}{g_1}, \text{ where } A = \frac{r_1 + m_1 - \alpha_{12} N_2^*}{2}, \tag{3}$$

and substituting this value of Q^* into eq. 2b gives the invasion growth rate as

$$\mathrm{IGR}_1 = \frac{r_1 - m_1 - \alpha_{12} N_2^*}{2} + \sqrt{\left(\frac{r_1 + m_1 - \alpha_{12} N_2^*}{2}\right)^2 + g_1 f_1}. \tag{4}$$

Inserting the particular parameter values used in Fig. 1, we find an invasion growth rate $\mathrm{IGR}_1 = 0.21$ matching the simulation result (Fig. 2b). This is no surprise, as direct simulation applies the same idea of a quasi-equilibrium stage distribution: we identify the time at which different starting states converge on a stable stage structure and evaluate the per-capita growth rate there (see general method in Appendix S3, Algorithm S3.1). Finally, we solve for the conditions under which $\mathrm{IGR}_1 > 0$ to find a simple condition for successful invasion by species 1:

$$\alpha_{12} N_2^* < r_1 + \frac{f_1 g_1}{m_1}. \tag{5}$$

In other words, species 1 can invade whenever the negative effect of the resident species 2 is outweighed by the sum of species 1's population growth from juveniles (r_1) and from adults (f_1), after correcting for the effects of maturation and mortality (g_1/m_1). This method can be applied to any ordinary differential equation model with population

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structure, and relies only on separation of timescales, a technique routinely covered in introductory curricula on theoretical ecology (e.g., Otto & Day, 2011). Thus, we emphasize that invasion analysis in structured populations need not be mathematically challenging.

Method 1.2: Strategic linear combination Though the final invasion condition in eq. 5 is simple, the intermediate calculations may require cumbersome algebra. However, knowing that stage structure rapidly approaches a quasi-equilibrium, we can derive conditions for invasion in a much simpler way. Consider defining a new variable as some linear combination of the juvenile and adult abundances, $X = N_1^{(1)} + b \cdot N_1^{(2)}$, where b is a positive constant. By linearity, $dX/dt = dN_1^{(1)}/dt + b \cdot dN_1^{(2)}/dt$. The key insight is that we can strategically choose b such that one stage variable is eliminated from dX/dt. As one option, we can choose $b = f_1/m_1$, which causes the $-m_1$ term in the adults' dynamics to cancel with the f_1 term in the juveniles' dynamics, leaving

$$\frac{dX}{dt} = \left(r_1 + \frac{f_1 g_1}{m_1} - \alpha_{12} N_2^* \right) N_1^{(1)}, \tag{6}$$

where we have again dropped the $\alpha_{11} N_1^{(2)}$ term due to the invasion scenario. Furthermore, thanks to the constant population structure at quasi-equilibrium, the remaining stage $N_1^{(1)}$ is a positive constant multiple of X: substituting $N_1^{(2)} = N_1^{(1)}/Q^*$ into the definition of X gives $N_1^{(1)} = kX$, where $k = \frac{Q^*}{Q^*+b} > 0$. Dividing eq. 6 by X and using the above expression, we obtain the invasion growth rate

$$\mathrm{IGR}_1 = k \cdot \left(r_1 + \frac{f_1 g_1}{m_1} - \alpha_{12} N_2^* \right), \tag{7}$$

which is equivalent to eq. 4 after multiplying out the full expression for k. However, since k is always positive, it never affects the sign of the invasion growth rate, and we can immediately see that the term within the parentheses leads to the same invasion condition we found earlier in eq. 5. In many models, a suitable positive linear combination can eliminate one or more stage variables from dX/dt, allowing the sign of the invasion growth rate to be determined without explicitly solving for the quasi-equilibrium stage distribution.

Box 2. Review: Population growth in matrix population models

In this Box, we review classic results for a family of models from population ecology known as matrix population models. In continuous time, the abundance of a single structured population can be represented by a vector $\vec{N}$, whose elements give the abundances in each state. When transition rates are constant, the

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Box 2 (continued)

population dynamics obey the vector differential equation

$$\frac{d\vec{N}}{dt} = \mathbf{C}\vec{N} \tag{B2.1}$$

where $\mathbf{C}$ is a constant *transition matrix* representing changes between states. Note that a similar analysis applies to the discrete-time analogue of this system, i.e., the classic Leslie and Lefkovitch models (Appendix S1).

Solving long-term dynamics using linear algebra A key result from matrix population theory is that differences in the initial population structure only affect the transient dynamics. After this transient phase, the relative abundances of the states approach a stable structure, and the whole population grows or declines exponentially with a common rate λ, the *long-term population growth rate*:

$$\vec{N}(t) \approx ae^{\lambda t}\vec{w}, \tag{B2.2}$$

where a is an arbitrary constant and $\vec{w}$ gives the relative proportions of different states, i.e., the *long-term state structure*. Basic tools from linear algebra allow us to solve for the long-term growth rate λ and the long-term state structure $\vec{w}$. Specifically, λ and $\vec{w}$ are mathematical properties of the transition matrix $\mathbf{C}$, obtained as solutions to the equation

$$\mathbf{C}\vec{w} = \lambda\vec{w}. \tag{B2.3}$$

Mathematically, $\vec{w}$ is an *eigenvector*, a vector whose elements' relative magnitudes are unchanged by matrix multiplication, but rather scaled by a scalar factor λ, known as the *eigenvalue*. For an $m \times m$ transition matrix $\mathbf{C}$, there are up to m distinct eigenvalues. The long-term growth rate of the population is given by the *leading eigenvalue*, i.e., the λ_i with the greatest real part, and the long-term population structure by the corresponding eigenvector $\vec{w}_i$. On the other hand, non-leading eigenvalues govern transient changes in population structure. Thus, eigenvalues provide a precise mathematical description of the timescale separation between fast changes in population structure and the long-term growth rate defining invasion success.

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4 Method 2: Applying classic theory from matrix population models

The conceptual insight above, that population structure has rapid dynamics during invasion, is closely related to classic results from the theory of matrix population models (Box 2). This connection provides a succinct and mathematically complete way to quantify invasion growth rates in structured populations. To see this, we first rewrite the dynamics of species 1 in matrix form as

$$\frac{d\vec{N}_1}{dt} = \begin{bmatrix} r_1 - \alpha_{11}N_1^{(2)} - \alpha_{12}N_2 & f_1 \\ g_1 & -m_1 \end{bmatrix} \vec{N}_1 \tag{8a}$$

$$\frac{dN_2}{dt} = \left(r_2 - \alpha_{21}N_1^{(2)} - \alpha_{22}N_2\right) \cdot N_2, \tag{8b}$$

where the matrix in the first equation is the transition matrix specifying changes between and within the two states of species 1. As we have discussed above, under the invasion scenario, species 1 is rare and species 2 is at its equilibrium abundance N_2^*; thus, as it invades, species 1 grows according to the constant transition matrix $\mathbf{C}$,

$$\mathbf{C} = \begin{bmatrix} r_1 - \alpha_{12}N_2^* & f_1 \\ g_1 & -m_1 \end{bmatrix}, \tag{9}$$

such that it obeys the linear vector differential equation $d\vec{N}_1/dt = \mathbf{C}\vec{N}_1$. Mathematically, this is the invader's Jacobian matrix evaluated at the resident equilibrium. Biologically, this matrix formulation shows that the rare invader behaves as a structured population growing in the fixed environment created by the resident. The eigenvalue perspective in Box 2 formalizes the timescale separation described above (Fig. 2): non-leading eigenvalues capture transient changes in stage composition, whereas the leading eigenvalue provides the invasion growth rate after the effects of arbitrary initial stage distribution have disappeared. This interpretation holds as long as the invader remains sufficiently rare for its dynamics to be governed by the constant $\mathbf{C}$. Accordingly, we can directly apply the results from Box 2 to quantify invasion growth rates in structured populations. We present two methods; as before, the first directly solves for the invasion growth rate (i.e., the leading eigenvalue), while the second derives conditions for invasion without explicitly solving for the eigenvalues.

Method 2.1: Direct solution for leading eigenvalue Following Box 2, after transient effects of the initial stage distribution have decayed, the long-term growth rate of the invading species 1 is given by the leading eigenvalue λ_1 of the transition matrix $\mathbf{C}$. The invader population grows exponentially at this rate, with a stable stage distribution

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given by the corresponding eigenvector. We can find the eigenvalues by solving the equation $\det(\mathbf{C} - \lambda \mathbf{I}) = 0$, giving the characteristic equation

$$\lambda^2 + (m_1 - r_1 + \alpha_{12} N_2^*)\,\lambda - f_1 g_1 - m_1\,(r_1 - \alpha_{12} N_2^*) = 0. \tag{10}$$

Solving this quadratic equation, we find that the dominant eigenvalue λ_1, and hence the invasion growth rate IGR_1, is

$$\lambda_1 = B + \sqrt{B^2 + f_1 g_1 + m_1\left(r_1 - \alpha_{12} N_2^*\right)}, \text{ where } B = \frac{r_1 - m_1 - \alpha_{12} N_2^*}{2}, \tag{11}$$

where we have taken the greater of the two roots. We can verify that this is equal to eq. 4 derived using timescale separation, confirming that this method recovers the correct invasion growth rate.

Method 2.2: Applying stability conditions For three or more stages, analytical expressions for eigenvalues can be biologically uninterpretable or may need to be solved numerically, limiting the usefulness of the above approach (Iritani & Day, 2026). Here, as in Method 1.2, we show that the sign of the leading eigenvalue can be determined without directly solving for its analytical expression, which greatly simplifies the algebra while providing the necessary information to predict invasion success. To do so, we apply the Routh–Hurwitz stability criteria (Appendix S1). For 2×2 matrices such as the one we consider here, the criteria reduce to two simple conditions: a system is stable—that is, all eigenvalues have negative real parts—if the matrix has a positive determinant $\det(\mathbf{C}) > 0$ and a negative trace (i.e., the sum of diagonal elements) $\mathrm{tr}(\mathbf{C}) < 0$. Thus, the resident equilibrium is unstable to invasion, and the invasion growth rate is positive, when at least one of the following conditions is true: (1) the determinant is negative, $\det(\mathbf{C}) < 0$, or (2) the trace is positive, $\mathrm{tr}(\mathbf{C}) > 0$, excluding boundary cases where each expression equals exactly zero. Applying this condition to $\mathbf{C}$ from eq. 9, we find that the determinant is $\det \mathbf{C} = -m_1\,(r_1 - \alpha_{12} N_2^*) - f_1 g_1$, and the condition for it to be negative exactly matches the invasion condition derived using earlier approaches (eq. 5). On the other hand, the trace is $\mathrm{tr}(\mathbf{C}) = r_1 - m_1 - \alpha_{12} N_2^*$, which is positive when $\alpha_{12} N_2^* < r_1 - m_1$. Because m_1, f_1, and g_1 are positive, we note that the trace condition always implies the determinant condition, making the latter necessary and sufficient for positive invasion growth. Together, these complementary approaches show that structured invasion analysis can often yield concise, interpretable invasion criteria without requiring explicit solutions for the invader's state distribution.

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5 Case studies: Population structure in diverse systems

Before turning to systems requiring fully numerical analysis, we illustrate how the analytical approaches developed above can be applied to answer questions from diverse systems and subfields of community ecology. Across our three case studies, population structure represents developmental stage, environmental conditions, and infection status, demonstrating that structured invasion analysis is not limited to age- or stage-structured life cycles. For each case, we derive an analytical invasion criterion and use it to identify how key biological parameters determine invasion success, thereby showing how structured invasion analysis can translate complex state-dependent dynamics into biologically interpretable results.

Case study 1: Stage-specific intraguild predation Stage-dependent interactions provide a classic example of how population structure can modify species invasion. In some insect species, for instance, individuals of later developmental stages are aggressive cannibalistic herbivores that consume younger conspecifics and other competing herbivores, in addition to their original plant resource (i.e., intraguild predation *sensu* Polis *et al.*, 1989). Here, we ask how aggression by older instars affects an aggressive herbivore species' ability to invade a resident competing herbivore population. We considered a model where the older juveniles of species 1 ($N_1^{(2)}$) consume younger conspecifics ($N_1^{(1)}$) and younger juveniles of the competitor ($N_2^{(1)}$). Both species' juveniles consume resource R to mature into the next stage, whereas adults ($N_1^{(A)}$ and $N_2^{(A)}$) reproduce without relying on consuming R. We studied how four key parameters affect the invasion growth rates of the aggressive species ($N_1^{(1)}$, $N_1^{(2)}$, and $N_1^{(A)}$; blue shaded box in Fig. 3a): the level of intraguild predation (a_{21}) and its effect on the older instar's mortality (b_{21}), as well as the cannibalism level (a_{11}) and its effect (b_{11}). Together, these parameters allow us to distinguish how often aggressive interactions occur from how strongly they affect the demographic performance of the aggressive species.

We derived the analytical invasion condition using all four methods described above. To isolate the effect of aggression, we assumed that the invader and the resident species are otherwise identical, so that all population differences between the two species are caused by aggression. We also assumed that adult-stage dynamics are fast compared with juvenile dynamics, allowing us to eliminate the adult stage and simplify the species to a two-stage population structure.

Under these assumptions, the invasion boundary for the aggressive species occurs simply at $b_{21} = 1$ (Appendix S4). Species 1 can invade only when consuming resident intraguild prey benefits its demographic performance ($b_{21} > 1$), whereas invasion fails when intraguild predation has a negative impact ($b_{21} < 1$). Our invasion analysis also separates the demographic consequence of a stage-specific interaction from the rate

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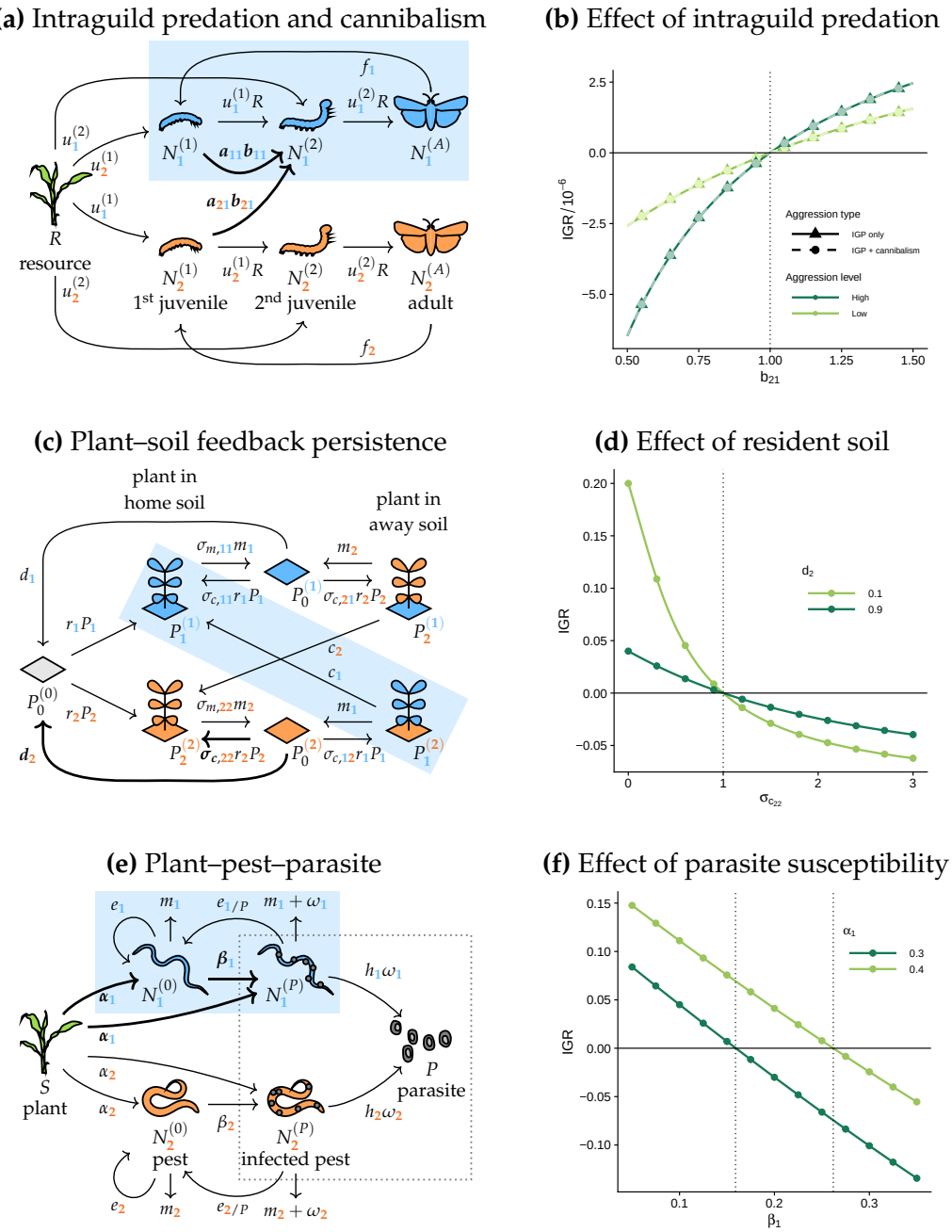


Figure 3: Applying invasion analysis to biologically distinct forms of population structure. The structured-population models include: **(a & b)** herbivores with developmental stage-dependent cannibalism and intraguild predation, for which we examine the effect of intraguild predation level (a_{21}) and its demographic impact (b_{21}); **(c & d)** a plant population structured across patches with different microbial legacies, for which we examine the strength of resident soil legacy ($\sigma_{c,22}$) and its persistence (d_2); and **(e & f)** pest populations structured by infection status in a plant–pest–parasite food chain, for which we examine the effects of the invading pest's plant consumption rate (α_1) and its susceptibility to the parasite (β_1). In each model diagram (left panels), blue shading highlights the structured population, while focal parameters are indicated in boldface and by thick arrows; blue and orange colors indicate different species. Right panels show the corresponding effects of these parameters on invasion growth rate: solid lines are analytical results and points are numerical estimates (Appendix S3, Algorithm S3.1). Vertical dotted lines show analytical invasion boundaries and colors distinguish values of the second focal parameter. See Appendices S4–S6 for detailed analytical derivations.

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at which that interaction occurs: while intraguild predation level (a_{21}) influences the magnitude of the invasion growth rate, its sign is determined solely by b_{21} (Fig. 3b). Moreover, aggressive behavior in the form of cannibalism does not affect the invasion outcome (overlapping circles and triangles in Fig. 3b). Although cannibalism may affect the equilibrium abundance once species 1 becomes common, it is biologically intuitive that cannibalism has little influence during invasion because encounters among conspecifics are negligible for the rare invader.

Case study 2: Resident soil legacy and rare-species invasion Environmental states provide a less conventional example of population structure, in which individuals are structured by the patch types they occupy. Plant–soil feedback considers populations structured in this way: plants modify soil microbial communities, and these microbial effects can persist after plant death as a distinct soil state affecting later-arriving plants (Ke & Levine, 2021). Here, we ask how the persistence and strength of resident-conditioned soil determine the ability of another plant species to invade its competitor's resident equilibrium. To address this question, we started from the fast-plant-conditioning patch occupancy model of Ke & Levine (2021), but extended it to a seven-variable model by allowing slow reconditioning when a plant grows on soil previously conditioned by another species. Specifically, state variables represent unoccupied unconditioned patches $P_0^{(0)}$, unoccupied patches conditioned by plant species 1 or 2 ($P_0^{(1)}$, $P_0^{(2)}$), and occupied patches in which plant i grows on soil conditioned by plant j ($P_i^{(j)}$). During plant 1's invasion, it may grow on its own conditioned soil or on resident-conditioned soil, thereby being structured across different soil patches ($P_1^{(1)}$ and $P_1^{(2)}$; blue shade in Fig. 3c). We focused on two key parameters: $\sigma_{c,22}$, the microbial effect modifying the resident plant 2's colonization on its own conditioned soil, and d_2, the decay rate of resident-conditioned soil. Together, $\sigma_{c,22}$ and d_2 allow us to examine how the strength and persistence of resident soil legacy shape the conditions for rare-species invasion.

Considering plant 1 invading the resident equilibrium of plant 2, we derived the analytical invasion condition using the methods described above (except for Method 2.1; Appendix S5). To isolate resident microbial effects on colonization, we set $\sigma_{m,ii} = \sigma_{m,ij}$ and $\sigma_{f,ii} = \sigma_{f,ij}$, assuming that conspecific and heterospecific soils have identical effects on a species' mortality and fecundity. We also assumed that the two species have identical values of $m_i\sigma_{m,ii}/r_i\sigma_{f,ii}$, thereby further removing species differences in microbe-mediated mortality and fecundity. Under these parameter assumptions, the remaining microbial effects are those on colonization, represented by $\sigma_{c,ii}$; values greater or less than one indicate that soil microbes enhance or reduce the resident's recolonization, respectively.

Under our simplified parameter assumptions, the analytical invasion criterion reduces to $\sigma_{c,12} > \sigma_{c,22}$, a classic result suggesting that the resident must limit its own

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growth more than it limits the invader (Appendix S5). By setting $\sigma_{c,12} = 1$, the invasion boundary occurs at $\sigma_{c,22} = 1$ (Fig. 3d). Plant 1 can invade only when resident-conditioned soil weakens the resident's ability to recolonize its own soil ($\sigma_{c,22} < 1$), and invasion growth rate increases when this soil legacy decays more slowly after host death (smaller d_2; Fig. 3d). Biologically, slow decay maintains patches less favorable for resident recovery, creating an opportunity for the rare invader. In contrast, if resident-conditioned soil enhances resident colonization ($\sigma_{c,22} > 1$), the invader cannot invade and slow decay of the microbial effect further prevents invasion.

Case study 3: Parasite-triggered stage structure Our final case study considers population structure based on individual status arising from species interactions. For example, parasite infection can divide a pest population into healthy and infected states, analogous to compartmental models in epidemiology. Here, we ask how parasite infection creates a structured population that shapes invasion between two competing pest species (Price *et al.*, 1986). We considered a tri-trophic food web in which plants (S) are consumed by two pest species and parasites (P) infect these pests, converting them from a healthy state ($N_i^{(0)}$) to an infected state ($N_i^{(P)}$; blue shade in Fig. 3e). As the infected pest lives for a period of time before eventually being killed by parasites, the two pest statuses differ in their mortality rates and conversion efficiencies. Interestingly, in this system the parasite can also be viewed as a stage-structured population, where the parasite is present either in the pest body or as free-living propagules (dotted box in Fig. 3e; Appendix S6). We studied how the parasite infection rate on pest species 1 (β_1) and its plant consumption rate (α_1) affect invasion of the two stage-structured populations described above. Together, these parameters allow us to examine how top-down control and resource competition jointly determine invasion in a system with interaction-generated population structure.

We examined the following two invasion scenarios: (i) the invasion criteria for the parasite invading a parasite-free equilibrium with just plants and one pest species, and (ii) the invasion criteria for pest species 1 ($N_1^{(0)}$ and $N_1^{(P)}$) invading a plant–pest–parasite system of resident pest species 2. For both scenarios, we derived the analytical invasion condition using the methods described above (except for Method 2.1; Appendix S6). To distinguish between the two pest species, we assumed that pest species 1 has a lower consumption rate ($\alpha_1 < \alpha_2$) and a higher mortality rate ($m_1 > m_2$) compared to pest species 2. In compensation, it experiences a lower parasite infection rate ($\beta_1 < \beta_2$), i.e., a virulence–susceptibility tradeoff. Moreover, we assumed that parasite infection reduces the conversion efficiency of pests ($e_{i/P} \leq e_i$). We present the invasion analysis for the parasite in Appendix S6 and focus on the invasion of pest species 1 into a resident equilibrium of species 2 below.

For the invasion of pest species 1, invasion growth rate decreases as the parasite

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infection rate β_1 increases, reflecting stronger top-down control on the rare invader (Fig. 3f). In contrast, a higher plant consumption rate (larger α_1) increases invasion growth rate by allowing pest species 1 to acquire more plant resources while rare. Consequently, increasing α_1 shifts the invasion boundary toward higher values of β_1; that is, pests with greater resource consumption can tolerate stronger parasite infection and still invade. Consistent with this interpretation, the analytical threshold for β_1 increases with plant abundance S^* but decreases with parasite abundance P^* of the resident equilibrium (Appendix S6). Thus, biologically, invasion success reflects a balance between the invader's advantage in resource acquisition and its susceptibility to top-down control.

6 Method 3: Simulation-based invasion analysis in complicated systems

So far, we have shown how to analytically solve invasion growth in systems with a stable resident equilibrium. However, theoretical (Chesson & Huntly, 1997) and empirical work (Letten *et al.*, 2018) have demonstrated that fluctuations in the conditions experienced by invading species can affect coexistence. In this section, we extend our framework to fluctuating resident dynamics, where the invaders' population transitions fluctuate over time and invasion growth rates are usually determined numerically. We first show how the concept of timescale separation (previously discussed in section 3) can guide numerical recipes for determining the invasion growth rate even when the resident population is not at equilibrium (Method 3.1). We then show how a standard mathematical technique known as Floquet theory rigorously extends the matrix population approach (introduced previously in section 4) to systems with cyclic resident dynamics (Method 3.2).

Model extension with cyclic dynamics As an example of non-equilibrium resident dynamics, we modified equation 1c so that species 2 also has two stages, juveniles $N_2^{(1)}$ and adults $N_2^{(2)}$:

$$\frac{d\vec{N}_2}{dt} = \begin{bmatrix} r_2 - \alpha_{21}N_1^{(2)} - \alpha_{22}N_2^{(2)} + \beta N_2^{(1)} & f_2 \\ g_2 + \gamma N_2^{(1)} & -m_2 \end{bmatrix} \vec{N}_2. \tag{12}$$

As highlighted in the model diagram (Fig. 4a, bolded terms), juveniles facilitate the juvenile-to-juvenile rate represented by β, and the maturation into adults represented by γ. When species 2 grows alone, these facilitative interactions can generate a stable limit cycle (Fig. 4b; see Appendix S7). Biologically, facilitation first allows juveniles to build up in abundance (dotted line), but population regulation is delayed because

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juveniles must mature into adults (dashed line) before exerting a negative effect on juveniles, which eventually causes both stages to decline. For our parameter values, the period of these oscillations is $T = 0.778$ (used in Fig. 4 as the unit of the x-axis).

The non-equilibrium dynamics of the resident species result in a further complication to invasion analysis. In addition to depending on initial population structure (Fig. 1b), species 1's instantaneous growth rate depends on when in the resident's population cycle it is introduced (shown for four different points in Fig. 4c). Because species 2's adult abundance ($N_2^{(2)}$) is responsible for regulating species 1's growth rate, species 1 grows rapidly if introduced when species 2's adults are at their lowest abundance ($t = 0$) but may initially decline if introduced at points in the cycle where adults are highly abundant ($t = T/4$ and $T/2$).

Method 3.1: Simulating convergent invasion dynamics Just as in the simple model (Fig. 1b), our simulations reveal that transient differences in the invader's population structure rapidly disappear during invasion (Fig. 5a). Here, however, trajectories do not approach a fixed population structure, but rather converge to the same fluctuating cycle as long as the invader remains rare (circa $2 \leq t \leq 10$). As a result, the invader's per-capita growth rate also converges to a stable cycle (Fig. 5b), and averaging this growth rate after the initial transient has decayed gives the long-term invasion growth rate that determines persistence. We can perform this calculation by identifying the interval during which species 1's stage structure has already converged but its population remains sufficiently rare (Appendix S3, Algorithm S3.1); for our example, we find that $\mathrm{IGR}_1 = 0.456$ (Fig. 5b; dashed line shows the moving average).

As an alternative, we can estimate the same quantity using a version of the standard algorithm for calculating the *dominant Lyapunov exponent* (Benettin *et al.*, 1980), of which the invasion growth rate is a special case. Briefly, we intermittently rescale the invader population size to a small value while allowing its population structure to evolve freely (see Appendix S3, Algorithm S3.2 and illustration in Appendix S8, Fig. S8.5). This numerical procedure has been used in structured population models (e.g., Werner *et al.*, 2025), but the biological reason why it recovers the appropriate invasion growth rate is often left implicit. Our framework clarifies the underlying logic: rescaling provides a practical way to allow transient effects of the initial population structure to decay while maintaining the invasion scenario, such that the invader's dynamics remain simple in the environment created by the resident. This numerical recipe is not restricted to cyclic resident dynamics. As we demonstrate in Appendix S8 for a chaotic Lotka–Volterra system (Sprott, 2010), the same approach can estimate the exponential growth rate of a rare species invading a resident community with chaotic dynamics, even when the resulting full community subsequently remains chaotic.

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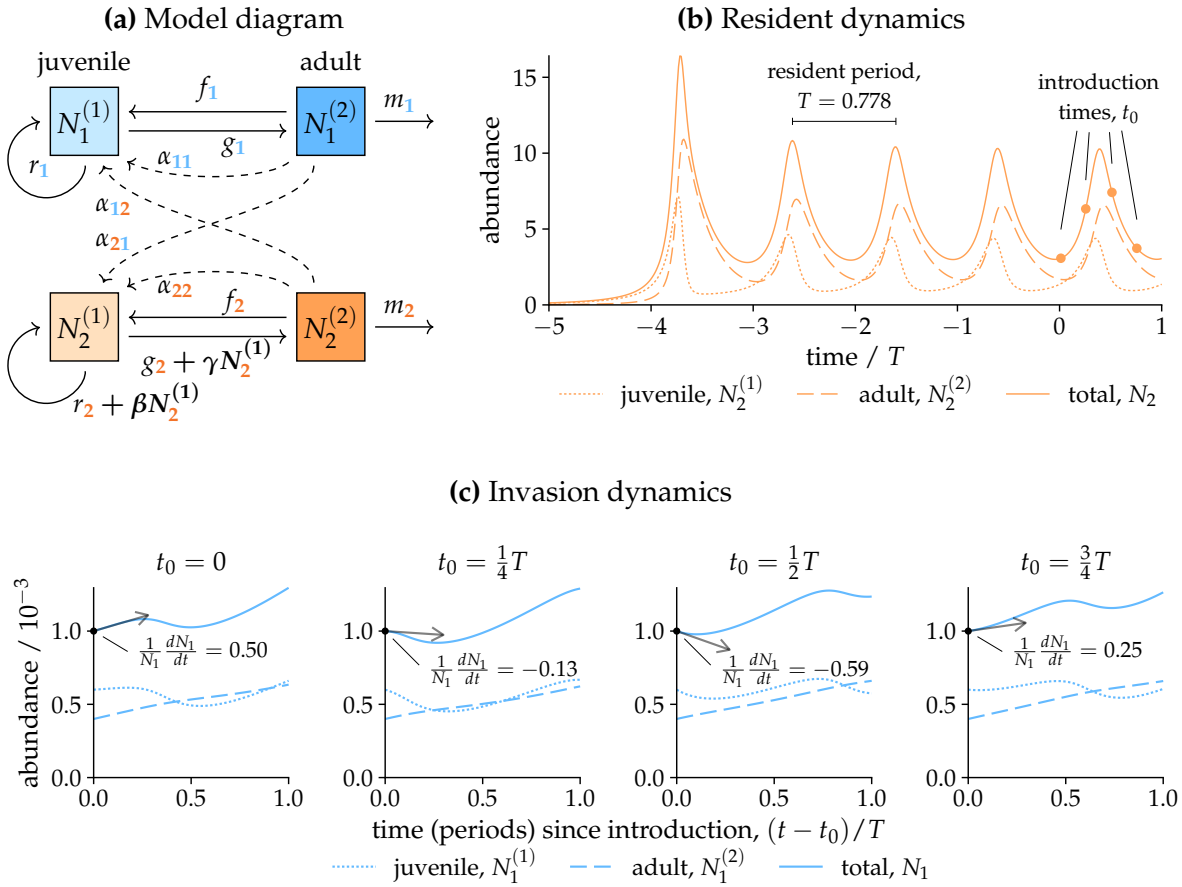


Figure 4: Model extension with cyclic resident dynamics. **(a)** Model diagram of the extended toy model in which both species 1 (blue) and 2 (orange) exhibit stage structure: juveniles $N_i^{(1)}$ and adults $N_i^{(2)}$ undergo maturation g_i, juvenile replenishment r_i, reproduction f_i, and mortality m_i (solid arrows). Each species' juvenile-to-juvenile transition is negatively affected by competition (α_{ij}) from adults of both species (dashed arrows). Additionally, juveniles of species 2 facilitate the rates of their juvenile-to-juvenile transitions and maturation according to coefficients β and γ, respectively. **(b)** Time series of species 2 when growing alone as the resident, exhibiting a stable limit cycle with period $T = 0.778$. Time (x-axis) is scaled by T, such that one unit represents one complete resident cycle, and zero indicates the beginning of invasion trials (line segments indicate the four introduction times used in panel (c)). **(c)** Initial dynamics of species 1's abundance (y-axis) when introduced at different points in the resident's cycle: $t_0 = 0$, $T/4$, $T/2$, or $3T/4$ (from left to right; gray tangent arrows indicate the initial rate of change of the total population). Here, the x-axis shows time since introduction, $(t - t_0)/T$, scaled by the resident period. In all time series, line types represent different population stages: juveniles $N_i^{(1)}$ (dotted), adults $N_i^{(2)}$ (dashed), and total N_i (solid). The initial population is as in Figure 1, $N_1^{(1)} + N_1^{(2)} = 10^{-3}$, and all simulations start with 40% adults. Parameter values: $f_2 = 3$, $g_2 = 1$, $m_2 = 5$, $\alpha_{12} = 0.5$, $\alpha_{21} = 1$, $\alpha_{22} = 5$, $\beta = 5$, $\gamma = 3$; all other parameters are as in Figure 1.

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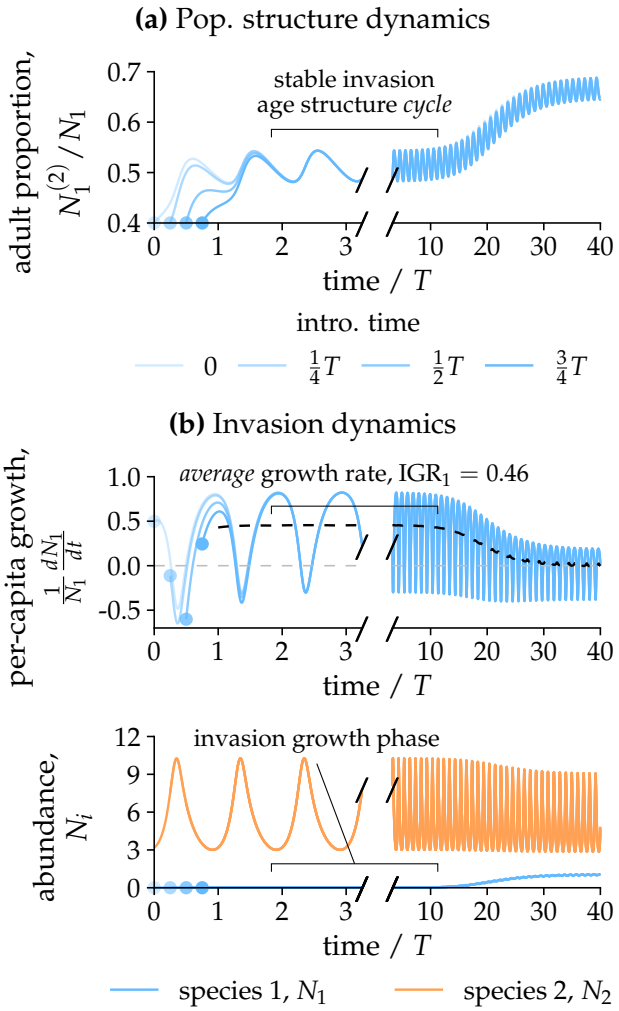


Figure 5: Fast and slow dynamics in the cyclic model. As in Figure 2, we plot the temporal trajectory of (a) population structure (the proportion of adults within the entire population) and (b) per-capita growth rate for the invader population (top) and the total species abundance (bottom). Parameters and invasion settings are as in Figure 4; blue colors indicate different invasion timing (light to dark blue lines: early to late invasion timing). Note that the resolution differs before (i.e., during the initial invasion) and after (i.e., subsequently approaching long-term cyclic dynamics) the break in the axis (after $t/T = 3$). The period when the invader reaches the same cyclic dynamics while remaining sufficiently rare gives the relevant invasion growth rate (IGR = 0.46; dashed line in (b) indicates the moving-window average of the per-capita growth rate).

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Rigorous theory for periodic systems: Floquet theory Just as in equilibrium systems, methods based on matrix population models (Method 2) provide an efficient and mathematically rigorous way to quantify invasion growth rates in systems with cyclic resident dynamics. In this case, transitions between the invader's stages now vary periodically as competitive pressure from the resident fluctuates over time. Fortunately, a mathematical technique known as Floquet theory shows that these periodic dynamics are closely related to the constant linear dynamics introduced in Box 2, providing a powerful technique for analyzing models of diverse natural phenomena (e.g., Barone *et al.*, 1977; Klausmeier, 2008). In Box 3 and Appendix S9, we introduce the key results of this theory and show how to apply them to calculate invasion growth rates. Crucially, Floquet theory shows that the long-term dynamics of an invader (Fig. 6a, left panel) are dominated by an exponential growth term (middle panel), modulated by periodic changes in population structure around this long-term growth (right panel). Thus, Floquet theory provides the periodic analog of the matrix-population result (Box 2) by showing why population structure converges to a stable limit cycle during invasion. Moreover, it reduces the complicated dynamics of the full system to two matrices: the monodromy matrix $\mathbf{M}$ (Fig. 6b, left panel), which describes the population's growth over discrete periods, and the Floquet exponent matrix $\mathbf{B}$ (right panel), which gives the corresponding continuous-time population growth. As we show in a worked numerical example below, calculating the leading eigenvalues of these matrices is a mathematically exact (and often much faster) method to find the invasion growth rates which we obtained by simulation in Method 3.1.

Box 3. Review: Floquet theory for growth in periodic environments

A well-established body of theory based on the work of Floquet (1883) allows us to extend the results of Box 2 to systems where transitions between population stages vary cyclically over time. Here, we summarize the application of *Floquet theory* to invasion analysis; see also Appendix S9 and Klausmeier (2008) for other ecological applications of the theory.

Periodic linear differential equation system Rather than a linear dynamical system with a constant transition matrix, we allow the transition matrix $\mathbf{C}(t)$ to vary cyclically with time such that dynamics obey a *periodic linear system*:

$$\frac{d\vec{N}}{dt} = \mathbf{C}(t)\vec{N}, \tag{B3.1}$$

for some period T such that $\mathbf{C}(t) = \mathbf{C}(t+T)$.

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Box 3 (continued)

Discrete-time reduction: the monodromy matrix and Floquet multipliers Floquet (1883) showed that the complicated dynamics of the time-dependent system (eq. B3.1) are closely related to the solutions of the constant linear systems we already know how to solve (Box 2). In particular, population change over entire period lengths (e.g., $t = 0, T, 2T, \ldots$) can be represented by a constant matrix $\mathbf{M}$ (Fig. 6b, left plot),

$$\vec{N}(t+T) = \mathbf{M}\,\vec{N}(t), \tag{B3.2}$$

where $t = kT$ for some whole number of periods k. Because the matrix $\mathbf{M}$ represents the combined effect of population transitions after the system has "run around" one full period, it is known as the *monodromy matrix* (from Greek roots meaning "single" and "run"). Effectively, eq. B3.2 represents dynamics in discrete time—much like classic discrete-time matrix population models (Lefkovitch, 1965; Leslie, 1945). Thus, the long-term population dynamics are determined by the leading eigenvalue ρ of $\mathbf{M}$ (known as the dominant *Floquet multiplier*), giving the factor by which the population grows over one period.

Continuous-time decomposition and the Floquet exponent Because eq. B3.2 dictates that solutions grow exponentially over entire periods, the continuous-time dynamics of eq. B3.1 also grow exponentially in the long run, but fluctuate within each period. The Floquet–Lyapunov theorem (Floquet, 1883; Lyapunov, 1907) shows that dynamics can be reduced to a continuous-time linear system $\frac{d\vec{y}}{dt} = \mathbf{B}\vec{y}$ (Fig. 6b, right plot), with transition matrix $\mathbf{B}$, modified by some cyclically varying transformation with period T. Accordingly, after transient dynamics decay, the long-term dynamics can be decomposed into two components (Fig. 6a): underlying exponential growth or decline at some long-term growth rate λ (analogous to eq. B2.2 in Box 2) times fluctuation in abundance captured by a periodically varying vector $\vec{p}(t)$:

$$\vec{N}(t) \approx a e^{\lambda t}\vec{p}(t). \tag{B3.3}$$

Here, λ is the leading eigenvalue of $\mathbf{B}$, known as the *Floquet exponent* of the system.

Relating the Floquet exponent to the Floquet multiplier For invasion analysis, we are primarily interested in the Floquet exponent λ, which gives the invader's long-term growth rate. Though it is possible to reconstruct the matrix $\mathbf{B}$ from the monodromy matrix $\mathbf{M}$ (Appendix S3, Algorithm S3.7), we can avoid this extra step if we are only interested in its eigenvalue λ. Since discrete-time growth with factor ρ per period describes the same system as continuous-time growth with

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Box 3 (continued)

rate λ (i.e., $\rho^k = e^{\lambda k T}$), we can directly relate the Floquet exponent λ to the Floquet multiplier ρ:

$$\lambda = \frac{\log \rho}{T}. \tag{B3.4}$$

Numerically solving for invasion growth rates Floquet theory provides a simple and efficient method to numerically compute the monodromy matrix $\mathbf{M}$ (Appendix S3, Algorithm S3.5). For a system with n stages, we need information from at least n initial conditions to fully reconstruct $\mathbf{M}$. The most convenient choice is to use a series of unit vectors $\vec{e}_1, \vec{e}_2, \ldots, \vec{e}_n$ (the standard basis vectors), where a single stage has population one, and the remaining stages are absent. We obtain the monodromy matrix by integrating each of these n starting conditions from time $t = 0$ to T and pasting together the resulting vectors:

$$\mathbf{M} = \begin{bmatrix} | & | & & | \\ \vec{N}_{\vec{e}_1}(T) & \vec{N}_{\vec{e}_2}(T) & \cdots & \vec{N}_{\vec{e}_n}(T) \\ | & | & & | \end{bmatrix}, \tag{B3.5}$$

where $\vec{N}_{\vec{e}_i}(T)$ denotes the outcome when starting with only stage i. The invasion growth rate is then obtained by calculating the leading eigenvalue of $\mathbf{M}$ (the dominant Floquet multiplier ρ) and converting it to the Floquet exponent λ (eq. B3.4; see also Appendix S3, Algorithm S3.6).

Method 3.2: Calculating invasion growth rates using Floquet theory For invasion analysis, we calculate the Floquet exponent λ in Box 3. Applying the recipe in Box 3 for the specific model and parameters in Fig. 6, we construct the monodromy matrix $\mathbf{M}$ from numerical simulations of the invasion system over one period (from $t = 0$ to T). Specifically, we define a new system where the invader does not affect the resident's dynamics (see Appendix S3, Algorithms S3.3–S3.4), and simulate this system for one period starting from two different initial conditions. With an initial population of only juveniles $\vec{N}(0) = [1\ 0]^\top$, we obtain the first column: $\vec{N}_{\vec{e}_1}(T) = [0.400\ 0.334]^\top$. Starting from only adults ($\vec{N}(0) = [0\ 1]^\top$), we obtain the second column $\vec{N}_{\vec{e}_2}(T) = [1.051\ 1.084]^\top$. Combining these columns (eq. B3.5), we find that the monodromy matrix for time $t = 0$ is:

$$\mathbf{M} = \begin{bmatrix} 0.400 & 1.051 \\ 0.334 & 1.084 \end{bmatrix}. \tag{13}$$

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The leading eigenvalue of $\mathbf{M}$ gives the dominant Floquet multiplier $\rho = 1.426$, which describes the population growth over one period. Finally, applying eq. B3.4, the continuous-time invasion growth rate is given by the associated Floquet exponent, $\lambda = \log \rho / T = 0.456$, which indeed matches the invasion growth rate estimated from direct simulation (Method 3.1; Fig. 5b). As calculating $\mathbf{M}$ only requires integration over a single period, this approach is faster and more accurate than simulating many periods until convergence (approximately $40\times$ faster in our code).

7 Discussion

Invasion analysis has been a cornerstone for studying species coexistence as it provides a direct criterion for species persistence: a species with a positive invasion growth rate can recover from low abundance in the environment created by its competitors. Here, we developed a general framework for applying invasion analysis when the invading species is structured into multiple states. The organizing principle is that invasion growth in structured populations should be treated as a long-term property of the rare structured population, rather than as the instantaneous growth of any particular initial state distribution. Building on this insight, we presented two complementary analytical approaches: (1) timescale separation and variable transformations, which provide a biologically intuitive route to invasion growth rates, and (2) matrix population theory, which provides an eigenvalue-based mathematical description of the same convergence process. We also showed that the same logic can guide numerical estimation of invasion growth rates under complex dynamics where analytical expressions are unavailable. Together, these approaches provide an integrated toolkit linking insights from matrix-population theory to community-level ecological questions about species coexistence.

The power of invasion analysis comes from simplifying the full community dynamics: otherwise complicated models are reduced to a smaller and much more tractable subsystem (i.e., the invasion scenario). The pieces of our analytical toolkit give mathematically equivalent answers, but researchers can choose an appropriate method based on familiarity, arithmetic convenience, and the particular information required to answer their questions. In many studies, the key information is not the exact magnitude of the invasion growth rate, but rather its sign (Ellner *et al.*, 2020; Pande *et al.*, 2020). In this spirit, both analytical approaches presented here include shortcuts for deriving invasion criteria directly, without solving the full expression of the invasion growth rate (see also Appendix S2, subsection S2.1.2 for another mathematical shortcut). Nonetheless, if the focal population has relatively few stages, and a researcher is specifically interested in the exact population structure realized during invasion, direct analytical timescale separation (Method 1.1) may be useful for its conceptual simplicity. Meanwhile, a researcher working on a population with many stages must focus on deriving a simple

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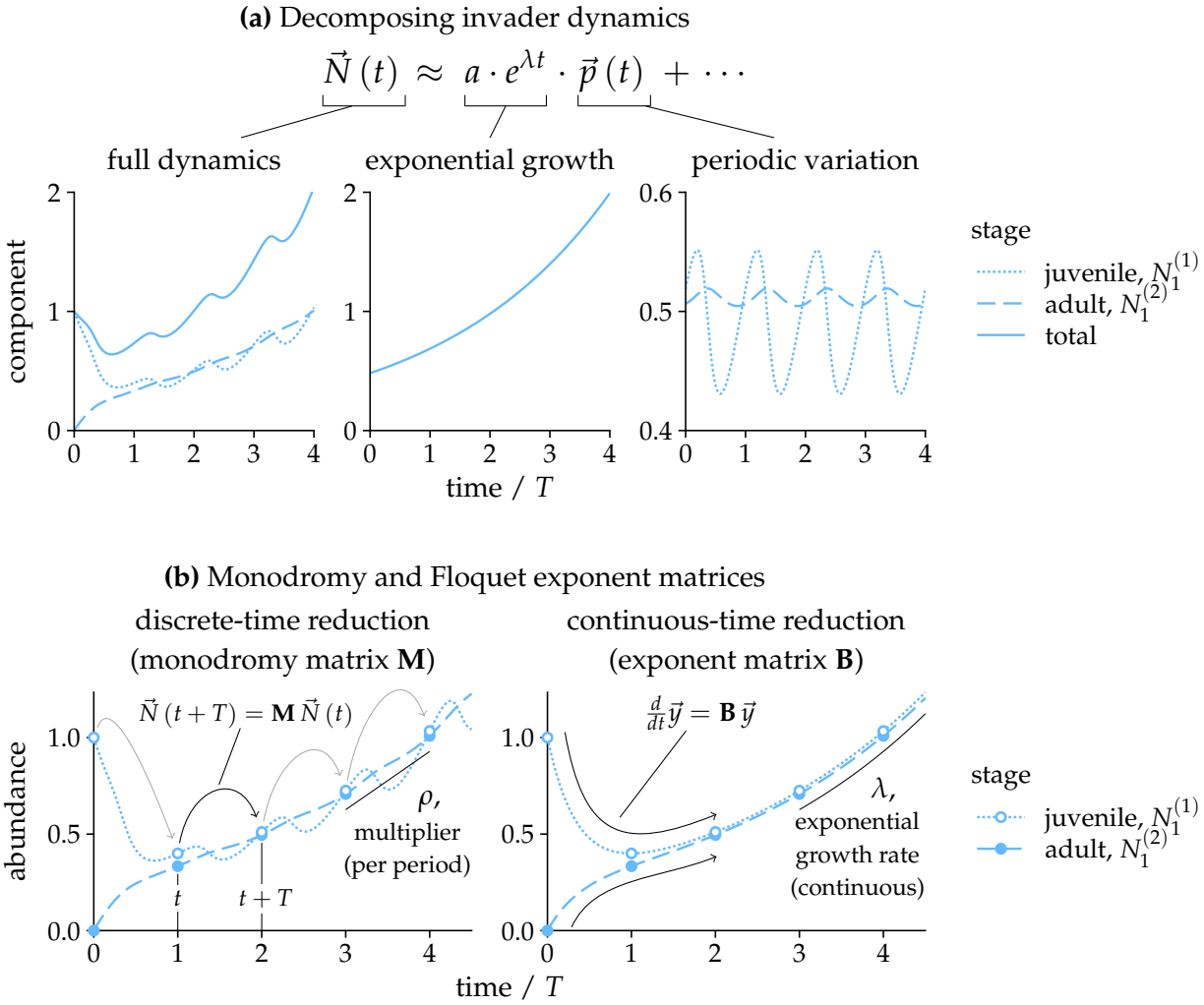


Figure 6: Applying Floquet theory to the cyclic model. Model and parameters as in Figure 5, except that the initial invader population ($t_0 = 0$) consists of juveniles only. **(a)** Using the procedure in Box 3 (eq. B3.3), we can decompose invasion dynamics (left panel) into exponential growth at a rate λ (middle panel) and a periodic component $\vec{p}(t)$ (right panel), which captures periodic population fluctuations around the overall exponential trend. **(b)** Visualization of the key mathematical objects in Floquet theory. The left plot shows how the full dynamics (lines; same as those shown in **(a)**) reduce to a simple recurrence relation (Box 3, eq. B3.2) when only considering abundances after a whole number of periods (points). Multiplication by the monodromy matrix **M** moves the system forward (arrows) by one period T. In the long term, each step multiplies abundances by ρ, the dominant Floquet multiplier (leading eigenvalue of **M**). The right plot shows the linear system (lines) associated with the matrix **B**. The system passes through the same points in the previous plot, but now advances in continuous time (arrows). In the long term, this linear system (and thus the original system as well) grows at a continuous exponential growth rate λ, the dominant Floquet exponent (leading eigenvalue of **B**). Components of the Floquet analysis are calculated according to Appendix S3, Algorithms S3.5–S3.8.

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invasion condition, since the exact expression for invasion growth rate may not be analytically solvable. Here, a reasonable approach would be to search first for a simple strategic linear combination (Method 1.2) and, if this fails, to fall back on the full stability conditions (Method 2.2). A well-chosen analytical approach pays off: our case studies highlight that even in complicated models, it is often possible to identify surprisingly simple rules for population persistence. For example, in the cannibalistic herbivores model, while the level of aggression matters for the magnitude of the invasion growth rate, invasion success itself is determined solely by the demographic effect of consuming intraguild prey (b_{21}). Thus, the power of analytical approaches lies in their ability to translate ecological complexity into interpretable invasion criteria.

By rigorously extending invasion analysis to structured populations, our framework reinforces the role of invasion growth rates as a common currency for ecological success and persistence (Grainger *et al.*, 2019). Indeed, the same approaches apply to systems beyond those in the main text, including chaotic dynamics (Appendix S8) and models in discrete time (Appendix S1). Appendix S10 illustrates the latter by deriving simple invasion conditions for plants experiencing discrete growth seasons, a workhorse system for studies on coexistence theory (e.g., Song *et al.*, 2021; Werner *et al.*, 2025). This opens the door to a variety of analyses. As a starting point, invasion analysis enables researchers to apply the well-known mutual invasibility criteria for coexistence (Box 1). Further decomposed, invasion growth rates can be used to partition coexistence outcomes into contributions from different ecological mechanisms (Barabás *et al.*, 2018; Ellner *et al.*, 2019; Letten *et al.*, 2018) or summarize them into niche and fitness differences (Chesson, 2000; Spaak & De Laender, 2020). As current work extends and compares alternative partitioning approaches across increasingly complex communities (Song & Spaak, 2024; Spaak *et al.*, 2023; Wan *et al.*, 2025), starting from simple, interpretable expressions for invasion growth rates will be particularly important. Analytical solutions for the invasion of structured populations may also help resolve long-standing questions in coexistence theory. For instance, while many well-known mechanisms contributing to coexistence involve population structure, e.g., annual–perennial life-history differences (Werner *et al.*, 2025), previous analytical treatments have relied on strong simplifying assumptions (DeMalach *et al.*, 2021; Song *et al.*, 2021). Finally, we caution that there are exceptional cases where the mutual invasibility criterion is not directly applicable. Despite this limitation, invasion analysis still forms the basis for newer approaches addressing these shortcomings, including in communities with positive density dependence (i.e., strong Allee effects: Schreiber *et al.*, 2019; Walker & Gilbert, 2025) or complicated assembly rules (i.e., the invasion graph approach: Brennan & Schreiber, 2026; Hofbauer & Schreiber, 2010; Spaak & Schreiber, 2023).

Our framework is closely related to established mathematical procedures in diverse fields, including complex systems, epidemiology, and evolutionary biology. Our nu-

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merical procedure for estimating invasion growth rate in cyclic or chaotic resident dynamics is closely related to standard algorithms for estimating dominant Lyapunov exponents (Benettin *et al.*, 1980), reflecting the fact that the invasion growth rate is a special case of this more general metric. In permanence theory, these exponents characterize the long-term growth rate of a rare structured population in a fluctuating environment and therefore provide a natural persistence criterion (Hofbauer & Schreiber, 2010; Roth & Schreiber, 2014). Meanwhile, our linear algebra-based approach mirrors a widely used approach from disease ecology and epidemiology, the next-generation matrix (Diekmann *et al.*, 1990, 2009). Developed to derive the basic reproduction number in compartmental epidemic models, this technique evaluates the matrix of transition rates between infected states at the disease-free equilibrium. Finally, when population composition closely tracks environmental change, timescale separation techniques have also been used to derive effective population-level responses of structured species to changing environments (Kussell & Leibler, 2005; Letten *et al.*, 2024).

By placing our work within a broader family of approaches that reduce complex state-dependent dynamics to interpretable criteria for long-term population growth, these connections point towards new syntheses between community ecology and other subdisciplines. For instance, because invasion growth rate is also the fundamental measure of fitness in evolutionary biology (Yamamichi *et al.*, 2022), invasion analysis of stage-structured mutants provides important general insight into eco-evolutionary feedbacks (Bukkuri & Brown, 2023; Iritani & Day, 2026). Beyond the core of population biology, invasion analysis is being applied to an ever-expanding set of questions, from biological invasions (Iles *et al.*, 2016; MacDougall *et al.*, 2009) to biogeochemical cycling (Smith & Wan, 2019; Xu *et al.*, 2026) and responses to climate change (Usinowicz & Levine, 2018). As such, we anticipate a growing number of research programs will incorporate population-level biological detail into models of species interactions. Amid this increasing complexity, ecologists must retain the interpretability that has made simple community models so powerful. We hope that general theory from population ecology, as synthesized here, will continue to provide ecologists with a guiding thread towards general understanding.

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References

Barabás, G., D'Andrea, R. & Stump, S.M. (2018) Chesson's coexistence theory. *Ecological Monographs*, **88**, 277–303. https://doi.org/10.1002/ecm.1302.

Barone, S.R., Narcowich, M.A. & Narcowich, F.J. (1977) Floquet theory and applications. *Physical Review A*, **15**, 1109–1125. https://doi.org/10.1103/PhysRevA.15.1109.

Benaïm, M. & Schreiber, S.J. (2009) Persistence of structured populations in random environments. *Theoretical Population Biology*, **76**, 19–34. https://doi.org/10.1016/j.tpb.2009.03.007.

Benettin, G., Galgani, L., Giorgilli, A. & Strelcyn, J.-M. (1980) Lyapunov Characteristic Exponents for smooth dynamical systems and for hamiltonian systems; A method for computing all of them. Part 2: Numerical application. *Meccanica*, **15**, 21–30. https://doi.org/10.1007/BF02128237.

Brennan, J. & Schreiber, S.J. (2026) Using Modern Coexistence Theory to Understand Community Disassembly. *Ecology Letters*, **29**, e70424. https://doi.org/10.1111/ele.70424.

Bukkuri, A. & Brown, J.S. (2023) Integrating eco-evolutionary dynamics into matrix population models for structured populations: Discrete and continuous frameworks. *Methods in Ecology and Evolution*, **14**, 1475–1488. https://doi.org/10.1111/2041-210X.14111.

Caswell, H. (2001) *Matrix Population Models: Construction, Analysis, and Interpretation*. Sinauer Associates, Sunderland, Mass, 2nd ed.

Chesson, P. (2000) Mechanisms of maintenance of species diversity. *Annual review of Ecology and Systematics*, **31**, 343–366.

Chesson, P. & Huntly, N. (1997) The Roles of Harsh and Fluctuating Conditions in the Dynamics of Ecological Communities. *The American Naturalist*, **150**, 519–553. https://doi.org/10.1086/286080.

De Roos, A.M. & Persson, L. (2013) *Population and Community Ecology of Ontogenetic Development*. Princeton University Press, 1st ed.

De Roos, A.M., Schellekens, T., Van Kooten, T., Van De Wolfshaar, K., Claessen, D. & Persson, L. (2008) Simplifying a physiologically structured population model to a stage-structured biomass model. *Theoretical Population Biology*, **73**, 47–62. https://doi.org/10.1016/j.tpb.2007.09.004.

DeMalach, N., Shnerb, N. & Fukami, T. (2021) Alternative States in Plant Communities Driven by a Life-History Trade-Off and Demographic Stochasticity. *The American Naturalist*, **198**, E27–E36. https://doi.org/10.1086/714418.

bioRxiv preprint doi: https://doi.org/10.64898/2026.11.04.763670; this version posted November 6, 2026. The copyright holder for this preprint (which was not certified by peer review) is the author/funder, who has granted bioRxiv a license to display the preprint in perpetuity. It is made available under a CC-BY 4.0 International license.

Diekmann, O., Heesterbeek, J.A.P. & Metz, J.A.J. (1990) On the definition and the computation of the basic reproduction ratio r 0 in models for infectious diseases in heterogeneous populations. *Journal of mathematical biology*, **28**, 365–382.

Diekmann, O., Heesterbeek, J.A.P. & Roberts, M.G. (2009) The construction of next-generation matrices for compartmental epidemic models. *Journal of the royal society interface*, **7**, 873.

Ellner, S.P., Snyder, R.E. & Adler, P.B. (2016) How to quantify the temporal storage effect using simulations instead of math. *Ecology Letters*, **19**, 1333–1342. https://doi.org/10.1111/ele.12672.

Ellner, S.P., Snyder, R.E., Adler, P.B. & Hooker, G. (2019) An expanded modern coexistence theory for empirical applications. *Ecology Letters*, **22**, 3–18. https://doi.org/10.1111/ele.13159.

Ellner, S.P., Snyder, R.E., Adler, P.B., Hooker, G. & Schreiber, S.J. (2020) Technical Comment on Pande et al. (2020): Why invasion analysis is important for understanding coexistence. *Ecology Letters*, **23**, 1721–1724. https://doi.org/10.1111/ele.13580.

Floquet, G. (1883) Sur les équations différentielles linéaires à coefficients périodiques. *Annales scientifiques de l'École Normale Supérieure*, **12**, 47–88. https://doi.org/10.24033/asens.220.

Grainger, T.N., Levine, J.M. & Gilbert, B. (2019) The invasion criterion: a common currency for ecological research. *Trends in ecology & evolution*, **34**, 925–935.

Gurney, W. & Nisbet, R. (1985) Fluctuation periodicity, generation separation, and the expression of larval competition. *Theoretical Population Biology*, **28**, 150–180. https://doi.org/10.1016/0040-5809(85)90026-7.

Hofbauer, J. & Schreiber, S.J. (2010) Robust permanence for interacting structured populations. *Journal of Differential Equations*, **248**, 1955–1971. https://doi.org/10.1016/j.jde.2009.11.010.

Iles, D.T., Salguero-Gómez, R., Adler, P.B. & Koons, D.N. (2016) Linking transient dynamics and life history to biological invasion success. *Journal of Ecology*, **104**, 399–408. https://doi.org/10.1111/1365-2745.12516.

Iritani, R. & Day, T. (2026) Evolutionary invasion analysis for structured populations: a synthesis. https://doi.org/10.64898/2026.03.23.713828.

Ke, P.-J. & Levine, J.M. (2021) The temporal dimension of plant-soil microbe interactions: mechanisms promoting feedback between generations. *The American Naturalist*, **198**, E80–E94. https://doi.org/10.1086/715577.

Klausmeier, C.A. (2008) Floquet theory: a useful tool for understanding nonequilibrium dynamics. *Theoretical Ecology*, **1**, 153–161. https://doi.org/10.1007/s12080-008-0016-2.

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Kussell, E. & Leibler, S. (2005) Phenotypic diversity, population growth, and information in fluctuating environments. *Science*, **309**, 2075–2078.

Lefkovitch, L.P. (1965) The Study of Population Growth in Organisms Grouped by Stages. *Biometrics*, **21**, 1. https://doi.org/10.2307/2528348.

Leslie, P.H. (1945) On the use of matrices in certain population mathematics. *Biometrika*, **33**, 183–212. https://doi.org/10.1093/biomet/33.3.183.

Letten, A.D., Dhami, M.K., Ke, P.-J. & Fukami, T. (2018) Species coexistence through simultaneous fluctuation-dependent mechanisms. *Proceedings of the National Academy of Sciences*, **115**, 6745–6750. https://doi.org/10.1073/pnas.1801846115.

Letten, A.D., Yamamichi, M., Richardson, J.A. & Ke, P.-J. (2024) Microbial Dormancy Supports Multi-Species Coexistence Under Resource Fluctuations. *Ecology Letters*, **27**, e14507. https://doi.org/10.1111/ele.14507.

Lyapunov, A.M. (1907) Problème général de la stabilité du mouvement. *Annales de la Faculté des sciences de l'Université de Toulouse pour les sciences mathématiques et les sciences physiques*, **9**, 203–474.

MacDougall, A.S., Gilbert, B. & Levine, J.M. (2009) Plant invasions and the niche. *Journal of Ecology*, **97**, 609–615. https://doi.org/10.1111/j.1365-2745.2009.01514.x.

Nakazawa, T. (2015) Ontogenetic niche shifts matter in community ecology: a review and future perspectives. *Population Ecology*, **57**, 347–354. https://doi.org/10.1007/s10144-014-0448-z.

Nakazawa, T. & Huang, C.-G. (2016) Two-species metacommunity dynamics mediated by habitat preference. *Oikos*, **125**, 1334–1341. https://doi.org/10.1111/oik.03101.

Otto, S.P. & Day, T. (2011) *A Biologist's Guide to Mathematical Modeling in Ecology and Evolution*. Princeton University Press.

Pande, J., Fung, T., Chisholm, R. & Shnerb, N.M. (2020) Mean growth rate when rare is not a reliable metric for persistence of species. *Ecology Letters*, **23**, 274–282. https://doi.org/10.1111/ele.13430.

Polis, G.A., Myers, C.A. & Holt, R.D. (1989) The Ecology and Evolution of Intraguild Predation: Potential Competitors That Eat Each Other. *Annual Review of Ecology and Systematics*, **20**, 297–330. https://doi.org/10.1146/annurev.es.20.110189.001501.

Price, P.W., Westoby, M., Rice, B., Atsatt, P.R., Fritz, R.S., Thompson, J.N. & Mobley, K. (1986) PARASITE MEDIATION IN ECOLOGICAL INTERACTIONS. *Annual Review of Ecology and Systematics*, **17**, 487–505. https://doi.org/10.1146/annurev.es.17.110186.002415.

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Roth, G. & Schreiber, S.J. (2014) Persistence in fluctuating environments for interacting structured populations. *Journal of Mathematical Biology*, **69**, 1267–1317. https://doi.org/10.1007/s00285-013-0739-6.

Schreiber, S.J., Yamamichi, M. & Strauss, S.Y. (2019) When rarity has costs: coexistence under positive frequency-dependence and environmental stochasticity. *Ecology*, **100**, e02664.

Shoemaker, L.G., Barner, A.K., Bittleston, L.S. & Teufel, A.I. (2020) Quantifying the relative importance of variation in predation and the environment for species coexistence. *Ecology Letters*, **23**, 939–950. https://doi.org/10.1111/ele.13482.

Smith, G.R. & Wan, J. (2019) Resource-ratio theory predicts mycorrhizal control of litter decomposition. *New Phytologist*, **223**, 1595–1606. https://doi.org/10.1111/nph.15884.

Song, C. & Spaak, J.W. (2024) Trophic tug-of-war: coexistence mechanisms within and across trophic levels. *Ecology Letters*, **27**, e14409.

Song, C., Uricchio, L.H., Mordecai, E.A. & Saavedra, S. (2021) Understanding the emergence of contingent and deterministic exclusion in multispecies communities. *Ecology Letters*, **24**, 2155–2168. https://doi.org/10.1111/ele.13846.

Spaak, J.W., Ke, P.-J., Letten, A.D. & De Laender, F. (2023) Different measures of niche and fitness differences tell different tales. *Oikos*, **2023**, e09573.

Spaak, J.W. & Schreiber, S.J. (2023) Building modern coexistence theory from the ground up: the role of community assembly. *Ecology Letters*, **26**, 1840–1861.

Spaak, J.W. & De Laender, F. (2020) Intuitive and broadly applicable definitions of niche and fitness differences. *Ecology Letters*, **23**, 1117–1128. https://doi.org/10.1111/ele.13511.

Sprott, J.C. (2010) *Elegant Chaos: Algebraically Simple Chaotic Flows*. World Scientific, Singapore.

Usinowicz, J. & Levine, J.M. (2018) Species persistence under climate change: a geographical scale coexistence problem. *Ecology Letters*, **21**, 1589–1603. https://doi.org/10.1111/ele.13108.

Walker, E.J. & Gilbert, B. (2025) Allee effects and coexistence. *The American Naturalist*, **205**, 559–571.

Wan, J., Ke, P.-J., Hordijk, I., Bialic-Murphy, L. & Crowther, T.W. (2025) Functional coexistence theory: Identifying mechanisms linking biodiversity and ecosystem function. *Ecological Monographs*, **95**, e70033. https://doi.org/10.1002/ecm.70033.

Werner, C.M., Hallett, L.M. & Shoemaker, L.G. (2025) Fluctuation-Dependent Coexistence of Stage-Structured Species. *The American Naturalist*, **205**, 327–341. https://doi.org/10.1086/733382.

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Xu, L., Sun, X. & Zakem, E.J. (2026) Dependency–competition tradeoffs structure microbial niches and nitrogen cycling. *ISME Communications*, **6**, ycag134. https://doi.org/10.1093/ismeco/ycag134.

Yamamichi, M., Gibbs, T.L. & Levine, J.M. (2022) Integrating eco-evolutionary dynamics and modern coexistence theory. *Ecology Letters*, **25**, 2091–2106. https://doi.org/10.1111/ele.14078.

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Statements and Declarations

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Competing interests

The authors declare no competing financial interests.

Author Contributions

JW and PJK conceived the study. JW led the analysis, with contributions from HCL, GYC, SCH, JZL, CTT, and PJK. JW and PJK wrote the manuscript with input from all authors.

Data Availability

No original data were presented in this manuscript. All code and inputs required for the simulations and re-analyses in this manuscript are publicly available at `https://github.com/joe-wan/structured-invasion-code/`.

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Supporting Information for

Invasion analysis for structured populations: analytical principles and numerical workflows

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S1 General framework: matrix population models

When individuals of a species differ from each other in some way, we must model the population in more detail than allowed by a single state variable (main text, Box 2). One way to do so is to consider transitions of species i between m_i different discrete states, representing age, size, habitat, disease status, or some other condition(s) of interest. In this Appendix, we describe the general application of matrix population models to these systems.

S1.1 Continuous-time dynamics

Representing the abundance of species i in state k as $N_i^{(k)}$, we can write a continuous-time system as

$$\frac{dN_i^{(k)}}{dt} = \sum_{\ell=1}^{m_i} c_i^{(k,\ell)} \left(N_1^{(1)}, N_1^{(2)}, \ldots, N_n^{(m_n-1)}, N_n^{(m_n)}\right) \cdot N_i^{(\ell)}, \tag{S1.1}$$

where $c_i^{(k,\ell)}$ is the transition rate from state ℓ to state k of species i, which may depend on the abundance of all states of all species in the system. We can also write the state distribution of species i as an m_i-dimensional vector

$$\vec{N}_i = \begin{bmatrix} N_i^{(1)} \\ \vdots \\ N_i^{(m_i)} \end{bmatrix}, \tag{S1.2}$$

such that the population dynamics of species i can be represented by the vector differential equation

$$\frac{d\vec{N}_i}{dt} = \mathbf{C}_i \vec{N}_i, \tag{S1.3}$$

where $\mathbf{C}_i$ is the $m_i \times m_i$ transition matrix between states of species i, dependent on the state distributions of all species in the system:

$$\mathbf{C}_i\left(\vec{N}_1, \ldots, \vec{N}_n\right) = \begin{bmatrix} c_i^{(1,1)} & \cdots & c_i^{(1,m_i)} \\ \vdots & \ddots & \vdots \\ c_i^{(m_i,1)} & \cdots & c_i^{(m_i,m_i)} \end{bmatrix}. \tag{S1.4}$$

When conducting invasion analysis, we consider some resident state where $\vec{N}_i = \vec{0}$ and the other species are at an equilibrium $\vec{N}_j = \vec{N}_j^*$ for all $j \neq i$. Evaluating the transition

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matrix at this resident equilibrium gives

$$\mathbf{C}_i^* = \mathbf{C}_i \left(\vec{N}_1^*, \ldots, \vec{N}_{i-1}^*, \vec{0}, \vec{N}_{i+1}^*, \ldots, \vec{N}_n^* \right). \tag{S1.5}$$

When we are near this invasion state, $\mathbf{C}_i$ is near some value which does not depend on the invader abundance, which we will denote $\mathbf{C}_i^*$. In other words, the transition matrix is independent of the invader's dynamics because it is too rare to exert feedbacks on its own population dynamics. Equivalently, $\mathbf{C}_i^*$ can be viewed as the Jacobian matrix for the invading species, evaluated at the resident equilibrium. In the following text, we will focus on a particular species i in the invasion condition, writing $\vec{N}$ for $\vec{N}_i$ and $\mathbf{C}$ for $\mathbf{C}_i^*$ in the interest of brevity.

S1.2 Discrete-time dynamics

Though the main text focuses on continuous-time systems, it is worth also considering the discrete-time analog of this problem. Writing $\vec{N}$ for $\vec{N}_i$ as above, we can write the abundances of a species' stages at the next time step $t+1$ in terms of its abundances at the current time t:

$$N^{(k)}(t+1) = \sum_{\ell=1}^{m} a^{(k,\ell)} \left(N_1^{(1)}, N_1^{(2)}, \ldots, N_n^{(m_n-1)}, N_n^{(m_n)} \right) \cdot N^{(\ell)}(t), \tag{S1.6}$$

or in matrix form,

$$\vec{N}(t+1) = \mathbf{A}\,\vec{N}(t), \tag{S1.7}$$

where $\mathbf{A}$ follows the notation used for this class of models in Caswell (2001). Here, considering the invasion scenario as in equation S1.5, $m = m_i$ is the number of stages of the invader, and $a^{(k,\ell)}$ is the expected contribution to stage k at the next time step per individual in stage ℓ at the current time step, evaluated at the resident equilibrium. This class of discrete-time models is well-studied in population ecology. In particular, if the classes $k = 1, 2, \ldots, m_i$ represent age groups, $\mathbf{A}$ is known as the Leslie matrix (after Leslie (1945)); more broadly, if k instead represents developmental, size, or life-history stages, $\mathbf{A}$ may be termed a Lefkovitch matrix (after Lefkovitch, 1965). Standard textbooks (e.g., Caswell, 2001) treat the analyses of these models in terms of $\mathbf{A}$ and its eigenvalues in depth.

Note that this $\mathbf{A}$ has a different interpretation from our continuous-time $\mathbf{C}$: elements on the diagonal represent the members of a particular class which *stay* in that class between times t and $t+1$, as opposed to in the continuous-time formulation above, where diagonal elements reflect change in a particular class. As a formulation more compatible with the continuous-time equation above, we can rewrite the system as a

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difference equation,

$$\underbrace{\vec{N}(t+1) - \vec{N}(t)}_{\Delta\vec{N}(t)} = \mathbf{A}'\,\vec{N}(t)\,, \tag{S1.8}$$

where $\mathbf{A}' = \mathbf{A} - \mathbf{I}$ is obtained by subtracting the identity matrix (i.e., subtracting one from each diagonal entry) to account for the members remaining in this class. Indeed, taking the limit of this difference equation model as the time step becomes small recovers the ordinary differential equation formulation above. To focus on connections between the discrete- and continuous-time formulations, it will be convenient to use discrete-time $\mathbf{A}'$ as an analogue of $\mathbf{C}$ to highlight similarities between analyses of the two types of models.

S1.3 Eigenvalue analysis of state transition matrices

The eigenvalues of $\mathbf{C}$ are the values $\lambda = \lambda_1, \lambda_2, \ldots$ such that

$$\mathbf{C}\vec{w} = \lambda\vec{w} \tag{S1.9}$$

for some non-zero vector $\vec{w}$. The value of $\vec{w}$ is associated with a particular eigenvalue λ and is known as its associated eigenvector. In general, if $\mathbf{C}$ is an $n \times n$ matrix, there will be up to n unique eigenvalues of $\mathbf{C}$, which may be complex-valued. To solve for the eigenvalues, a convenient procedure is to solve the equation

$$\det(\mathbf{C} - \lambda\mathbf{I}) = 0, \tag{S1.10}$$

which gives a polynomial equation of degree n in λ, known as the *characteristic polynomial* of $\mathbf{C}$. Biologically, an eigenvalue/eigenvector pair $\lambda_i, \vec{w}_i$ represents a solution to equation S1.3 where the elements of the population vector $\vec{N}$ remain in a fixed proportion determined by $\vec{w}_i$, but its magnitude grows or shrinks at an exponential rate λ_i; i.e.,

$$\vec{N}(0) = \vec{w}_i; \quad \vec{N}(t) = e^{\lambda_i t}\vec{w}_i. \tag{S1.11}$$

Even if our starting population state $\vec{N}(0)$ is not an eigenvector of $\mathbf{C}$, we will usually be able to represent it as some combination of eigenvectors. The fact that equation S1.3 is a linear system under the invasion scenario in equation S1.5 means that the time-dependent population trajectory is also a combination of the trajectories determined by individual eigenvalues/eigenvectors,

$$\vec{N}(0) = \sum_i k_i\vec{w}_i; \quad \vec{N}(t) = \sum_i k_i e^{\lambda_i t}\vec{w}_i. \tag{S1.12}$$

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In fact, if we assume the eigenvalues are ordered by decreasing real parts, with λ_1 being the eigenvalue with the greatest real part (known as the leading eigenvalue), we see that

$$\vec{N}(t) = k_1 e^{\lambda_1 t}\vec{w}_1 + \underbrace{k_2 e^{\lambda_2 t}\vec{w}_2 + \cdots}_{\text{transient terms}} \tag{S1.13}$$

$$= k_1 e^{\lambda_1 t}\vec{w}_1 + o\left(e^{\lambda_1 t}\right), \tag{S1.14}$$

where the terms associated with the non-leading eigenvalues are transient, because their influence becomes small relative to the first term as time t becomes large. Here, the little-o notation $o(e^{\lambda_1 t})$ indicates that the transient terms grow more slowly than $e^{\lambda_1 t}$. Accordingly, the long-term growth rate is determined by the leading eigenvalue λ_1 (for brevity, we denote it as λ when the context makes it clear, as in the main text) and the long-term population structure approaches $\vec{w}_1$ (similarly abbreviated $\vec{w}$ in the main text), giving the result in eq. B2.2 of the main text. If λ_1 is positive, the population maintains positive growth.

There is an important mathematical complication: generally, some eigenvectors may have components with negative signs. Does this imply the long-term stage structure is not biologically meaningful because it contains negative abundances? Fortunately, it turns out that the special properties of $\mathbf{C}$ guarantee that this problem cannot arise: because the off-diagonal elements of $\mathbf{C}$ are nonnegative, solutions which start positive will remain positive (Box S1.B1). Furthermore, due to the Perron–Frobenius theorem (see Box S1.B2), the leading eigenvalue λ_1 is real and its eigenvector $\vec{w}_1$ has nonnegative components. For sensible population models, $\mathbf{C}$ is irreducible (Box S1.B3) and we furthermore have that $\vec{w}_1$ is uniquely defined up to a constant; i.e., the population's stages approach a single well-defined set of proportions.

In discrete time, we can write

$$\vec{N}(t) = k_1\lambda_1^t\vec{w}_1 + k_2\lambda_2^t\vec{w}_2 + \cdots \tag{S1.15}$$

$$= k_1\left(1+\lambda_1'\right)^t\vec{w}_1 + k_2\left(1+\lambda_2'\right)^t\vec{w}_2 + \cdots, \tag{S1.16}$$

where $\lambda_1, \lambda_2, \ldots$ are the eigenvalues of $\mathbf{A}$, $\lambda_1', \lambda_2', \ldots$ are the eigenvalues of $\mathbf{A}'$, and $\vec{w}_1, \vec{w}_2, \ldots$ are the associated eigenvectors. In an exact analog to the continuous-time result, the population maintains positive growth if the leading eigenvalue λ_1' of $\mathbf{A}'$ is positive (or equivalently, if $\mathbf{A}$'s dominant eigenvalue λ_1 is greater than one).

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Box S1.B1. Special properties of **C**

In biologically realistic models, a particular individual cannot give rise to a negative abundance of individuals in a different stage. This means the off-diagonal elements of the matrix **C**, $c^{(k,\ell)}$ for $k \neq \ell$, should be nonnegative. Biologically, this condition ensures that trajectories starting from nonnegative populations remain nonnegative for all stages. This follows from the fact that **C** is a member of a special class of matrices known as Metzler matrices.

Metzler matrices A matrix **M** where all off-diagonal elements are nonnegative (i.e., $(\mathbf{M})_{ij} \geq 0$ for all $i \neq j$) is known as a Metzler matrix. Named after Metzler (1945), who first applied them in economics, this class of matrices naturally describes the evolution of many systems whose elements take nonnegative values. Consider the system of linear ordinary differential equations associated with a matrix **M**,

$$\frac{d\vec{x}}{dt} = \mathbf{M}\vec{x}. \tag{S1.B1.1}$$

Ważewski (1950) proved that all solutions starting from nonnegative $\vec{x}$ remain nonnegative if and only if **M** is a Metzler matrix. To see this, consider the following necessary and sufficient condition for the system to avoid becoming negative: when x_i, the ith element of $\vec{x}$, is zero, its derivative dx_i/dt must be nonnegative. Writing out the element of the matrix-vector product, this condition is $\sum_j (\mathbf{M})_{ij} x_j \geq 0$; furthermore, since $x_i = 0$, we can write $\sum_{j\neq i} (\mathbf{M})_{ij} x_j \geq 0$. This holds generally for all nonnegative $\vec{x}$ if and only if all non-diagonal entries of **M** are nonnegative; $(\mathbf{M})_{ij} \geq 0$ for $i \neq j$.

Discrete-time nonnegativity In discrete time, let $\vec{x}\,(t+1) = \mathbf{A}\,\vec{x}\,(t)$. Here, $\vec{x}\,(t+1)$ remains nonnegative for all nonnegative $\vec{x}\,(t)$ if and only if **A** is a nonnegative matrix (i.e., all elements of **A** are nonnegative). Equivalently, since $\mathbf{A}' = \mathbf{A} - \mathbf{I}$, all non-diagonal elements of $\mathbf{A}'$ must be nonnegative, and the diagonal elements must be at least -1, i.e., $(\mathbf{A}')_{ii} \geq -1$ for all i. Thus, for the system to remain nonnegative, it is a necessary (but not sufficient) condition that $\mathbf{A}'$ is a Metzler matrix.

Box S1.B2. The Perron–Frobenius theorem

A foundational result in linear algebra, known as the Perron–Frobenius theorem (Frobenius, 1912; Perron, 1907), states that a matrix with nonnegative entries has a real, nonnegative eigenvalue λ^* (1) whose magnitude is greater than or equal to the magnitude of any other eigenvalue, and (2) whose associated eigenvector $\vec{w}^*$

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Box S1.B2 (continued)

can be chosen to have all nonnegative components. Furthermore, if the matrix is irreducible (Box S1.B3), $\vec{w}^*$ is uniquely defined up to a constant.

Leading eigenvalues We apply this to show our result for $\mathbf{C}$, which can have negative elements on its diagonal. Define a new matrix $\mathbf{D}$ whose elements are all nonnegative by letting $\mathbf{D} = \mathbf{C} + b\mathbf{I}$, where $b > 0$ is a large enough positive constant. Because $\mathbf{D}\vec{w} = (\mathbf{C} + b\mathbf{I})\,\vec{w} = (\lambda + b)\,\vec{w}$, each eigenvalue λ of $\mathbf{C}$ corresponds to an eigenvalue λ' of $\mathbf{D}$ according to $\lambda' - b = \lambda$; moreover, if $\vec{w}$ is an eigenvector of $\mathbf{C}$ associated with eigenvalue λ (i.e., $\mathbf{C}\vec{w} = \lambda\vec{w}$), then it is also an eigenvector of $\mathbf{D}$ associated with the eigenvalue λ'.

Because $\mathbf{D}$ has all nonnegative elements, the Perron–Frobenius theorem guarantees there is a real nonnegative *dominant* eigenvalue $\lambda^{*\prime}$ whose magnitude is at least as great as that of any other eigenvalue λ, i.e., $\lambda^{*\prime} \geq |\lambda|$. Because λ and $\lambda^{*\prime}$ are distinct eigenvalues, any other eigenvalue λ must have a strictly lesser real part than this dominant eigenvalue, $\mathrm{Re}\,(\lambda) < \lambda^{*\prime}$. Thus, $\lambda^{*\prime}$ is also the *leading* eigenvalue of $\mathbf{D}$. Moreover, it is guaranteed that we can choose an eigenvector $\vec{w}^*$ associated with $\lambda^{*\prime}$, whose components are all nonnegative.

Since the eigenvalues of $\mathbf{C}$ have simply been shifted by $-b$, $\mathbf{C}$ is also guaranteed to have a leading eigenvalue $\lambda^* = \lambda^{*\prime} - b$. (Note that, unlike $\lambda^{*\prime}$, this eigenvalue may be negative, and need not be the dominant eigenvalue.) Furthermore, the nonnegative eigenvector $\vec{w}^*$ of $\mathbf{D}$ is also the eigenvector of $\mathbf{C}$ associated with the leading eigenvalue λ^*.

Box S1.B3. Irreducible matrices

An $n \times n$ matrix $\mathbf{M}$ is irreducible if, for any two distinct indices i and j such that $1 \leq i, j \leq n$, we can find a sequence of distinct indices $s_1, s_2, \ldots, s_m$ such that $s_1 = i$, $s_m = j$, and $(\mathbf{M})_{s_{k+1}, s_k} \neq 0$ for all k such that $1 \leq k \leq m - 1$. In other words, if we view each element of the matrix $(\mathbf{M})_{s_{k+1}, s_k}$ as representing the transition of individuals from stage s_k to stage s_{k+1}, irreducibility of $\mathbf{M}$ is equivalent to saying that individuals of any stage can eventually end up in any other stage after some sequence of population transitions—a natural condition for considering all stages as part of the same population. Thus, this condition will naturally be satisfied in biologically realistic models of transitions within a single population.

Properties of irreducible Metzler matrices If the continuous-time population transition matrix $\mathbf{C}$ is irreducible, the shifted matrix $\mathbf{D}$ is also irreducible, since

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Box S1.B3 (continued)

only the off-diagonal entries (which remain unchanged) matter for the definition of irreducibility above. In this case, a stronger version of the Perron–Frobenius theorem states that the leading eigenvalues of $\mathbf{C}$ and $\mathbf{D}$ are *algebraically simple*. Among other properties, this means that they are associated with a positive eigenvector which is unique up to a constant. Precisely stated, if λ^* is the leading eigenvalue of $\mathbf{C}$, then there is an eigenvector $\vec{w}^*$ whose elements are all positive satisfying $\mathbf{C}\vec{w}^* = \lambda^*\vec{w}^*$; furthermore, the only positive vectors satisfying this condition are positive multiples of $\vec{w}^*$. The same argument holds for discrete-time $\mathbf{A}'$. Biologically, this means that population dynamics converge to the same long-term set of non-zero state proportions, regardless of the starting conditions.

S1.4 Eigenvalue analysis with stability conditions

To evaluate whether continuous-time $\mathbf{C}$ or discrete-time $\mathbf{A}'$ admits long-term population growth (i.e., persistence of the invader population), we need to know the sign of the leading eigenvalue. In systems with three or more stages, however, directly solving for this eigenvalue may often be infeasible. As an alternative, we can apply mathematical conditions from *stability analysis* which diagnose whether the matrix has a nonnegative leading eigenvalue. This is closely related to invasion analysis: a system where invasion is possible is an unstable system, because the small perturbation caused by introducing the invader grows in magnitude.

In this section, we present several recipes for stability analysis in systems of varying complexity. Box S1.B4 presents frequently used stability conditions which can directly be applied to small systems with two to five stages, while Box S1.B5 gives the general Routh–Hurwitz criteria from which these conditions are derived.

Finally, in very complex cases, it may be preferable to only focus on the region of parameter space near the invasion boundary. In this case, the invasion condition can be derived using the *invasion determinant*, summarized in Box S1.B6. This approach is naturally suited for evolutionary invasion analyses, where the system is always near the invasion boundary because the traits of the invader and resident differ only infinitesimally (Iritani & Day, 2026). More generally, however, caution is warranted because this approach may recover spurious boundaries in addition to the correct invasion condition.

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Box S1.B4. Stability conditions for small matrices

For an $n \times n$ matrix $\mathbf{M}$, find the characteristic polynomial using eq. S1.10. Then, we can write the coefficients as $a_0, a_1, a_2, \ldots, a_n$ such that the polynomial is

$$a_0\lambda^n + a_1\lambda^{n-1} + \cdots + a_{n-1}\lambda + a_n = 0. \tag{S1.B4.1}$$

Note that some texts use the opposite convention for the subscripts of a, where a_i is the coefficient of the λ^i term. Assume that the leading coefficient is one, $a_0 = 1$. We can always rewrite the polynomial in this form by dividing the polynomial through by the highest-order coefficient (alternatively, solving $\det(\lambda\mathbf{I} - \mathbf{M}) = 0$ will always give a highest-order coefficient of 1). Routh (1877) and Hurwitz (1895) provided a simple general test to determine when all eigenvalues of $\mathbf{M}$ have negative real parts, summarized in Box S1.B5. Here, we give the frequently-encountered special cases for $n = 2$ through $n = 5$.

Two-dimensional matrices In two-dimensional systems, the conditions for stability are particularly simple. In particular, all eigenvalues have negative real parts if and only if

$$a_1 > 0 \tag{S1.B4.2}$$

$$a_2 > 0. \tag{S1.B4.3}$$

Furthermore, by expanding the determinant in $\det(\lambda\mathbf{I} - \mathbf{M})$, we see that $a_1 = -\operatorname{tr}\mathbf{M}$ and $a_2 = \det\mathbf{M}$. Noting that the trace is the sum of the eigenvalues and that the determinant is the product, the stability conditions can also be rewritten as $\lambda_1 + \lambda_2 = \operatorname{tr}\mathbf{M} < 0$ and $\lambda_1\lambda_2 = \det\mathbf{M} > 0$.

Three-dimensional matrices In three-dimensional systems, the eigenvalues all have negative real parts if and only if the following three conditions hold:

$$a_1 > 0 \tag{S1.B4.4}$$

$$a_3 > 0 \tag{S1.B4.5}$$

$$a_1a_2 - a_3 > 0. \tag{S1.B4.6}$$

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Box S1.B4 (continued)

Four-dimensional matrices In four dimensions, the conditions become

$$a_1 > 0 \tag{S1.B4.7}$$
$$a_2 > 0 \text{ or } a_3 > 0 \tag{S1.B4.8}$$
$$a_4 > 0 \tag{S1.B4.9}$$
$$(a_1 a_2 - a_3)\, a_3 - a_1^2 a_4 > 0, \tag{S1.B4.10}$$

where we may choose $a_2 > 0$ or $a_3 > 0$ depending on which one is easier to calculate, and the last condition may also be written as $a_1 a_2 a_3 > a_3^2 + a_1^2 a_4$.

Five-dimensional matrices In five dimensions, it suffices to show that the following conditions hold

$$a_1, a_3 > 0 \text{ or } a_2, a_4 > 0 \tag{S1.B4.11}$$
$$a_5 > 0 \tag{S1.B4.12}$$
$$a_1 a_2 - a_3 > 0 \tag{S1.B4.13}$$
$$(a_1 a_2 - a_3)\,(a_3 a_4 - a_2 a_5) > (a_1 a_4 - a_5)^2, \tag{S1.B4.14}$$

where we can choose whether to check positivity of a_1, a_3 or of a_2, a_4.

Box S1.B5. General Routh–Hurwitz conditions

Say we have a general characteristic polynomial

$$a_0 \lambda^n + a_1 \lambda^{n-1} + \cdots + a_{n-1}\lambda + a_n = 0, \tag{S1.B5.1}$$

where $a_0 > 0$. Define the Hurwitz matrix $\mathbf{H}$, an $n \times n$ matrix such that

$$(\mathbf{H})_{ij} = \begin{cases} a_{2i-j} & \text{for } 0 \le 2i - j \le n \\ 0 & \text{otherwise.} \end{cases} \tag{S1.B5.2}$$

The resulting matrix has the structure

$$\mathbf{H} = \begin{bmatrix} a_1 & a_0 & 0 & 0 & 0 & \cdots \\ a_3 & a_2 & a_1 & a_0 & 0 & \cdots \\ a_5 & a_4 & a_3 & a_2 & a_1 & \cdots \\ \vdots & \vdots & \vdots & \vdots & \vdots & \ddots \end{bmatrix}, \tag{S1.B5.3}$$

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Box S1.B5 (continued)

where we note that the indices in each row decrease by one, the indices in each column increase by two, and entries where the index would be outside the range from 0 to n are replaced with zeros. Note that some texts (e.g., the original work of Hurwitz (1895)) define $\mathbf{H}$ as the transpose of the above matrix instead, but this does not affect the stability calculation.

Now, let $\mathbf{H}_k$ denote the $k \times k$ square matrix obtained by only taking the first k rows and columns of $\mathbf{H}$ (i.e., the kth leading principal submatrix of $\mathbf{H}$). Then, assuming that $a_0 > 0$, stability occurs if and only if all of these submatrices have positive determinants, i.e.,

$$\det \mathbf{H}_1 > 0,\ \det \mathbf{H}_2 > 0,\ \ldots,\ \det \mathbf{H}_n > 0. \tag{S1.B5.4}$$

Generally, it is a necessary condition for stability that all a_i are positive. Indeed, Liénard & Chipart (1914) showed that the number of determinants which need to be calculated can be greatly reduced if we know the signs of some of the a_i coefficients. The simplified Liénard–Chipart conditions show that stability can be proven by showing that approximately half of the a_i coefficients and half of the determinants are positive. In particular, stability occurs if and only if two conditions hold: first, at least one of the following coefficient conditions is satisfied:

$$a_n > 0,\ a_{n-2} > 0,\ a_{n-4} > 0,\ a_{n-6} > 0,\ \ldots \tag{S1.B5.5}$$

$$\text{or } a_n > 0,\ a_{n-1} > 0,\ a_{n-3} > 0,\ a_{n-5} > 0,\ \ldots; \tag{S1.B5.6}$$

and second, at least one of the following determinant conditions is satisfied:

$$\det \mathbf{H}_1 > 0,\ \det \mathbf{H}_3 > 0,\ \det \mathbf{H}_5 > 0,\ \ldots \tag{S1.B5.7}$$

$$\text{or } \det \mathbf{H}_2 > 0,\ \det \mathbf{H}_4 > 0,\ \det \mathbf{H}_6 > 0,\ \ldots. \tag{S1.B5.8}$$

In other words, we only need to check if (1) all odd-numbered or all even-numbered coefficients, as well as a_n, are positive, and (2) all odd-numbered or all even-numbered determinants are positive. Applying these conditions (always choosing the set of determinants with the smallest maximum order) and letting $a_0 = 1$, we straightforwardly obtain the three-, four-, and five-dimensional criteria in Box S1.B4.

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Box S1.B6. Iritani & Day (2026)'s invasion determinant

A necessary condition for stability is that all a_i coefficients are positive in the characteristic polynomial

$$\det(\lambda \mathbf{I} - \mathbf{M}) = \lambda^n + a_1\lambda^{n-1} + \cdots + a_{n-1}\lambda + a_n = 0. \tag{S1.B6.1}$$

Furthermore, note that $a_1 = \operatorname{tr}(-\mathbf{M})$ and $a_n = \det(-\mathbf{M})$ for any n. Thus, two necessary conditions for stability are

$$\operatorname{tr}(-\mathbf{M}) > 0 \tag{S1.B6.2}$$

$$\det(-\mathbf{M}) > 0. \tag{S1.B6.3}$$

In the two-stage case above (eqs. S1.B4.2 and S1.B4.3), these conditions are also sufficient for stability. More generally, the trace and determinant conditions alone do not suffice to show stability. When conducting invasion analysis, however, there are special cases where it may suffice to only consider the determinant condition (eq. S1.B6.3) instead of fully evaluating all stability conditions (Boxes S1.B4 and S1.B5).

Invasibility near the invasion boundary Say we are only interested in deriving the boundary where changes in some underlying parameter(s) cause the invasion growth rate to cross continuously from negative to positive. In other words, we assume that the leading eigenvalue of the matrix $\mathbf{C}$ is near zero. Furthermore, as before, assume the special properties of $\mathbf{C}$ required for biological realism hold as this transition occurs: that it has only nonnegative off-diagonal elements (i.e., is a Metzler matrix; Box S1.B1) and that it is irreducible (Box S1.B3). Then the transition from negative to positive invasion growth (i.e., stability to instability) occurs when the leading eigenvalue λ^* of $\mathbf{C}$ guaranteed by the Perron–Frobenius theorem (Box S1.B2) crosses from negative to positive, with the real part of all other eigenvalues remaining negative. Since λ^* is an algebraically simple leading eigenvalue and the determinant can be understood as the product of all eigenvalues, this implies that $\det(-\mathbf{C})$ changes sign from positive (a necessary condition for stability) to negative at the invasion boundary.

The invasion determinant condition Thus, the condition for invasibility *near the invasion boundary* under our assumptions is

$$-\det(-\mathbf{C}) > 0. \tag{S1.B6.4}$$

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Box S1.B6 (continued)

The left-hand side of this expression, termed the *invasion determinant* by Iritani & Day (2026), serves as a useful proxy when performing analytical investigations: near the invasion boundary, it shares the same sign as the true invasion growth rate. Nonetheless, some caution is warranted in general invasion analyses: while a positive value of the left-hand side always implies invasibility (i.e., the system must be unstable because $\det(-\mathbf{C}) = a_n < 0$), negative values may also occur in an invasible system due to a non-leading eigenvalue crossing zero. Thus, solving the boundaries defined by inequality S1.B6.4 may recover spurious boundaries in addition to the correct invasion boundary. In these cases, comparison to numerical simulations or additional analyses may be needed to confirm the results of determinant-based calculations.

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S2 Detailed analyses for simple stage-structured model

We consider the simple model of a stage-structured invader species from the main text (eq. 1 and Fig. 1).

$$\frac{d\vec{N}_1}{dt} = \begin{bmatrix} r_1 - \alpha_{11}N_1^{(2)} - \alpha_{12}N_2 & f_1 \\ g_1 & -m_1 \end{bmatrix} \vec{N}_1 \tag{S2.1}$$

$$\frac{dN_2}{dt} = \left(r_2 - \alpha_{21}N_1^{(2)} - \alpha_{22}N_2\right) \cdot N_2. \tag{S2.2}$$

The resident equilibrium of species 2 is

$$N_2^* = \frac{r_2}{\alpha_{22}}. \tag{S2.3}$$

Since we are interested in species 1's invasion condition, we set $N_2 = N_2^*$ for all subsequent derivations in this Appendix.

S2.1 Method 1.1: Direct timescale separation

S2.1.1 Timescale separation using transformations in the main text

We will choose a new pair of variables: N, representing the slow dynamics of population growth, and Q, representing the fast dynamics of population stage structure. Since competition only depends on adults (stage 2), we choose $N = N_1^{(2)}$ as a convenient choice for the population (slow) variable. Accordingly, a convenient choice of the stage structure (fast) variable is the ratio of juveniles (stage 1) to adults (stage 2), $Q = N_1^{(1)}/N_1^{(2)}$. Then $N_1^{(1)} = QN$ and we can rewrite the system as:

$$\frac{dN}{dt} = g_1 QN - m_1 N \tag{S2.4}$$

$$\frac{dQ}{dt} = \frac{dN_1^{(1)}}{dt} \cdot N^{-1} - \frac{dN_1^{(2)}}{dt} \cdot Q \cdot N^{-1} \tag{S2.5}$$

$$= \left(r_1 - \alpha_{11}N - \alpha_{12}N_2^*\right) Q + f_1 - g_1 Q^2 + m_1 Q \tag{S2.6}$$

$$= -g_1 Q^2 + \left(r_1 + m_1 - \alpha_{11}N - \alpha_{12}N_2^*\right) Q + f_1. \tag{S2.7}$$

Note that many other choices of N and Q would be valid here. While the choice above simplifies the algebra considerably, we could also have selected N to be the total population and Q to be the proportion of adults within the total population (demonstrated in subsection S2.1.3).

First, we solve for the quasi-equilibrium of Q in the invasion condition. In this case, $N \approx 0$, allowing us to neglect the term $-\alpha_{11}N$. Solving using the quadratic formula

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gives the single biologically meaningful root,

$$Q^* = \frac{A + \sqrt{A^2 + g_1 f_1}}{g_1}, \text{ with } A = \frac{r_1 + m_1 - \alpha_{12} N_2^*}{2}, \tag{S2.8}$$

where we note that the other root is necessarily negative and therefore not biologically meaningful (since Q is a ratio of populations). Next, we find the exponential growth rate of N when $Q = Q^*$. We find that the growth rate of N is expressed as:

$$\frac{1}{N} \cdot \frac{dN}{dt} = g_1 Q - m_1. \tag{S2.9}$$

Substituting $Q = Q^*$ into the per-capita growth rate, we find

$$\text{IGR}_1 = g_1 Q^* - m_1 \tag{S2.10}$$

$$= \frac{r_1 - m_1 - \alpha_{12} N_2^*}{2} + \sqrt{\left(\frac{r_1 + m_1 - \alpha_{12} N_2^*}{2}\right)^2 + g_1 f_1}. \tag{S2.11}$$

Finally, we solve for the conditions under which the invasion growth rate is positive. Noting that

$$\left(\frac{r_1 + m_1 - \alpha_{12} N_2^*}{2}\right)^2 = \left(\frac{r_1 - m_1 - \alpha_{12} N_2^*}{2}\right)^2 + m_1 \left(r_1 - \alpha_{12} N_2^*\right), \tag{S2.12}$$

we can rewrite

$$\text{IGR}_1 = B + \sqrt{B^2 + f_1 g_1 + m_1 \left(r_1 - \alpha_{12} N_2^*\right)}, \tag{S2.13}$$

with

$$B = A - m_1 = \frac{r_1 - m_1 - \alpha_{12} N_2^*}{2}. \tag{S2.14}$$

We note that the invasion growth rate is positive (1) when $f_1 g_1 + m_1 \left(r_1 - \alpha_{12} N_2^*\right)$ is positive, which rearranges to

$$\alpha_{12} N_2^* < r_1 + \frac{f_1 g_1}{m_1}, \tag{S2.15}$$

or (2) when B is positive, which occurs when $\alpha_{12} N_2^* < r_1 - m_1$. Since the second condition also implies the first, the first condition is necessary and sufficient for positive invasion growth.

S2.1.2 Simplified derivation: the λ-trick

There is also a convenient alternative to the full calculation of the quasi-equilibrium stage ratio above. Applying the chain rule to differentiate $Q = N_1^{(1)}/N$ (eq. S2.5), we see that the result can be written in terms of the per-capita growth rate of N, i.e.,

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$(1/N) \cdot dN/dt$:

$$\frac{dQ}{dt} = \frac{1}{N} \cdot \frac{dN_1^{(1)}}{dt} - Q \cdot \left(\frac{1}{N} \cdot \frac{dN}{dt} \right) \tag{S2.16}$$

$$= Q \cdot (r_1 - \alpha_{12} N_2^*) + f_1 - Q \cdot \left(\frac{1}{N} \cdot \frac{dN}{dt} \right) \tag{S2.17}$$

$$= Q \left(r_1 - \alpha_{12} N_2^* - \lambda \right) + f_1, \tag{S2.18}$$

where we applied the invasion condition in the second line, and in the last line, we have denoted $\lambda = (1/N) \cdot dN/dt$ for convenience. We note that this per-capita growth λ is precisely the invasion growth rate in which we are interested. Accordingly, we can solve the quasi-equilibrium Q^* in terms of the unknown λ, finding that

$$Q^* = \frac{f_1}{\alpha_{12} N_2^* - r_1 + \lambda}. \tag{S2.19}$$

Substituting this into the expression for invasion growth rate (eq. S2.9), we find that

$$\lambda = \frac{g_1 f_1}{\alpha_{12} N_2^* - r_1 + \lambda} - m_1, \tag{S2.20}$$

where we have also re-expressed the per-capita growth rate on the left-hand side as λ. Now we could rearrange this equation to a quadratic in λ and solve to obtain the same result as above. However, in the case where we are only interested in the boundary between successful and unsuccessful invasion, we can take the shortcut of considering when the parameters in the system and the value of N_2^* cause the system to be on this invasion boundary. To do so, we substitute $\lambda = 0$ into the above expression and rearrange to find that the boundary occurs when:

$$0 = \frac{g_1 f_1}{\alpha_{12} N_2^* - r_1} - m_1 \tag{S2.21}$$

$$0 = \frac{g_1 f_1}{m_1} - \alpha_{12} N_2^* + r_1 \tag{S2.22}$$

$$\alpha_{12} N_2^* = r_1 + \frac{g_1 f_1}{m_1}, \tag{S2.23}$$

where in the second line we have multiplied through by the quantity $(\alpha_{12} N_2^* - r_1)/m_1$. Biologically, we note that increasing the term on the left-hand side corresponds to increasing competitive pressure from residents, and therefore harms the invader's growth, while increasing the terms on the right-hand side benefits the invader (e.g., increasing its reproduction and maturation, or decreasing its mortality). Accordingly,

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we conclude that the invasion condition is

$$\alpha_{12} N_2^* < r_1 + \frac{f_1 g_1}{m_1}, \tag{S2.24}$$

matching the result from the full calculation.

Alternatively, we can reason about the inequality from the mathematical properties of the invasion growth rate expression. We can write the invasion growth rate condition in λ (eq. S2.20) as $p(\lambda) = 0$, where

$$p(x) = -(x + m_1)(\alpha_{12} N_2^* - r_1 + x) + g_1 f_1. \tag{S2.25}$$

Here, $p(x)$ is a quadratic polynomial with negative leading coefficient, such that λ is the biologically meaningful root of p. We note that the only biologically relevant root of p is the one corresponding to greater Q^* (as shown above) and thus is the greater of the two possible roots. Therefore, due to p's negative leading coefficient, a positive value of $p(x)$ for some value x implies that the root λ is greater than x. Thus, $p(0) > 0$ implies $\lambda > 0$ and means that species 1 can successfully invade, which we can rearrange (noting that $\alpha_{12} N_2^* - r_1 > 0$ near the invasion boundary) to give the inequality for the invasion condition above.

S2.1.3 Derivation with alternate variable transformations

To illustrate that there are multiple valid options for N and Q, we use the alternate and somewhat more intuitive transformations, as mentioned in the main text:

$$N = N_1^{(1)} + N_1^{(2)} \tag{S2.26}$$

$$Q = \frac{N_1^{(2)}}{N}, \tag{S2.27}$$

where N is the total abundance of both stages, and Q is the fraction of that total occupied by the adult stage $N_1^{(2)}$. Though the algebra is significantly messier, we find the same results. First, using the transformations $N_1^{(1)} = (1 - Q)N$, $N_1^{(2)} = QN$, we find that the per-capita dynamics of N follow

$$\frac{1}{N} \cdot \frac{dN}{dt} = (1 - Q)\left(r_1 - \alpha_{11} QN - \alpha_{12} N_2^*\right) + g_1(1 - Q) + f_1 Q - m_1 Q, \tag{S2.28}$$

where we can remove the $-\alpha_{11} QN$ term when considering the invasion dynamics ($N \approx 0$). Next, using the chain rule, we find that

$$\frac{dQ}{dt} = \frac{1}{N} \cdot \frac{dN_1^{(2)}}{dt} - Q \cdot \frac{1}{N} \cdot \frac{dN}{dt}, \tag{S2.29}$$

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and substituting the expressions for $dN_1^{(2)}/dt$, $(1/N)\,dN/dt$, and $N_1^{(1)}, N_1^{(2)}$, we find (again neglecting the α_{11} term due to the invasion condition):

$$\begin{aligned}\frac{dQ}{dt} &= \left[g_1(1-Q) - m_1 Q\right] - Q\Big[(1-Q)(r_1 - \alpha_{12}N_2^*) + g_1(1-Q) + f_1 Q - m_1 Q\Big] \\ &= \left(r_1 + m_1 + g_1 - f_1 - \alpha_{12}N_2^*\right)Q^2 - \left(r_1 + m_1 + 2g_1 - \alpha_{12}N_2^*\right)Q + g_1.\end{aligned} \tag{S2.30}$$

Solving the quadratic, we find that

$$Q^* = \frac{(A + g_1/2) - \sqrt{(A+g_1/2)^2 - (2A - f_1)\,g_1}}{(2A - f_1)}, \text{ with } A = \frac{r_1 + m_1 + g_1 - \alpha_{12}N_2^*}{2}, \tag{S2.31}$$

and it is now the root with $-$ which is biologically meaningful (since Q cannot be greater than 1). Finally, we substitute back into the per-capita growth rate of N to find the invasion growth rate. Simplifying $\frac{1}{N}\frac{dN}{dt}$ and applying the invasion condition (i.e., dropping the α_{11} term as $N \approx 0$ and setting species 2 to the resident equilibrium N_2^*), we find:

$$\frac{1}{N}\cdot\frac{dN}{dt} = \left(r_1 - \alpha_{12}N_2^*\right)(1-Q) + g_1(1-Q) + f_1 Q - m_1 Q \tag{S2.32}$$

$$= r_1 + g_1 - \alpha_{12}N_2^* - \left(r_1 + g_1 + m_1 - f_1 - \alpha_{12}N_2^*\right)Q \tag{S2.33}$$

$$= r_1 + g_1 - \alpha_{12}N_2^* - (2A - f_1)\,Q. \tag{S2.34}$$

Thus, substituting the quasi-equilibrium stage structure Q^*, we find that

$$\text{IGR}_1 = r_1 + g_1 - \alpha_{12}N_2^* - (2A - f_1)\,Q^* \tag{S2.35}$$

$$= r_1 + g_1 - \alpha_{12}N_2^* - (A + g_1/2) + \sqrt{(A+g_1/2)^2 - (2A - f_1)\,g_1} \tag{S2.36}$$

$$= \frac{r_1 - m_1 - \alpha_{12}N_2^*}{2} + \sqrt{(A - g_1/2)^2 + g_1 f_1} \tag{S2.37}$$

$$= \frac{r_1 - m_1 - \alpha_{12}N_2^*}{2} + \sqrt{\left(\frac{r_1 + m_1 - \alpha_{12}N_2^*}{2}\right)^2 + g_1 f_1}, \tag{S2.38}$$

matching eq. S2.11 above. Thus, these alternate transformations give identical results, though the algebra is notably more tedious.

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S2.2 Method 1.2: Strategic linear combination

Let $X = N_1^{(1)} + \frac{f_1}{m_1} N_1^{(2)}$. Then

$$\frac{dX}{dt} = \left(r_1 + \frac{f_1 g_1}{m_1} - \alpha_{11} N_1^{(2)} - \alpha_{12} N_2^* \right) N_1^{(1)} \tag{S2.39}$$

$$= \left(r_1 + \frac{f_1 g_1}{m_1} - \alpha_{11} N_1^{(2)} - \alpha_{12} N_2^* \right) kX. \tag{S2.40}$$

Then the invasion growth rate is

$$\text{IGR}_1 = k \left(r_1 + \frac{f_1 g_1}{m_1} - \alpha_{12} N_2^* \right). \tag{S2.41}$$

Given that k is a positive constant, the criterion for a positive IGR matches the expression in eq. S2.15:

$$\alpha_{12} N_2^* < r_1 + \frac{f_1 g_1}{m_1} \tag{S2.42}$$

S2.3 Method 2.1: Direct solution for leading eigenvalue

Via linear algebra, we consider the eigenvalues of the matrix

$$\mathbf{C} = \begin{bmatrix} r_1 - \alpha_{12} N_2^* & f_1 \\ g_1 & -m_1 \end{bmatrix}. \tag{S2.43}$$

Solving the characteristic equation $\det(\mathbf{C} - \lambda \mathbf{I}) = 0$, we obtain

$$(r_1 - \alpha_{12} N_2^* - \lambda)(-m_1 - \lambda) - f_1 g_1 = 0 \tag{S2.44}$$

$$\lambda^2 + (m_1 - r_1 + \alpha_{12} N_2^*) \lambda - f_1 g_1 - m_1 (r_1 - \alpha_{12} N_2^*) = 0, \tag{S2.45}$$

which has the leading eigenvalue:

$$\lambda = B + \sqrt{B^2 + f_1 g_1 + m_1 (r_1 - \alpha_{12} N_2^*)}, \text{ where } B = \frac{r_1 - m_1 - \alpha_{12} N_2^*}{2}, \tag{S2.46}$$

and we note that, following the logic in Method 1.1, λ is positive when the term $f_1 g_1 + m_1 (r_1 - \alpha_{12} N_2^*)$ is positive, which rearranges to the same expression in eq. S2.15:

$$\alpha_{12} N_2^* < r_1 + \frac{f_1 g_1}{m_1}. \tag{S2.47}$$

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S2.4 Method 2.2: Applying stability conditions

Applying the Routh–Hurwitz stability conditions, we find that λ is positive if one of the following conditions holds. First, a negative determinant implies exactly one eigenvalue with a positive real part,

$$\det \mathbf{C} < 0 \tag{S2.48}$$

$$-m_1 \left(r_1 - \alpha_{12} N_2^*\right) - f_1 g_1 < 0 \tag{S2.49}$$

$$\alpha_{12} N_2^* < r_1 + \frac{f_1 g_1}{m_1}, \tag{S2.50}$$

which matches the expression in eq. S2.15. Second, if this condition does not hold, the eigenvalues share a sign and a positive trace implies that both eigenvalues are positive:

$$\operatorname{tr} \mathbf{C} > 0 \tag{S2.51}$$

$$r_1 - m_1 - \alpha_{12} N_2^* > 0 \tag{S2.52}$$

$$\alpha_{12} N_2^* < r_1 - m_1, \tag{S2.53}$$

but as we noted above, this condition also implies the first.

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S3 General numerical recipes

S3.1 Simulation-based algorithms for invasion growth rates

Algorithm S3.1. Direct simulation for invasion growth

```
function simulation.IGR(system, population.structures, times,
                        maximum.abundance)
    R⃗ ← resident state at equilibrium
    growth.trajectories ← empty list
    abundance.trajectories ← empty list

    # Simulate invasion from several initial stage structures
    for N⃗ in population.structures
        y⃗₀ ← concatenate(N⃗, R⃗)
        result ← integrate.ODE(system, y⃗₀, times) # Entire trajectory

        # Calculate per-capita growth rate of combined population
        growth.rates ← empty list
        abundances ← empty list
        for each time t and state y⃗(t) in result
            N⃗ ← invader stages of y⃗(t)
            N⃗′ ← invader derivatives calculated from t,y⃗(t)
            growth.rates[t] ← sum(N⃗′) / sum(N⃗)
            abundances[t] ← sum(N⃗)
        append growth.rates to growth.trajectories
        append abundances to abundance.trajectories

    # Identify time of convergence of growth rates
    SDs ← empty list
    for each t in times
        if abundances[t] < maximum.abundance for each
                abundances in abundance.trajectories
            rates ← trajectory[t] for each
                trajectory in growth.trajectories
            SDs[t] ← sd(rates)
    t* ← time at which SDs is minimized

    # Return the average growth rate at the best time
    growth.rates ← trajectory[t*] for each
        trajectory in growth.trajectories
    return mean(growth.rates)
```

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Algorithm S3.2. Rescaling (Lyapunov exponent) algorithm for invasion growth

```
function rescaling.IGR(full.system, t_end, Δt, N_0, t_burn-in)
    N⃗ ← (N_0, 0, ..., 0) # Invader starts in first stage
    R⃗ ← resident state at equilibrium (or on attractor)
    y⃗ ← concatenate(N⃗, R⃗) # Current state of the full system
    growth.rates ← empty list
    t ← 0

    while t + Δt <= t_end
        # Simulate the dynamics normally for Δt time units
        y⃗ ← integrate.ODE(full.system, y⃗, [t, t + Δt])[end]
        N⃗ ← invader stages of y⃗
        R⃗ ← residents from y⃗

        # Calculate the invader's growth rate
        N_total ← sum(N⃗)
        GR_i ← log(N_total/N_0)/Δt
        # Only use growth rates after the burn-in period
        if t > t_burn-in
            append GR_i to growth.rates

        # Rescale the invader's total abundance
        N⃗ ← (N_0/N_total) · N⃗
        y⃗ ← concatenate(N⃗, R⃗)
        t ← t + Δt

    IGR ← mean(growth.rates)
    return IGR
```

S3.2 Linearization of invader dynamics

Algorithm S3.3. Linearization of invader dynamics

```
function linearize.invader.derivatives(full.system.derivatives, ϵ)
    R⃗(t) ← function giving resident abundance at equilibrium
           or on time-dependent attractor
    indices ← indices of invader stages

    function result.function(t, N⃗)
        R⃗ ← R⃗(t)
        N⃗_0 ← vector of zeros of length n
        d⃗_0 ← full.system.derivatives(t, concatenate(N⃗_0, R⃗))[indices]
        J ← empty matrix of size n × n # Jacobian matrix
        for i in 1 to n
```

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Algorithm S3.3 (continued)

```
            # Small perturbation around zero in the i-th invader stage
            N⃗_i ← vector of zeros of length n
            N⃗_i[i] ← ϵ
            # Calculate model derivative
            d⃗_i ← full.system.derivatives(t, concatenate(N⃗_i, R⃗))[indices]
            # Numerically approximate partial derivative
            J[:,i] ← (d⃗_i - d⃗_0) / ϵ
        # Return the linearized invader dynamics
        return J N⃗

    # Return the generated model derivative function
    return result.function
```

Algorithm S3.4. Invasion growth with linearized invader dynamics

```
function linearized.IGR(linearized.invader.model, t_end, N_0, t_burn-in)
    N⃗ ← (N_0, 0, ..., 0) # Invader starts in first stage

    # Simulate through the initial transient in population structure
    N⃗ ← integrate.ODE(linearized.invader.model, N⃗, [0, t_burn-in])[end]
    N_burn-in ← sum(N⃗)

    # Continue the simulation and measure long-term population growth
    N⃗ ← integrate.ODE(linearized.invader.model, N⃗, [t_burn-in, t_end])[end]
    N_end ← sum(N⃗)

    IGR ← log(N_end / N_burn-in) / (t_end − t_burn-in)
    return IGR
```

S3.3 Floquet theory

Algorithm S3.5. Monodromy matrix

```
function monodromy.matrix(invader.system, T, t_0 = 0)
    n ← number of invader stages
    M ← empty matrix of size n × n

    for i in 1 to n
        # Construct the i-th standard basis vector
        e⃗ ← zero vector of length n
        e⃗[i] ← 1
```

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Algorithm S3.5 (continued)

```
        # Integrate the invader system for one period T
        N⃗ ← integrate.ODE(invader.system, e⃗, [t0, t0 + T])[end]

        # Set the i-th column of M to the final state
        M[:, i] ← N⃗

    return M
```

Algorithm S3.6. Floquet calculation of invasion growth

```
function floquet.IGR(invader.system, T)
    M ← monodromy.matrix(invader.system, T)
    spectrum ← eigenvalues(M)
    ρ ← max(abs(spectrum)) # Dominant Floquet multiplier
    λ ← log(ρ)/T # Dominant Floquet exponent
    return λ
```

Algorithm S3.7. Finding the Floquet exponent matrix B

```
function floquet.B(invader.system, T, t0 = 0)
    n ← number of invader stages
    I ← identity matrix of size n × n

    # Get fundamental matrix solution at start time and one period later
    Φ0 ← integrate.matrix.ODE(invader.system, I, [0, t0])
    ΦT ← integrate.matrix.ODE(invader.system, I, [0, t0 + T])

    # Compute the matrix mapping state at t0 to one period later
    M ← ΦT Φ0^-1 # This is the monodromy matrix for phase t0

    # Use the matrix logarithm to find and return B
    B ← matrix.logarithm(M) / T
    return B

function integrate.matrix.ODE(system, X0, time.interval)
    n ← number of rows of X0
    m ← number of columns of X0
    X ← empty matrix of size n × m

    # Integrate each column of the initial matrix independently
    for i in 1 to m
        x⃗0 ← X0[:, i]
```

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Algorithm S3.7 (continued)

```
        x⃗ ← integrate.ODE(system, x⃗_0, time.interval)[end]

        # Set the i-th column to the final state
        X[:, i] ← x⃗

    return X
```

Algorithm S3.8. Periodic Floquet transformation

```
function floquet.transformation(invader.system, T, times, t_0 = 0)
    n ← number of invader stages
    I ← identity matrix of size n × n
    B ← floquet.B(invader.system, T, t_0)
    result.list ← empty list

    for each t in times
        # Fundamental matrix solution at current time
        Φ_t ← integrate.matrix.ODE(invader.system, I, [t_0, t])
        # State of the linear system at same time
        Y_t ← matrix.exponential(B(t − t_0))

        # Compute transformation mapping linear system to Φ_t
        P_t ← Φ_t Y_t^{-1}
        append P_t to result.list

    return result.list
```

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S4 Intraguild predation and cannibalism model

We consider a set of insect populations with three stages $N_i^{(1)}, N_i^{(2)}, N_i^{(A)}$ competing for a shared logistically growing resource R.

$$\frac{dN_i^{(A)}}{dt} = g_i^{(2)} N_i^{(2)} - m_i^{(A)} N_i^{(A)} \tag{S4.1}$$

$$\frac{dN_i^{(2)}}{dt} = g_i^{(1)} N_i^{(1)} - \left(g_i^{(2)} + m_i^{(2)}\right) N_i^{(2)} \tag{S4.2}$$

$$\frac{dN_i^{(1)}}{dt} = f_i N_i^{(A)} - \left(g_i^{(1)} + m_i^{(1)}\right) N_i^{(1)} \tag{S4.3}$$

$$\frac{dR}{dt} = rR\left(1 - \frac{R}{K}\right) - \sum_i \sum_j u_i^{(j)} N_i^{(j)} R. \tag{S4.4}$$

In this general model, assume development $g_i^{(j)}$, mortality $m_i^{(j)}$, and fecundity f_i are functions of the state variables (superscript $j = 1, 2, A$). In particular, they may be coupled to resource uptake $u_i^{(j)} R$. On the other hand, stages may die due to predation by other stages, or the development of predatory stages may change as a result of their consumption of prey stages.

Three-stage intraguild predation model From this general framework, we analyze a specific version of the model with intraguild predation (IGP) in two species whose development depends on resource uptake. Assume species 1's second juvenile stage $N_1^{(2)}$ performs predation at density-dependent rate a_{i1} on first-stage juveniles $N_i^{(1)}$ of both species ($i = 1, 2$, representing cannibalism and IGP, respectively) which (i) affects the mortality of the predator individual, and (ii) causes mortality of the prey individual. Thus, the mortality of the first stage of both species is given by a baseline mortality $\mu_i^{(1)}$ plus mortality due to predation by $N_1^{(2)}$,

$$m_i^{(1)} = \mu_i^{(1)} + a_{i1} N_1^{(2)}, \tag{S4.5}$$

where $i = 1, 2$. Furthermore, for the predatory stage $N_1^{(2)}$, assume that its mortality depends not only on resource uptake ($u_1^{(2)} R$) but also on the rate at which it predates prey species i, i.e., $a_{i1} N_i^{(1)}$ according to the following function

$$m_1^{(2)} = \frac{u_1^{(2)} R + a_{11} N_1^{(1)} + a_{21} N_2^{(1)}}{u_1^{(2)} R + b_{11} a_{11} N_1^{(1)} + b_{21} a_{21} N_2^{(1)}} \times \mu_1^{(2)}. \tag{S4.6}$$

Now, consider resource dynamics. Assume that juvenile states' development rates

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$g_i^{(j)}$ increase linearly with resource level according to the uptake constant $u_i^{(j)}$. Then,

$$g_i^{(j)} = u_i^{(j)} R. \tag{S4.7}$$

Finally, we assume fecundity is a constant rate f_i.

Putting these together, the dynamics of the two species, where species 2 is preyed upon by its competitor but does not itself engage in predation, are given by:

$$\frac{dN_1^{(1)}}{dt} = f_1 N_1^{(A)} - \left(u_1^{(1)} R + \mu_1^{(1)} + a_{11} N_1^{(2)}\right) N_1^{(1)} \tag{S4.8}$$

$$\frac{dN_1^{(2)}}{dt} = u_1^{(1)} R N_1^{(1)} - \left[u_1^{(2)} R + m_1^{(2)}\right] N_1^{(2)} \tag{S4.9}$$

$$\frac{dN_1^{(A)}}{dt} = u_1^{(2)} R N_1^{(2)} - \mu_1^{(A)} N_1^{(A)} \tag{S4.10}$$

$$\frac{dN_2^{(1)}}{dt} = f_2 N_2^{(A)} - \left(u_2^{(1)} R + \mu_2^{(1)} + a_{21} N_1^{(2)}\right) N_2^{(1)} \tag{S4.11}$$

$$\frac{dN_2^{(2)}}{dt} = u_2^{(1)} R N_2^{(1)} - \left(u_2^{(2)} R + \mu_2^{(2)}\right) N_2^{(2)} \tag{S4.12}$$

$$\frac{dN_2^{(A)}}{dt} = u_2^{(2)} R N_2^{(2)} - \mu_2^{(A)} N_2^{(A)}, \tag{S4.13}$$

with $m_1^{(2)}$ as the function defined above in eq. S4.6, and resource dynamics given by the same equation as in the general model above.

Two-stage simplification Note that the dynamics of adults are relatively simple: assuming the other variables are at equilibrium or changing slowly, adult abundance of species i reaches the quasi-equilibrium value,

$$N_i^{(A)*} = \frac{u_i^{(2)} R}{\mu_i^{(A)}} N_i^{(2)}. \tag{S4.14}$$

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Then, with the assumption that adult population dynamics are fast, we can simplify the dynamics of the two species to:

$$\frac{dN_1^{(1)}}{dt} = k_1 f_1 u_1^{(2)} R N_1^{(2)} - \left(u_1^{(1)} R + \mu_1^{(1)} + a_{11} N_1^{(2)}\right) N_1^{(1)} \tag{S4.15}$$

$$\frac{dN_1^{(2)}}{dt} = u_1^{(1)} R N_1^{(1)} - \left[u_1^{(2)} R + m_1^{(2)}\right] N_1^{(2)} \tag{S4.16}$$

$$\frac{dN_2^{(1)}}{dt} = k_2 f_2 u_2^{(2)} R N_2^{(2)} - \left(u_2^{(1)} R + \mu_2^{(1)} + a_{21} N_1^{(2)}\right) N_2^{(1)} \tag{S4.17}$$

$$\frac{dN_2^{(2)}}{dt} = u_2^{(1)} R N_2^{(1)} - \left(u_2^{(2)} R + \mu_2^{(2)}\right) N_2^{(2)}, \tag{S4.18}$$

where $k_i = \frac{1}{\mu_i^{(A)}}$ and $m_1^{(2)}$ is a function of resource and $N_1^{(2)}$'s consumption of first-stage juveniles (eq. S4.6); resource dynamics are as previously given.

Monoculture equilibrium of species 2 Consider the invasion of species 1 (the predator species) into a monoculture of species 2 (the prey). We first solve for the monoculture equilibrium of species 2 alone, thereby ignoring the $a_{21} N_1^{(2)}$ term. The resulting equilibrium (R^*, $N_2^{(1)*}$, $N_2^{(2)*}$, $N_2^{(A)*}$) fulfills the following:

$$N_2^{(A)*} = \frac{u_2^{(2)} R^*}{\mu_2^{(A)}} N_2^{(2)*} \tag{S4.19}$$

$$N_2^{(1)*} = \frac{k_2 f_2 u_2^{(2)} R^* N_2^{(2)*}}{u_2^{(1)} R^* + \mu_2^{(1)}} \tag{S4.20}$$

$$0 = \frac{u_2^{(1)} k_2 f_2 u_2^{(2)} N_2^{(2)*} R^{*2}}{u_2^{(1)} R^* + \mu_2^{(1)}} - \left(u_2^{(2)} R^* + \mu_2^{(2)}\right) N_2^{(2)*}. \tag{S4.21}$$

Dividing the third equation above by $N_2^{(2)*}$ and multiplying by the denominator, we arrive at:

$$u_2^{(1)} k_2 f_2 u_2^{(2)} R^{*2} - \left(u_2^{(2)} R^* + \mu_2^{(2)}\right)\left(u_2^{(1)} R^* + \mu_2^{(1)}\right) = 0. \tag{S4.22}$$

S4.1 Method 1.1: Direct timescale separation

We will choose a new pair of variables: N, representing the slow dynamics of population growth, and Q, representing the fast dynamics of population stage structure. Since direct predation only depends on the second juvenile stage, we choose $N = N_1^{(2)}$ as a convenient choice for the population (slow) variable. Accordingly, a convenient choice of the stage structure (fast) variable is the ratio of first- to second-stage juveniles,

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$Q = N_1^{(1)}/N_1^{(2)}$. Then, $N_1^{(1)} = QN$, and we can rewrite the system considering the invasion scenario:

$$\frac{dN}{dt} = u_1^{(1)} R^* QN - \left[u_1^{(2)} R^* + m_1^{(2)}\right] N \tag{S4.23}$$

$$\begin{aligned}\frac{dQ}{dt} &= \frac{dN_1^{(1)}}{dt}\frac{1}{N} - \frac{dN}{dt}Q\frac{1}{N} \\ &= k_1 f_1 u_1^{(2)} R^* - \left(u_1^{(1)} R^* + \mu_1^{(1)} + a_{11}N\right) Q - u_1^{(1)} R^* Q^2 + \left[u_1^{(2)} R^* + m_1^{(2)}\right] Q \\ &= -u_1^{(1)} R^* Q^2 - \left(u_1^{(1)} R^* + \mu_1^{(1)} + a_{11}N - u_1^{(2)} R^* - m_1^{(2)}\right) Q + k_1 f_1 u_1^{(2)} R^*. \end{aligned} \tag{S4.24}$$

We proceed to solve for the quasi-equilibrium of Q in the invasion condition. In this case, $N \approx 0$, allowing us to neglect the term $a_{11}N$ (i.e., the cannibalism term representing the consumption of first-stage juveniles by second-stage juveniles). Solving using the quadratic formula gives the single biologically meaningful root,

$$Q^* = \frac{-B + \sqrt{B^2 + 4AC}}{2A}, \tag{S4.25}$$

where

$$A = u_1^{(1)} R^*,\ B = u_1^{(1)} R^* + \mu_1^{(1)} - u_1^{(2)} R^* - m_1^{(2)},\ C = k_1 f_1 u_1^{(2)} R^*. \tag{S4.26}$$

We note that the other root is necessarily negative and therefore not biologically meaningful (since Q is a ratio of populations). Next, we find the per-capita growth rate of N when $Q = Q^*$, i.e., the invasion growth rate IGR_1:

$$\mathrm{IGR}_1 = u_1^{(1)} R^* Q^* - \left[u_1^{(2)} R^* + m_1^{(2)}\right] \tag{S4.27}$$

$$= \frac{-B + \sqrt{B^2 + 4AC}}{2} - \left[u_1^{(2)} R^* + m_1^{(2)}\right]. \tag{S4.28}$$

Finally, we solve for the conditions under which the invasion growth rate is positive:

$$\sqrt{B^2 + 4AC} > B + 2\left[u_1^{(2)} R^* + m_1^{(2)}\right]. \tag{S4.29}$$

Noting that $\sqrt{B^2 + 4AC} > 0$ and the right-hand side of the above inequality $B + 2\left[u_1^{(2)} R^* + m_1^{(2)}\right] = u_1^{(1)} R^* + \mu_1^{(1)} + u_1^{(2)} R^* + m_1^{(2)}$ is also positive, we can rewrite:

$$\begin{aligned} B^2 + 4AC &> \left\{B + 2\left[u_1^{(2)} R^* + m_1^{(2)}\right]\right\}^2 \\ AC &> \left\{B + \left[u_1^{(2)} R^* + m_1^{(2)}\right]\right\}\left[u_1^{(2)} R^* + m_1^{(2)}\right] \\ AC &> \left(u_1^{(1)} R^* + \mu_1^{(1)}\right)\left[u_1^{(2)} R^* + m_1^{(2)}\right]. \end{aligned} \tag{S4.30}$$

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Substituting the expressions for A and C, followed by subtracting $\mu_1^{(2)}$, and finally dividing the whole inequality by $\mu_1^{(2)}$, we get:

$$\frac{u_1^{(1)}u_1^{(2)}k_1f_1R^{*2} - \left(u_1^{(2)}R^* + \mu_1^{(2)}\right)\left(u_1^{(1)}R^* + \mu_1^{(1)}\right)}{\mu_1^{(2)}\left(u_1^{(1)}R^* + \mu_1^{(1)}\right)} > \frac{m_1^{(2)}}{\mu_1^{(2)}} - 1. \tag{S4.31}$$

S4.2 Method 1.2: Strategic linear combination

At the invasion equilibrium, let $X = N_1^{(1)} + bN_1^{(2)}$, with $b = \frac{u_1^{(1)}R^* + \mu_1^{(1)}}{u_1^{(1)}R^*}$. Then:

$$\frac{dX}{dt} = b\left[\frac{u_1^{(1)}u_1^{(2)}k_1f_1R^{*2}}{u_1^{(1)}R^* + \mu_1^{(1)}} - \left(u_1^{(2)}R^* + m_1^{(2)}\right)\right]N_1^{(2)} \tag{S4.32}$$

$$= b\left[\frac{u_1^{(1)}u_1^{(2)}k_1f_1R^{*2}}{u_1^{(1)}R^* + \mu_1^{(1)}} - \left(u_1^{(2)}R^* + m_1^{(2)}\right)\right]\kappa X, \tag{S4.33}$$

with $\kappa > 0$. The invasion growth rate is thus $\frac{1}{X}\frac{dX}{dt}$, which is positive when:

$$\frac{u_1^{(1)}u_1^{(2)}k_1f_1R^{*2}}{u_1^{(1)}R^* + \mu_1^{(1)}} - \left(u_1^{(2)}R^* + m_1^{(2)}\right) > 0. \tag{S4.34}$$

With the same analytical trick as above, this criterion can be written as follows, which is identical to the condition in eq. S4.31:

$$\frac{u_1^{(1)}u_1^{(2)}k_1f_1R^{*2} - \left(u_1^{(2)}R^* + \mu_1^{(2)}\right)\left(u_1^{(1)}R^* + \mu_1^{(1)}\right)}{\mu_1^{(2)}\left(u_1^{(1)}R^* + \mu_1^{(1)}\right)} > \frac{m_1^{(2)}}{\mu_1^{(2)}} - 1. \tag{S4.35}$$

S4.3 Method 2.1: Direct solution for leading eigenvalue

For the same two-stage simplification as before, the dynamics of species 1 during invasion can be written in the linear matrix form:

$$\frac{d}{dt}\begin{bmatrix} N_1^{(1)} \\ N_1^{(2)} \end{bmatrix} = \mathbf{C}\begin{bmatrix} N_1^{(1)} \\ N_1^{(2)} \end{bmatrix}, \tag{S4.36}$$

where the matrix $\mathbf{C}$ is given by (note that the term $a_{11}N_1^{(2)}$ is omitted due to low abundance during invasion):

$$\mathbf{C} = \begin{bmatrix} -\left(u_1^{(1)}R^* + \mu_1^{(1)}\right) & k_1f_1u_1^{(2)}R^* \\ u_1^{(1)}R^* & -\left(u_1^{(2)}R^* + m_1^{(2)}\right) \end{bmatrix}. \tag{S4.37}$$

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Writing the characteristic equation $\det(\mathbf{C} - \lambda \mathbf{I}) = 0$, we obtain

$$\lambda^2 + \left[\left(u_1^{(1)} R^* + \mu_1^{(1)}\right) + \left(u_1^{(2)} R^* + m_1^{(2)}\right)\right] \lambda \tag{S4.38}$$

$$+ \left(u_1^{(1)} R^* + \mu_1^{(1)}\right)\left(u_1^{(2)} R^* + m_1^{(2)}\right) - u_1^{(1)} u_1^{(2)} k_1 f_1 R^{*2} = 0, \tag{S4.39}$$

which has the leading eigenvalue,

$$\lambda = \frac{-D + \sqrt{D^2 - 4E}}{2}, \tag{S4.40}$$

where

$$D = \left(u_1^{(1)} R^* + \mu_1^{(1)}\right) + \left(u_1^{(2)} R^* + m_1^{(2)}\right) > 0 \tag{S4.41}$$

$$E = \left(u_1^{(1)} R^* + \mu_1^{(1)}\right)\left(u_1^{(2)} R^* + m_1^{(2)}\right) - u_1^{(1)} u_1^{(2)} k_1 f_1 R^{*2}. \tag{S4.42}$$

Given that $D > 0$, the leading eigenvalue, and thus the invasion growth rate, is positive if E is negative, giving

$$\frac{u_1^{(1)} u_1^{(2)} k_1 f_1 R^{*2}}{u_1^{(1)} R^* + \mu_1^{(1)}} - \left(u_1^{(2)} R^* + m_1^{(2)}\right) > 0, \tag{S4.43}$$

which is identical to the previous invasion criterion in eq. S4.34 and can be further arranged to give eq. S4.31.

S4.4 Method 2.2: Applying stability conditions

Applying the Routh–Hurwitz stability conditions, we find that λ is positive if one of the following conditions holds. First, a negative determinant implies a strictly positive dominant eigenvalue,

$$\left(u_1^{(1)} R^* + \mu_1^{(1)}\right)\left(u_1^{(2)} R^* + m_1^{(2)}\right) - u_1^{(1)} u_1^{(2)} k_1 f_1 R^{*2} < 0, \tag{S4.44}$$

which matches the expression in eq. S4.31 and eq. S4.34. Second, if the determinant is positive, the eigenvalues share a sign and a positive trace implies that both eigenvalues are positive:

$$-\left(u_1^{(1)} R^* + \mu_1^{(1)}\right) - \left(u_1^{(2)} R^* + m_1^{(2)}\right) > 0, \tag{S4.45}$$

which cannot be fulfilled since all the parameters are strictly positive. Therefore, the determinant criterion solved above is the necessary and sufficient condition for the invasion of species 1.

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S4.5 Parameter effects on invasion

From the invasion criterion in eq. S4.31, we can also derive a pure parameter-based invasion condition by substituting the expression for $m_1^{(2)}$ into the invasion condition. Considering the scenario that the two species are identical except for the aggressive behavior of species 1, the numerator of the left-hand side of eq. S4.31, i.e.,

$$u_1^{(1)} u_1^{(2)} k_1 f_1 R^{*2} - \left(u_1^{(2)} R^* + \mu_1^{(2)}\right)\left(u_1^{(1)} R^* + \mu_1^{(1)}\right), \tag{S4.46}$$

equals

$$u_2^{(1)} u_2^{(2)} k_2 f_2 R^{*2} - \left(u_2^{(2)} R^* + \mu_2^{(2)}\right)\left(u_2^{(1)} R^* + \mu_2^{(1)}\right), \tag{S4.47}$$

which equals zero given the monoculture condition in eq. S4.22. Therefore, the necessary and sufficient condition for species 1 to invade is $\frac{m_1^{(2)}}{\mu_1^{(2)}} < 1$, which can be written out in full at invasion (i.e., $a_{11} N_1^{(1)} \approx 0$) as:

$$\frac{u_1^{(2)} R^* + a_{21} N_2^{(1)*}}{u_1^{(2)} R^* + b_{21} a_{21} N_2^{(1)*}} < 1. \tag{S4.48}$$

Therefore, if IGP is beneficial to species 1 ($b_{21} > 1$), which reduces the mortality of the predator, species 1 can always invade. Alternatively, if IGP is not beneficial (e.g., species 2 is toxic and increases species 1's mortality), then species 1 cannot invade. Importantly, one can also note that cannibalism does not affect the invasion criterion, which is understandable as cannibalism is inconsequential when species 1's abundance is low.

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S5 Plant–soil feedback model

We consider a simplification of the patch occupancy model used by Ke & Levine (2021) to study the effects of soil legacy persistence on plant competition. As in their "fast conditioning" models, we consider the transition to the conditioned state to be instantaneous when a plant grows on unconditioned soil (termed "conditioning" for short). However, we allow the transition to the conditioned state to be slow when a plant grows on a patch which already has a conditioned microbial community (termed "reconditioning"). This reflects the assumption that reconditioning should take longer than conditioning because it requires regression of the existing microbial legacy in addition to conditioning of the plant's own microbial community.

Our model treats seven state variables, representing the dynamics of the proportion of patches occupied by combinations of plants and soil states. Unoccupied patches may host no microbial community $P_0^{(0)}$ or be conditioned by either plant species $P_0^{(j)}$ (where $j = 1, 2$). However, patches with living plants ($i = 1, 2$) always have a conditioned community ($j = 1, 2$), and patches with the soil community of the opposite species $P_i^{(j)}$ ($j \neq i$) transition to patches where the plant species has conditioned its own community $P_i^{(i)}$. Thus, the model can be written as the system of differential equations

$$\frac{dP_0^{(0)}}{dt} = \sum_j d_j P_0^{(j)} - \sum_i r_i P_i P_0^{(0)} \tag{S5.1}$$

$$\frac{dP_i^{(i)}}{dt} = r_i P_i P_0^{(0)} + r_i \sigma_{c,ii} P_i P_0^{(i)} + \sum_{j \neq i} c_i P_i^{(j)} - m_i \sigma_{m,ii} P_i^{(i)} \tag{S5.2}$$

$$\frac{dP_i^{(j)}}{dt} = r_i \sigma_{c,ij} P_i P_0^{(j)} - c_i P_i^{(j)} - m_i \sigma_{m,ij} P_i^{(j)} \tag{S5.3}$$

$$\frac{dP_0^{(j)}}{dt} = \sum_i m_i \sigma_{m,ij} P_i^{(j)} - \sum_i r_i \sigma_{c,ij} P_i P_0^{(j)} - d_j P_0^{(j)}, \tag{S5.4}$$

where $i, j \in \{1, 2\}$.

Here, r_i is the colonization rate of plant i, which is modified by a colonization feedback factor of $\sigma_{c,ij}$ for patches with microbial community $j \neq 0$. Colonization furthermore depends on the total fecundity of species i, which we define as $P_i = \sum_j \sigma_{f,ij} P_i^{(j)}$, where $\sigma_{f,ij}$ is the fecundity feedback factor of plant i when growing in soil j. Upon colonization, unconditioned patches ($P_0^{(0)}$) immediately transition to the $P_i^{(i)}$ state, while patches with conditioned microbial communities ($P_0^{(j)}$) transition to the $P_i^{(j)}$ state. Meanwhile, plants in occupied patches die at a mortality rate m_i, which is modified by a mortality factor $\sigma_{m,ij}$. Finally, plants of species i gradually recondition soil to their own community at a rate c_i, while unoccupied patches of soil type j (i.e., $P_0^{(j)}$) slowly

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decay to unconditioned soil $P_0^{(0)}$ at a rate d_j.

Written out in full for two species, the model is:

$$\frac{dP_0^{(0)}}{dt} = d_1 P_0^{(1)} + d_2 P_0^{(2)} - r_1 P_1 P_0^{(0)} - r_2 P_2 P_0^{(0)} \tag{S5.5}$$

$$\frac{dP_0^{(1)}}{dt} = m_1 \sigma_{m,11} P_1^{(1)} + m_2 \sigma_{m,21} P_2^{(1)} - \left(r_1 \sigma_{c,11} P_1 + r_2 \sigma_{c,21} P_2\right) P_0^{(1)} - d_1 P_0^{(1)} \tag{S5.6}$$

$$\frac{dP_0^{(2)}}{dt} = m_1 \sigma_{m,12} P_1^{(2)} + m_2 \sigma_{m,22} P_2^{(2)} - \left(r_1 \sigma_{c,12} P_1 + r_2 \sigma_{c,22} P_2\right) P_0^{(2)} - d_2 P_0^{(2)} \tag{S5.7}$$

$$\frac{dP_1^{(1)}}{dt} = r_1 P_1 P_0^{(0)} + r_1 \sigma_{c,11} P_1 P_0^{(1)} + c_1 P_1^{(2)} - m_1 \sigma_{m,11} P_1^{(1)} \tag{S5.8}$$

$$\frac{dP_1^{(2)}}{dt} = r_1 \sigma_{c,12} P_1 P_0^{(2)} - c_1 P_1^{(2)} - m_1 \sigma_{m,12} P_1^{(2)} \tag{S5.9}$$

$$\frac{dP_2^{(2)}}{dt} = r_2 P_2 P_0^{(0)} + r_2 \sigma_{c,22} P_2 P_0^{(2)} + c_2 P_2^{(1)} - m_2 \sigma_{m,22} P_2^{(2)} \tag{S5.10}$$

$$\frac{dP_2^{(1)}}{dt} = r_2 \sigma_{c,21} P_2 P_0^{(1)} - c_2 P_2^{(1)} - m_2 \sigma_{m,21} P_2^{(1)}, \tag{S5.11}$$

where the fecundity-weighted total abundance terms P_1, P_2 used for calculating colonization are given by

$$P_1 = \sigma_{f,11} P_1^{(1)} + \sigma_{f,12} P_1^{(2)} \tag{S5.12}$$

$$P_2 = \sigma_{f,22} P_2^{(2)} + \sigma_{f,21} P_2^{(1)}. \tag{S5.13}$$

Note that we have written c_i instead of c_{ij} because there is only one reconditioning rate per species.

Monoculture equilibrium We first solve for the monoculture equilibrium of plant i growing alone. Dynamics reduce to:

$$\frac{dP_0^{(0)}}{dt} = d_i P_0^{(i)} - r_i P_i P_0^{(0)} \tag{S5.14}$$

$$\frac{dP_i^{(i)}}{dt} = r_i P_i P_0^{(0)} + r_i \sigma_{c,ii} P_i P_0^{(i)} - m_i \sigma_{m,ii} P_i^{(i)} \tag{S5.15}$$

$$\frac{dP_0^{(i)}}{dt} = m_i \sigma_{m,ii} P_i^{(i)} - r_i \sigma_{c,ii} P_i P_0^{(i)} - d_i P_0^{(i)}, \tag{S5.16}$$

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where we have dropped any terms involving other species j. At equilibrium,

$$0 = d_i P_0^{(i)*} - r_i P_i^* P_0^{(0)*} \tag{S5.17}$$

$$0 = r_i P_i^* P_0^{(0)*} + r_i \sigma_{c,ii} P_i^* P_0^{(i)*} - m_i \sigma_{m,ii} P_i^{(i)*} \tag{S5.18}$$

$$0 = m_i \sigma_{m,ii} P_i^{(i)*} - r_i \sigma_{c,ii} P_i^* P_0^{(i)*} - d_i P_0^{(i)*}, \tag{S5.19}$$

$P_i^{(i)}$ can be factored out of $dP_i^{(i)}/dt$, so we solve $\frac{1}{P_i^{(i)}} \frac{dP_i^{(i)}}{dt} = 0$ for $P_0^{(0)}$ to find:

$$P_0^{(0)*} = \frac{m_i \sigma_{m,ii} - r_i \sigma_{f,ii} \sigma_{c,ii} P_0^{(i)*}}{r_i \sigma_{f,ii}}. \tag{S5.20}$$

Letting $\beta_i = \sigma_{m,ii}/\sigma_{m,ij}$, $\gamma_i = \sigma_{f,ii}/\sigma_{f,ij}$, $\tilde{r}_i = r_i \sigma_{f,ij}$ and $\mu_i = m_i \sigma_{m,ij}/\tilde{r}_i$, with $j \neq i$, we can simplify this to:

$$P_0^{(0)*} = \frac{\beta_i \mu_i - \gamma_i \sigma_{c,ii} P_0^{(i)*}}{\gamma_i}. \tag{S5.21}$$

Using $P_i^{(i)} + P_0^{(i)} + P_0^{(0)} = 1$, and letting $\delta_i = \frac{d_i}{\tilde{r}_i}$, we can set $dP_0^{(0)}/dt = 0$ to obtain a quadratic equation for $P_0^{(i)*}$:

$$P_0^{(i)*2} \gamma_i \sigma_{c,ii} (1 - \sigma_{c,ii}) - P_0^{(i)*} [\delta_i + \beta_i \mu_i (1 - \sigma_{c,ii}) + \sigma_{c,ii} (\gamma_i - \beta_i \mu_i)] + \beta_i \mu_i \left(1 - \frac{\beta_i \mu_i}{\gamma_i}\right) = 0, \tag{S5.22}$$

which is a quadratic function of $P_0^{(i)}$. To ensure feasibility, we constrain $r_i \sigma_{f,ii} > m_i \sigma_{m,ii}$ in further calculations. Substituting $P_0^{(i)}$ with 0 gives $\beta_i \mu_i \left(1 - \frac{\beta_i \mu_i}{\gamma_i}\right)$, which is positive, and substituting $P_0^{(i)}$ with 1, we obtain $-\delta_i - \gamma_i \left(\sigma_{c,ii} - \frac{\beta_i \mu_i}{\gamma_i}\right)^2$, which is negative. Thus, the quadratic function must have exactly one root in the interval $(0, 1)$. Assuming negative colonization feedback ($\sigma_{c,ii} < 1$), the polynomial is a concave-up quadratic equation, with the biologically meaningful root

$$P_0^{(i)*} = G - \sqrt{G^2 - \frac{\beta_i \mu_i \left(1 - \frac{\beta_i \mu_i}{\gamma_i}\right)}{\gamma_i \sigma_{c,ii} (1 - \sigma_{c,ii})}}, \tag{S5.23}$$

where

$$G = \frac{1}{2} \left[\frac{\delta_i}{\gamma_i \sigma_{c,ii} (1 - \sigma_{c,ii})} + \frac{\beta_i \mu_i}{\gamma_i \sigma_{c,ii}} + \frac{\gamma_i - \beta_i \mu_i}{\gamma_i (1 - \sigma_{c,ii})} \right]. \tag{S5.24}$$

Note that we have taken the root with $-$ because there are two positive roots, but only one in the interval $(0, 1)$, so the root with $+$ must be greater than 1 and thus not biologically meaningful. In contrast, when assuming positive colonization feedback ($\sigma_{c,ii} > 1$), the polynomial becomes a concave-down quadratic. Here, we should take the root with $+$ since there is only one positive root, and the criterion makes $\gamma_i \sigma_{c,ii} (1 - \sigma_{c,ii})$

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negative, so the root with $-$ must be negative.

S5.1 Method 1.1: Solve for the invasion growth rate using the λ trick

The easiest way to solve for the invasion growth rate using separation of timescales is to use the "λ trick" from the simple model analysis (section S2.1.2). We consider the case where species 1 is invading the resident equilibrium of species 2 ($P_0^{(0)*}, P_0^{(2)*}, P_2^{(2)*}$). We define the ratio variable:

$$Q = \frac{P_1^{(1)}}{P_1^{(2)}}. \tag{S5.25}$$

Taking the time derivative gives:

$$\frac{dQ}{dt} = \frac{1}{P_1^{(2)}}\frac{dP_1^{(1)}}{dt} - Q\frac{1}{P_1^{(2)}}\frac{dP_1^{(2)}}{dt}. \tag{S5.26}$$

Substituting the dynamical equations for $P_1^{(1)}$ and $P_1^{(2)}$ and considering the invasion scenario yields (with $k_1 = c_1/\tilde{r}_1$):

$$\begin{aligned}\frac{dQ}{dt} = \tilde{r}_1\frac{1}{P_1^{(2)}}\Big[&P_1^{(1)}(\gamma_1 P_0^{(0)*} - \mu_1\beta_1) + P_1^{(2)}(P_0^{(0)*} + k_1) \\ &- Q\big(P_1^{(1)}\gamma_1\sigma_{c,12}P_0^{(2)*} + P_1^{(2)}(\sigma_{c,12}P_0^{(2)*} - k_1 - \mu_1)\big)\Big].\end{aligned} \tag{S5.27}$$

Rearranging terms gives:

$$\frac{dQ}{dt} = \tilde{r}_1\big[-\gamma_1\sigma_{c,12}P_0^{(2)*}Q^2 + \big(\gamma_1 P_0^{(0)*} - \mu_1\beta_1 - \sigma_{c,12}P_0^{(2)*} + k_1 + \mu_1\big)Q + P_0^{(0)*} + k_1\big]. \tag{S5.28}$$

To apply the λ trick, we define:

$$\lambda = \frac{1}{P_1^{(2)}}\frac{dP_1^{(2)}}{dt}. \tag{S5.29}$$

Then the equation for Q becomes:

$$\begin{aligned}\frac{dQ}{dt} &= \frac{1}{P_1^{(2)}}\frac{dP_1^{(1)}}{dt} - Q\lambda \\ &= \tilde{r}_1\frac{1}{P_1^{(2)}}\Big[P_1^{(1)}(\gamma_1 P_0^{(0)*} - \mu_1\beta_1) + P_1^{(2)}(P_0^{(0)*} + k_1)\Big] - Q\lambda \\ &= \tilde{r}_1\big[Q(\gamma_1 P_0^{(0)*} - \mu_1\beta_1 - \frac{\lambda}{\tilde{r}_1}) + P_0^{(0)*} + k_1\big].\end{aligned} \tag{S5.30}$$

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Setting $\frac{dQ}{dt} = 0$ yields the quasi-steady-state solution

$$Q^* = \frac{P_0^{(0)*} + k_1}{\mu_1\beta_1 - \gamma_1 P_0^{(0)*} + \frac{\lambda}{\tilde{r}_1}}. \tag{S5.31}$$

The growth rate λ is given by

$$\begin{aligned}\lambda &= \frac{1}{P_1^{(2)}}\frac{dP_1^{(2)}}{dt} \\ &= \tilde{r}_1\left(Q\gamma_1\sigma_{c,12}P_0^{(2)*} + \sigma_{c,12}P_0^{(2)*} - k_1 - \mu_1\right).\end{aligned} \tag{S5.32}$$

Substituting Q^* into λ gives

$$\lambda = \tilde{r}_1\left[\left(\frac{P_0^{(0)*} + k_1}{\mu_1\beta_1 - \gamma_1 P_0^{(0)*} + \frac{\lambda}{\tilde{r}_1}}\right)\left(\gamma_1\sigma_{c,12}P_0^{(2)*}\right) + \sigma_{c,12}P_0^{(2)*} - k_1 - \mu_1\right]. \tag{S5.33}$$

Setting $\lambda = 0$ at the invasion threshold yields

$$(P_0^{(0)*} + k_1)(\gamma_1\sigma_{c,12}P_0^{(2)*}) + (\mu_1\beta_1 - \gamma_1 P_0^{(0)*})(\sigma_{c,12}P_0^{(2)*} - k_1 - \mu_1) = 0. \tag{S5.34}$$

Expanding and rearranging terms leads to the following equation for the invasion threshold:

$$P_0^{(0)*}\gamma_1(k_1 + \mu_1) + P_0^{(2)*}\sigma_{c,12}(\gamma_1 k_1 + \mu_1\beta_1) - \mu_1\beta_1(k_1 + \mu_1) = 0. \tag{S5.35}$$

By substituting $P_0^{(0)*}$ with the relationship in eq. S5.21 and solving the above equation at zero, we can find the critical value of $P_0^{(2)*}$ that would be needed to make $\mathrm{IGR}_1 = 0$ (i.e., $\left(P_0^{(2)*}\right)_{\mathrm{crit}}$):

$$\left(P_0^{(2)*}\right)_{\mathrm{crit}} = \frac{\beta_1(k_1 + \mu_1)\mu_1 - \frac{\gamma_1}{\gamma_2}\beta_2(k_1 + \mu_1)\mu_2}{(\gamma_1 k_1 + \beta_1\mu_1)\sigma_{c,12} - \gamma_1(k_1 + \mu_1)\sigma_{c,22}} \equiv \frac{\Delta\mu}{\Delta\sigma}. \tag{S5.36}$$

Biologically, we note that soil 2 favors the invader when $\Delta\sigma > 0$ and the invasion condition is $P_0^{(2)*} > \left(P_0^{(2)*}\right)_{\mathrm{crit}}$; alternatively, soil 2 disfavors the invader when $\Delta\sigma < 0$ and the invasion condition is $P_0^{(2)*} < \left(P_0^{(2)*}\right)_{\mathrm{crit}}$. Both cases give the invasion condition for species 1 in terms of the resident equilibrium of species 2.

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S5.2 Method 1.2: Strategic linear combination

Let $X = P_1^{(1)} + bP_1^{(2)}$, with $b = \frac{\beta_1\mu_1 - \gamma_1 P_0^{(0)*}}{\gamma_1\sigma_{c,12}P_0^{(2)*}}$. Then, we can write the dynamics of X under the invasion scenario as:

$$\frac{dX}{dt} = \tilde{r}_1\left[P_1^{(2)}\left(P_0^{(0)*} + k_1 + \frac{\beta_1\mu_1 - \gamma_1 P_0^{(0)*}}{\gamma_1} - \frac{\beta_1\mu_1 - \gamma_1 P_0^{(0)*}}{\gamma_1\sigma_{c,12}P_0^{(2)*}}(k_1+\mu_1)\right)\right]. \quad \text{(S5.37)}$$

By substituting $P_1^{(2)}$ with $\frac{X}{Q^*+b}$, which is derived from eq. S5.25, we arrive at

$$\frac{dX}{dt} = \frac{X\tilde{r}_1}{(Q^*+b)\gamma_1\sigma_{c,12}P_0^{(2)*}}\left[\gamma_1 P_0^{(0)*}(k_1+\mu_1) + \gamma_1 k_1\sigma_{c,12}P_0^{(2)*} + \mu_1\beta_1\sigma_{c,12}P_0^{(2)*} - \mu_1\beta_1(k_1+\mu_1)\right]. \quad \text{(S5.38)}$$

The invasion growth rate is:

$$\text{IGR}_1 = \frac{\tilde{r}_1}{Q^*+b}\left[P_0^{(0)*} + k_1 + \frac{\beta_1\mu_1 - \gamma_1 P_0^{(0)*}}{\gamma_1} - \frac{\left(\beta_1\mu_1 - \gamma_1 P_0^{(0)*}\right)(k_1+\mu_1)}{\gamma_1\sigma_{c,12}P_0^{(2)*}}\right], \quad \text{(S5.39)}$$

where Q^* is obtained from eq. S5.28 as

$$Q^* = \frac{-B - \sqrt{B^2 - 4AC}}{2A}, \quad \text{(S5.40)}$$

where

$$A = -\gamma_1\sigma_{c,12}P_0^{(2)*} \quad \text{(S5.41)}$$

$$B = \gamma_1 P_0^{(0)*} - \beta_1\mu_1 - \sigma_{c,12}P_0^{(2)*} + k_1 + \mu_1 \quad \text{(S5.42)}$$

$$C = P_0^{(0)*} + k_1. \quad \text{(S5.43)}$$

Assuming $\Delta\sigma$ in eq. S5.36 is positive, we can derive the condition for $\text{IGR}_1 > 0$ as:

$$P_0^{(2)*} > \frac{\beta_1(k_1+\mu_1)\mu_1 - \frac{\gamma_1}{\gamma_2}\beta_2(k_1+\mu_1)\mu_2}{(\gamma_1 k_1 + \beta_1\mu_1)\sigma_{c,12} - \gamma_1(k_1+\mu_1)\sigma_{c,22}}. \quad \text{(S5.44)}$$

S5.3 Method 2.2: Applying stability conditions

To determine whether species 1 can invade when rare, we linearize the dynamics of species 1 around the resident equilibrium of species 2. Throughout this analysis, all quantities with a superscript $*$ denote values evaluated at the resident equilibrium. Using the notation introduced above, we define nondimensionalized parameters as follows: $\beta_1 = \sigma_{m,11}/\sigma_{m,12}$, $\gamma_1 = \sigma_{f,11}/\sigma_{f,12}$, $\tilde{r}_1 = r_1\sigma_{f,12}$, $\mu_1 = m_1\sigma_{m,12}/\tilde{r}_1$, $\delta_1 = d_1/\tilde{r}_1$,

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and $k_1 = c_1/\tilde{r}_1$. Furthermore, we rescale time to remove a factor of $\tilde{r}_1$ from the dynamics by letting $\tau = \tilde{r}_1 t$. Thus, when species 1 is rare, its dynamics are governed by the linearized subsystem:

$$\frac{dP_1^{(1)}}{d\tau} = P_1^{(1)}(\gamma_1 P_0^{(0)*} - \mu_1\beta_1) + P_1^{(2)}(P_0^{(0)*} + k_1) \tag{S5.45}$$

$$\frac{dP_1^{(2)}}{d\tau} = P_1^{(1)}(\gamma_1\sigma_{c,12}P_0^{(2)*}) + P_1^{(2)}(\sigma_{c,12}P_0^{(2)*} - k_1 - \mu_1). \tag{S5.46}$$

Rewriting these equations in matrix form yields

$$\frac{d}{d\tau}\begin{bmatrix} P_1^{(1)} \\ P_1^{(2)} \end{bmatrix} = \mathbf{C}\begin{bmatrix} P_1^{(1)} \\ P_1^{(2)} \end{bmatrix}, \tag{S5.47}$$

where the Jacobian matrix $\mathbf{C}$ is given by

$$\mathbf{C} = \begin{bmatrix} \gamma_1 P_0^{(0)*} - \mu_1\beta_1 & P_0^{(0)*} + k_1 \\ \gamma_1\sigma_{c,12}P_0^{(2)*} & \sigma_{c,12}P_0^{(2)*} - k_1 - \mu_1 \end{bmatrix}. \tag{S5.48}$$

The invasion growth rate of species 1 is determined by the dominant eigenvalue of $\mathbf{C}$. Rather than solving explicitly for the eigenvalues, we apply the standard stability conditions for 2×2 linear systems. In particular, invasion is possible if and only if the dominant eigenvalue of $\mathbf{C}$ is positive, which occurs when either

$$\mathrm{tr}(\mathbf{C}) > 0 \quad \text{or} \quad \det(\mathbf{C}) < 0. \tag{S5.49}$$

We first consider the determinant:

$$\det(\mathbf{C}) = \left(\gamma_1 P_0^{(0)*} - \mu_1\beta_1\right)\left(\sigma_{c,12}P_0^{(2)*} - k_1 - \mu_1\right) - \left(P_0^{(0)*} + k_1\right)\left(\gamma_1\sigma_{c,12}P_0^{(2)*}\right) < 0. \tag{S5.50}$$

Expanding and rearranging the determinant condition $\det(\mathbf{C}) < 0$ gives:

$$(\gamma_1 k_1 + \gamma_1\mu_1)P_0^{(0)*} + (\mu_1\beta_1\sigma_{c,12} + k_1\gamma_1\sigma_{c,12})P_0^{(2)*} > \mu_1\beta_1(k_1 + \mu_1). \tag{S5.51}$$

This inequality defines a linear constraint on the resident equilibrium state $(P_0^{(0)*}, P_0^{(2)*})$ from eq. S5.21, under which the determinant of the Jacobian becomes negative. Biologically, this corresponds to the regime in which the resident equilibrium loses linear stability to perturbations introduced by species 1. Again, assuming $\Delta\sigma > 0$ in eq. S5.36 and solving inequality S5.51 explicitly for the conditioned resident patch fraction $P_0^{(2)*}$

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yields the invasion threshold:

$$P_0^{(2)*} > \frac{\beta_1(k_1+\mu_1)\mu_1 - \frac{\gamma_1}{\gamma_2}\beta_2(k_1+\mu_1)\mu_2}{(\gamma_1 k_1 + \beta_1\mu_1)\sigma_{c,12} - \gamma_1(k_1+\mu_1)\sigma_{c,22}}. \tag{S5.52}$$

S5.4 Parameter effects on invasion

From the above invasion criterion, we can also derive a pure parameter-based invasion condition by substituting the equilibrium expressions for $P_0^{(2)*}$ (with square roots) into the invasion condition. However, this is messy to solve. For simpler algebra, consider the boundary where the invader can just invade, i.e., $\text{IGR}_1 = 0$. As previously mentioned, the critical value of $P_0^{(2)*}$ that would be needed to make $\text{IGR}_1 = 0$ is:

$$\left(P_0^{(2)*}\right)_{\text{crit}} = \frac{\beta_1(k_1+\mu_1)\mu_1 - \frac{\gamma_1}{\gamma_2}\beta_2(k_1+\mu_1)\mu_2}{(\gamma_1 k_1 + \beta_1\mu_1)\sigma_{c,12} - \gamma_1(k_1+\mu_1)\sigma_{c,22}} \equiv \frac{\Delta\mu}{\Delta\sigma}. \tag{S5.53}$$

Substituting $P_0^{(2)*}$ with this critical value into the quadratic expression in eq. S5.22 derived above gives:

$$(P_0^{(2)*})^2_{\text{crit}}\gamma_2\sigma_{c,22}\left(1-\sigma_{c,22}\right) - (P_0^{(2)*})_{\text{crit}}\left[\delta_2 + \beta_2\mu_2\left(1-\sigma_{c,22}\right) + \sigma_{c,22}\left(\gamma_2 - \beta_2\mu_2\right)\right] + \beta_2\mu_2\left(1-\frac{\beta_2\mu_2}{\gamma_2}\right). \tag{S5.54}$$

Note that this polynomial has a positive quadratic coefficient as $\sigma_{c,22} < 1$, is positive at 0, and the biologically relevant root is the smaller one. Thus, if the expression is positive at $\left(P_0^{(2)*}\right)_{\text{crit}}$, the actual equilibrium $P_0^{(2)*}$ is higher than $\left(P_0^{(2)*}\right)_{\text{crit}}$, and if the expression is negative, $P_0^{(2)*}$ is lower than $\left(P_0^{(2)*}\right)_{\text{crit}}$. The above two situations correspond to the cases where $\Delta\sigma > 0$ and $\Delta\sigma < 0$, respectively. After some algebra, we can simplify the expression to:

$$-\frac{\Delta\mu}{\Delta\sigma}\delta_2 + \frac{\omega(\Delta\mu - \Delta\sigma - \frac{\omega}{\gamma_2})}{\Delta\sigma^2}, \tag{S5.55}$$

where

$$\begin{aligned}
\Delta\mu &= \beta_1(k_1+\mu_1)\mu_1 - \frac{\gamma_1}{\gamma_2}\beta_2(k_1+\mu_1)\mu_2 \\
\Delta\sigma &= (\gamma_1 k_1 + \beta_1\mu_1)\sigma_{c,12} - \gamma_1(k_1+\mu_1)\sigma_{c,22} \\
\omega &= \gamma_2\sigma_{c,22}\Delta\mu - \beta_2\mu_2\Delta\sigma.
\end{aligned} \tag{S5.56}$$

To get the final invasion condition, consider that if soil 2 favors the invader ($\Delta\sigma > 0$), then invasion requires $P_0^{(2)*} > \left(P_0^{(2)*}\right)_{\text{crit}}$, which implies expression S5.55 is positive; multiplying through once by $\Delta\sigma$ gives our final answer: species 1 can invade if and only if

$$\Delta\mu \cdot \delta_2 < \frac{\omega(\Delta\mu - \Delta\sigma - \frac{\omega}{\gamma_2})}{\Delta\sigma}. \tag{S5.57}$$

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Finally, we note that if soil 2 disfavors the invader ($\Delta\sigma < 0$), then invasion requires $P_0^{(2)*} < \left(P_0^{(2)*}\right)_{\text{crit}}$ and the expression in eq. S5.55 to be negative. However, since $\Delta\sigma < 0$, multiplying through once by $\Delta\sigma$ reverses the inequality and results in the same invasion criterion as above. To gain further insights, we consider the following special cases. First, if we assume that species receive identical microbial effects from both conspecifics and heterospecifics (i.e., $\beta_i = \gamma_i = 1$), the inequality S5.57 can be simplified to:

$$(\mu_1 - \mu_2) \cdot \delta_2 < (\mu_1\sigma_{c,22} - \mu_2\sigma_{c,12})[(\frac{\mu_1 - \mu_2}{\sigma_{c,12} - \sigma_{c,22}}) - 1 - (\frac{\mu_1\sigma_{c,22} - \mu_2\sigma_{c,12}}{\sigma_{c,12} - \sigma_{c,22}})]. \tag{S5.58}$$

Furthermore, if we assume the realized mortality-to-colonization ratios of species 1 and 2 are identical ($\mu_1 = \mu_2 \equiv \mu_i$), the inequality can be simplified to:

$$0 > \mu_i(\sigma_{c,22} - \sigma_{c,12})(1 - \mu_i). \tag{S5.59}$$

Imposing the feasibility condition, we constrain $\mu_i < 1$ (i.e., $r_i\sigma_{f,ij} > m_i\sigma_{m,ij}$) to obtain the final criterion:

$$\sigma_{c,12} > \sigma_{c,22}. \tag{S5.60}$$

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S6 Plant–pest–parasite model

We model a plant–pest–parasite system where multiple pest species consume a shared host plant S and are infected by a common parasite P. The parasite infects healthy pests ($N_i^{(0)}$) and transitions them to parasitized pests ($N_i^{(P)}$), which can survive for a period of time during the infection phase before eventually dying as parasite offspring emerge from their bodies. We assume that the parasite is not vertically transmitted. This system can be described as follows:

$$\frac{dS}{dt} = rS\left(1 - \frac{S}{K}\right) - \sum_{i=1}^{n}(\alpha_i N_i^{(0)} + \alpha_i N_i^{(P)})S \tag{S6.1}$$

$$\frac{dN_i^{(0)}}{dt} = e_i\alpha_i N_i^{(0)} S + e_{i/P}\alpha_i N_i^{(P)} S - \beta_i N_i^{(0)} P - m_i N_i^{(0)} \tag{S6.2}$$

$$\frac{dN_i^{(P)}}{dt} = \beta_i N_i^{(0)} P - (\omega_i + m_i) N_i^{(P)} \tag{S6.3}$$

$$\frac{dP}{dt} = \sum_{i=1}^{n}(h_i\omega_i N_i^{(P)}) - \sum_{i=1}^{n}(c_i\beta_i N_i^{(0)})P - dP. \tag{S6.4}$$

The plant (S) grows logistically and is consumed by uninfected pests ($N_i^{(0)}$; we consider two pest species, so $i = 1, 2$) and parasitized pests ($N_i^{(P)}$) with consumption efficiency α_i. As the parasite is not vertically transmitted, host plant consumption by both pest statuses contributes to the uninfected pest population size (with conversion efficiencies e_i and $e_{i/P}$, respectively). Pest population decreases due to natural mortality (with intrinsic mortality rate m_i) and parasite infection (with efficiency β_i), converting pests into the parasitized state. In addition to their natural mortality, parasitized pests experience an additional parasite-induced mortality ω_i. Finally, the free-living parasite population increases following the death of parasitized pests (h_i parasites emerge upon the death of a single parasitized pest), and decreases as they infect pests (with conversion parameter c_i) and when they naturally die (with mortality rate d).

S6.1 Invasion analysis for parasite

In this host–pest–parasite system, there are two stage-structured population dynamics: one involves the transition of parasite stages (i.e., free-living versus inhabiting within the pest body) and the other involves the transition of the pest stages (i.e., uninfected versus infected). In this section, we will perform the invasion analysis for the first stage-structured population dynamics; namely, the parasite can be either (i) free-living, represented by the original state variable P, or (ii) within the pest body, represented by the original state variable $N_i^{(P)}$. To be consistent with the notation used throughout

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this study, we represent the free-living parasite as $P^{(0)}$ and the pest-infecting stage as $P^{(N)}$. For simplicity, we consider the invasion of the parasite into a host–single pest equilibrium (S^*, N^*) and rewrite the system as follows (note we rename $N^{(0)}$ as N to avoid cluttering the notation with too many superscripts):

$$\frac{dS}{dt} = rS\left(1 - \frac{S}{K}\right) - (\alpha N + \alpha P^{(N)})S \tag{S6.5}$$

$$\frac{dN}{dt} = e\alpha NS + e_P\alpha P^{(N)}S - \beta N P^{(0)} - mN \tag{S6.6}$$

$$\frac{dP^{(N)}}{dt} = \beta N P^{(0)} - (\omega + m)P^{(N)} \tag{S6.7}$$

$$\frac{dP^{(0)}}{dt} = h\omega P^{(N)} - c\beta N P^{(0)} - dP^{(0)}. \tag{S6.8}$$

S6.1.1 Method 1.1: Direct timescale separation

We will choose a new pair of variables: P, representing the slow dynamics of population growth, and Q_P, representing the fast dynamics of population stage structure. We choose $P = P^{(0)}$ for the population (slow) variable. The stage structure (fast) variable is the ratio of parasitized pests to free-living parasites, $Q_P = P^{(N)}/P^{(0)}$. Then $P^{(N)} = PQ_P$, and we can rewrite the system under the invasion condition as

$$\frac{dP}{dt} = h\omega PQ_P - c\beta N^* P - dP \tag{S6.9}$$

$$\frac{dQ_P}{dt} = P^{-2}\left(\frac{dP^{(N)}}{dt}P - \frac{dP}{dt}P^{(N)}\right) \tag{S6.10}$$

$$= -h\omega Q_P^2 + c\beta N^* Q_P + dQ_P - (m + \omega)Q_P + \beta N^*. \tag{S6.11}$$

First, we solve for the quasi-equilibrium of Q_P in the invasion condition. Solving using the quadratic formula gives the single biologically meaningful root,

$$Q_P^* = \frac{-(A + M) + \sqrt{(A + M)^2 + 4h\omega\beta N^*}}{2h\omega}, \text{ where } A = -c\beta N^* - d, \text{ and } M = m + \omega. \tag{S6.12}$$

Next, we find the per-capita growth rate of P when $Q_P = Q_P^*$, i.e., the invasion growth rate for the parasite:

$$\begin{aligned} \text{IGR}_P &= h\omega Q_P^* - c\beta N^* - d \\ &= \frac{-(A + M) + \sqrt{(A + M)^2 + 4h\omega\beta N^*}}{2} + A \\ &= \frac{A - M + \sqrt{(A + M)^2 + 4h\omega\beta N^*}}{2}. \end{aligned} \tag{S6.13}$$

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Finally, we solve for the conditions under which the invasion growth rate is positive:

$$A - M + \sqrt{(A+M)^2 + 4h\omega\beta N^*} > 0, \tag{S6.14}$$

and we note that $M - A$ is always positive, from which the following inequality can be derived:

$$(A+M)^2 + 4h\omega\beta N^* > (M-A)^2, \tag{S6.15}$$

and we get the condition for the parasite to invade:

$$d < \frac{h\omega\beta N^*}{m+\omega} - c\beta N^*. \tag{S6.16}$$

S6.1.2 Method 1.2: Strategic linear combination

Let $X = P^{(0)} + bP^{(N)}$, with $b = \frac{h\omega}{m+\omega}$. Then

$$\frac{dX}{dt} = \left(\frac{h\omega\beta N^*}{m+\omega} - c\beta N^* - d\right) P^{(0)} \tag{S6.17}$$

$$= \left(\frac{h\omega\beta N^*}{m+\omega} - c\beta N^* - d\right) \kappa X, \tag{S6.18}$$

with $\kappa > 0$. Then the invasion growth rate is

$$\mathrm{IGR}_P = \kappa \left(\frac{h\omega\beta N^*}{m+\omega} - c\beta N^* - d\right), \tag{S6.19}$$

and the invasion criterion is identical to the condition in eq. S6.16:

$$d < \frac{h\omega\beta N^*}{m+\omega} - c\beta N^*. \tag{S6.20}$$

S6.1.3 Method 2.2: Applying stability conditions

The dynamics of the parasite during invasion can be written in the linear matrix form:

$$\frac{d}{dt}\begin{bmatrix} P^{(0)} \\ P^{(N)} \end{bmatrix} = \mathbf{C}\begin{bmatrix} P^{(0)} \\ P^{(N)} \end{bmatrix}, \tag{S6.21}$$

where the matrix $\mathbf{C}$ is given by

$$\mathbf{C} = \begin{bmatrix} -(c\beta N^* + d) & h\omega \\ \beta N^* & -(m+\omega) \end{bmatrix}. \tag{S6.22}$$

Considering the eigenvalues (λ) of $\mathbf{C}$ and applying the Routh–Hurwitz stability conditions, we find that the leading eigenvalue is positive if one of the following conditions

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holds. First, a negative determinant implies a positive dominant eigenvalue,

$$(c\beta N^* + d)(m + \omega) - h\omega\beta N^* < 0, \tag{S6.23}$$

which matches the expression in eq. S6.16. Second, if this condition does not hold, the eigenvalues share a sign and a positive trace implies that both eigenvalues are positive:

$$-(c\beta N^* + d) - (m + \omega) > 0, \tag{S6.24}$$

which cannot be fulfilled since all the parameters are strictly positive. Therefore, the determinant criterion solved above is the necessary and sufficient condition for the invasion of the parasite.

S6.1.4 Parameter effects on parasite invasion

The equilibrium abundances of the host–pest system (S^*, N^*) can easily be obtained by setting $\frac{dS}{dt} = 0$ and $\frac{dN}{dt} = 0$, which gives $S^* = \frac{m}{e\alpha}$ and $N^* = \frac{r}{\alpha}\left(1 - \frac{m}{e\alpha K}\right)$. Substituting N^* into the inequality in eq. S6.16, the invasion criterion of the parasite can be obtained as:

$$\beta > \frac{d}{\frac{r}{\alpha}\left(1 - \frac{m}{e\alpha K}\right)\left(\frac{h\omega}{m+\omega} - c\right)}. \tag{S6.25}$$

We find that positive invasion growth of the parasite requires that the infection rate of the parasite must be higher than a threshold determined by host, pest, and parasite traits. In other words, the higher the infection rate of the parasite, the more likely it is that the parasite can invade and persist when introduced to the host–pest system.

S6.2 Invasion analysis for pest

Now, we turn our focus to the second stage-structured dynamics in the general model: pests can be either healthy ($N_i^{(0)}$) or infected ($N_i^{(P)}$) by parasites. Considering two pest

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species, and reverting back to the notation in the general model, the system is

$$\frac{dS}{dt} = rS\left(1 - \frac{S}{K}\right) - (\alpha_1 N_1^{(0)} + \alpha_1 N_1^{(P)} + \alpha_2 N_2^{(0)} + \alpha_2 N_2^{(P)})S \tag{S6.26}$$

$$\frac{dN_1^{(0)}}{dt} = e_1\alpha_1 N_1^{(0)} S + e_{1/P}\alpha_1 N_1^{(P)} S - \beta_1 N_1^{(0)} P - m_1 N_1^{(0)} \tag{S6.27}$$

$$\frac{dN_2^{(0)}}{dt} = e_2\alpha_2 N_2^{(0)} S + e_{2/P}\alpha_2 N_2^{(P)} S - \beta_2 N_2^{(0)} P - m_2 N_2^{(0)} \tag{S6.28}$$

$$\frac{dN_1^{(P)}}{dt} = \beta_1 N_1^{(0)} P - (\omega_1 + m_1) N_1^{(P)} \tag{S6.29}$$

$$\frac{dN_2^{(P)}}{dt} = \beta_2 N_2^{(0)} P - (\omega_2 + m_2) N_2^{(P)} \tag{S6.30}$$

$$\frac{dP}{dt} = h_1\omega_1 N_1^{(P)} + h_2\omega_2 N_2^{(P)} - c_1\beta_1 N_1^{(0)} P - c_2\beta_2 N_2^{(0)} P - dP. \tag{S6.31}$$

Below, we perform the invasion analysis for pest species 1, with stage-structured state variables $N_1^{(0)}$ and $N_1^{(P)}$, to invade the resident equilibrium of pest species 2 (S^*, $N_2^{(0)*}$, $N_2^{(P)*}$, P^*).

S6.2.1 Method 1.1: Direct timescale separation

We will choose a new pair of variables: N_1, representing the slow dynamics of population growth, and Q_{N_1}, representing the fast dynamics of population stage structure. We choose $N_1 = N_1^{(P)}$ for the population (slow) variable. Accordingly, the stage structure (fast) variable is the ratio of uninfected to parasitized pests, $Q_{N_1} = N_1^{(0)}/N_1^{(P)}$. Then $N_1^{(0)} = N_1 Q_{N_1}$ and we can rewrite the system considering the invasion scenario:

$$\frac{dN_1}{dt} = \beta_1 Q_{N_1} N_1 P^* - (m_1 + \omega_1) N_1 \tag{S6.32}$$

$$\frac{dQ_{N_1}}{dt} = -\beta_1 P^* Q_{N_1}^2 - \left(\beta_1 P^* + m_1 - (m_1 + \omega_1) - e_1\alpha_1 S^*\right) Q_{N_1} + e_{1/P}\alpha_1 S^*. \tag{S6.33}$$

First, we solve for the quasi-equilibrium of Q_{N_1} in the invasion condition. Solving using the quadratic formula gives the single biologically meaningful root,

$$Q_{N_1}^* = \frac{-(B - M) + \sqrt{(B - M)^2 + 4\beta_1 P^* e_{1/P}\alpha_1 S^*}}{2\beta_1 P^*}, \tag{S6.34}$$

where $M = m_1 + \omega_1$ and $B = \beta_1 P^* + m_1 - e_1\alpha_1 S^*$. Next, we find the per-capita growth rate of N_1 when the resident species are at equilibrium (S^*, $N_2^{(0)*}$, $N_2^{(P)*}$, and P^*). We

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find that the invasion growth rate of N_1 is expressed as

$$\begin{aligned}\text{IGR}_{N_1} &= \beta_1 P^* Q^*_{N_1} - M \\ &= \frac{-(B-M) + \sqrt{(B-M)^2 + 4\beta_1 P^* e_{1/P}\alpha_1 S^*}}{2} - M \\ &= \frac{-(B+M) + \sqrt{(B-M)^2 + 4\beta_1 P^* e_{1/P}\alpha_1 S^*}}{2}.\end{aligned} \tag{S6.35}$$

Finally, we solve for the conditions under which the invasion growth rate is positive:

$$-(B+M) + \sqrt{(B-M)^2 + 4\beta_1 P^* e_{1/P}\alpha_1 S^*} > 0. \tag{S6.36}$$

We now discuss the sign of IGR_{N_1} based on the sign of $B+M$. If $B+M$ is negative (i.e., $e_1\alpha_1 S^* - \beta_1 P^* - m_1 > m_1 + \omega_1$), then IGR_{N_1} must be positive. Alternatively, if $B+M$ is positive (i.e., $e_1\alpha_1 S^* - \beta_1 P^* - m_1 < m_1 + \omega_1$), we derive the following condition for positive invasion growth rate:

$$(B-M)^2 + 4\beta_1 P^* e_{1/P}\alpha_1 S^* > (B+M)^2, \tag{S6.37}$$

which can be simplified to the following:

$$e_1\alpha_1 S^* + \frac{\beta_1 e_{1/P}\alpha_1 S^* P^*}{m_1+\omega_1} - \beta_1 P^* - m_1 > 0. \tag{S6.38}$$

Given that the condition for $B+M<0$ also implies the above condition, eq. S6.38 is the necessary and sufficient condition for pest species 1 to invade the monoculture of pest species 2 along with its associated host plant and parasite abundances.

S6.2.2 Method 1.2: Strategic linear combination

Let $X = N_1^{(0)} + bN_1^{(P)}$, with $b = \frac{e_{1/P}\alpha_1 S^*}{m_1+\omega_1}$. Then

$$\begin{aligned}\frac{dX}{dt} &= \left(e_1\alpha_1 S^* - \beta_1 P^* - m_1 + \frac{\beta_1 e_{1/P}\alpha_1 S^* P^*}{m_1+\omega_1}\right) N_1^{(0)} \\ &= \left(e_1\alpha_1 S^* - \beta_1 P^* - m_1 + \frac{\beta_1 e_{1/P}\alpha_1 S^* P^*}{m_1+\omega_1}\right) \kappa X,\end{aligned} \tag{S6.39}$$

with $\kappa > 0$. Then the invasion growth rate is

$$\text{IGR}_{N_1} = \left(e_1\alpha_1 S^* - \beta_1 P^* - m_1 + \frac{\beta_1 e_{1/P}\alpha_1 S^* P^*}{m_1+\omega_1}\right) \kappa, \tag{S6.40}$$

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and the invasion criterion is the same as that in eq. S6.38:

$$e_1\alpha_1 S^* + \frac{\beta_1 e_{1/P}\alpha_1 S^* P^*}{m_1+\omega_1} - \beta_1 P^* - m_1 > 0. \tag{S6.41}$$

S6.2.3 Method 2.2: Applying stability conditions

The dynamics of pest species 1 during invasion can be written in the linear matrix form:

$$\frac{d}{dt}\begin{bmatrix} N_1^{(P)} \\ N_1^{(0)} \end{bmatrix} = \mathbf{C}\begin{bmatrix} N_1^{(P)} \\ N_1^{(0)} \end{bmatrix}, \tag{S6.42}$$

where the matrix $\mathbf{C}$ is given by

$$\mathbf{C} = \begin{bmatrix} -(m_1+\omega_1) & \beta_1 P^* \\ e_{1/P}\alpha_1 S^* & e_1\alpha_1 S^* - \beta_1 P^* - m_1 \end{bmatrix}. \tag{S6.43}$$

Considering the eigenvalues (λ) of $\mathbf{C}$ and applying the Routh–Hurwitz stability conditions, we find that the leading eigenvalue is positive if one of the following conditions holds. First, a negative determinant implies a dominant eigenvalue with nonnegative real part,

$$-(m_1+\omega_1)(e_1\alpha_1 S^* - \beta_1 P^* - m_1) - \beta_1 P^* e_{1/P}\alpha_1 S^* < 0, \tag{S6.44}$$

which matches the expression in eq. S6.38. Second, if this condition does not hold, the eigenvalues share a sign and a positive trace implies that both eigenvalues are positive:

$$-(m_1+\omega_1) + e_1\alpha_1 S^* - \beta_1 P^* - m_1 > 0, \tag{S6.45}$$

but as we noted above, this condition also implies the first; therefore, the determinant criterion solved above is the necessary and sufficient condition for the invasion of species 1.

S6.2.4 Parameter effects on pest invasion

From eq. S6.38, we find that for pest species 1 to invade, the parameters must satisfy

$$\beta_1 P^*(e_{1/P}\alpha_1 S^* - m_1 - \omega_1) > (m_1 - e_1\alpha_1 S^*)(m_1+\omega_1). \tag{S6.46}$$

Let $M_1 = m_1 + \omega_1$, representing the mortality of infected individuals. If $e_{1/P}\alpha_1 S^* > M_1$, we divide the above expression by the positive term with the parenthesis on the left-

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hand side, and rewrite the inequality as

$$\beta_1 > \frac{(m_1 - e_1\alpha_1 S^*)M_1}{(e_{1/P}\alpha_1 S^* - M_1)P^*}. \tag{S6.47}$$

Since $e_{1/P} \leq e_1$ (i.e., parasite infection reduces the conversion efficiency of pests),

$$m_1 - e_1\alpha_1 S^* \leq m_1 - e_{1/P}\alpha_1 S^* < -\omega_1 < 0. \tag{S6.48}$$

It follows that the right-hand side of equation S6.47 is less than zero, and the invasion criterion for N_1 is always fulfilled since $\beta_1 > 0$. Biologically speaking, when $e_{1/P}\alpha_1 S^* > M_1$, the mortality of infected individuals is relatively low while their energy gain from host consumption is relatively high; therefore, N_1 can always invade regardless of the pest's susceptibility to parasites (β_1).

On the other hand, if $e_{1/P}\alpha_1 S^* < M_1$, i.e., the resource level is low and the price of infection is high, species 1 can only invade if the infection efficiency of parasites satisfies

$$\beta_1 < \frac{(e_1\alpha_1 S^* - m_1)M_1}{(M_1 - e_{1/P}\alpha_1 S^*)P^*}. \tag{S6.49}$$

One can see that the above criterion is a decreasing function of P^*, meaning that a top-down control from higher resident parasite population makes invasion harder, and an increasing function of S^*, meaning that a bottom-up control from higher resident plant population provides more resource to make invasion easier. We can also rewrite the invasion criterion as

$$\beta_1 < \frac{e_1\alpha_1 S^* - m_1}{P^*} \times \frac{M_1}{M_1 - e_{1/P}\alpha_1 S^*}, \tag{S6.50}$$

where the latter term increases when M_1 decreases; thus, a lower mortality of infected individuals allows species 1 to endure a higher infection efficiency.

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S7 Stage-structured model with limit cycle

We consider the stage-structured resident species N_2 growing alone in the cyclic model introduced in eq. 12 of the main text. Dropping the species subscripts (i.e., $\vec{N} = \vec{N}_2$), the subsystem is given by

$$\frac{d\vec{N}}{dt} = \begin{bmatrix} r - \alpha N^{(2)} + \beta N^{(1)} & f \\ g + \gamma N^{(1)} & -m \end{bmatrix} \vec{N}. \tag{S7.1}$$

In this Appendix, we derive the conditions on facilitation β, γ for a limit cycle via a Hopf bifurcation.

S7.1 Equilibrium

Let $\vec{N}^*$ represent the resident equilibrium of the system and $\mathbf{C}^*$ represent the resident's stage transition matrix (i.e., the matrix in eq. S7.1) evaluated at this resident equilibrium. Since $\mathbf{C}^*\vec{N}^* = 0$ for a nonzero vector $\vec{N}^*$, it must be the case that $\mathbf{C}^*$ is singular. Thus a necessary condition for equilibrium is $\det \mathbf{C}^* = 0$, which means the abundances lie on the line:

$$mr + fg - \alpha m N^{(2)*} + (\beta m + f\gamma)\, N^{(1)*} = 0. \tag{S7.2}$$

Next, setting $dN^{(2)}/dt = 0$, we also have

$$mN^{(2)*} = \gamma \left(N^{(1)*}\right)^2 + gN^{(1)*}. \tag{S7.3}$$

Substituting the expression for $N^{(2)*}$ into the equilibrium condition gives

$$\alpha\gamma \left(N^{(1)*}\right)^2 - (\beta m + f\gamma - \alpha g)\, N^{(1)*} - (mr + fg) = 0. \tag{S7.4}$$

Then, we obtain

$$N^{(1)*} = \frac{(\beta m + f\gamma - \alpha g) \pm \sqrt{(\beta m + f\gamma - \alpha g)^2 + 4\alpha\gamma\,(mr + fg)}}{2\alpha\gamma} \tag{S7.5}$$

$$= B \pm \sqrt{B^2 + \frac{mr + fg}{\alpha\gamma}}, \text{ where } B = \frac{\beta m + f\gamma - \alpha g}{2\alpha\gamma}, \tag{S7.6}$$

where we can ignore the $-$ root because it is always negative and hence not biologically meaningful.

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S7.2 Hopf bifurcation

The first step to showing the emergence of a limit cycle is showing that the equilibrium loses stability through a Hopf bifurcation; i.e., as parameters are varied, the real parts of a complex conjugate pair of leading eigenvalues cross from negative to positive through zero.

Letting $dN^{(k)}/dt = f_k(\vec{N})$, the Jacobian matrix is the matrix $\mathbf{J}$ such that the element in row k and column l is $(\mathbf{J})_{kl} = \partial f_k/\partial N^{(l)}$. We have $f_1 = \begin{bmatrix} r - \alpha N^{(2)} + \beta N^{(1)} & f \end{bmatrix} \vec{N} = \left(r - \alpha N^{(2)} + \beta N^{(1)}\right) N^{(1)} + f N^{(2)}$ and $f_2 = \begin{bmatrix} g + \gamma N^{(1)} & -m \end{bmatrix} \vec{N} = \left(g + \gamma N^{(1)}\right) N^{(1)} - m N^{(2)}$ and thus, at equilibrium $\vec{N}^*$,

$$\mathbf{J}^* = \mathbf{C}^* + N^{(1)*} \begin{bmatrix} \beta & -\alpha \\ \gamma & 0 \end{bmatrix} \tag{S7.7}$$

$$= \begin{bmatrix} r - \alpha N^{(2)*} + 2\beta N^{(1)*} & f - \alpha N^{(1)*} \\ g + 2\gamma N^{(1)*} & -m \end{bmatrix}. \tag{S7.8}$$

First, we derive the general criterion on $\mathbf{J}^*$ for a Hopf bifurcation in a two-dimensional system. Generally speaking, the characteristic equation is $\lambda^2 - (\operatorname{tr}\mathbf{J}^*)\lambda + \det\mathbf{J}^* = 0$. A Hopf bifurcation occurs when the eigenvalues are a complex conjugate pair and the real part changes sign (i.e., they cross the imaginary axis while their imaginary parts remain nonzero). Thus, we have two conditions: (1) the real part changes sign at $\operatorname{tr}\mathbf{J}^* = 0$ for some parameter values, and (2) at those parameter values, the roots are complex, which occurs whenever $\det\mathbf{J}^* > 0$ (note that $\operatorname{tr}\mathbf{J}^* = 0$ here, so the sign of the polynomial's discriminant is fully determined by the matrix's determinant).

Now we consider our specific expression for the Jacobian. Leaving the elements of $\mathbf{C}^*$ unsubstituted at first,

$$\det\mathbf{J}^* = \det \begin{bmatrix} c_{11}^* + \beta N^{(1)*} & c_{12}^* - \alpha N^{(1)*} \\ c_{21}^* + \gamma N^{(1)*} & c_{22}^* \end{bmatrix} \tag{S7.9}$$

$$= \det\mathbf{C}^* + c_{22}^*\beta N^{(1)*} - c_{12}^*\gamma N^{(1)*} + c_{21}^*\alpha N^{(1)*} + \alpha\gamma\left(N^{(1)*}\right)^2 \tag{S7.10}$$

$$= N^{(1)*}\left(c_{22}^*\beta - c_{12}^*\gamma + c_{21}^*\alpha + \alpha\gamma N^{(1)*}\right) \tag{S7.11}$$

$$= N^{(1)*}\left(\alpha g - \beta m - f\gamma + 2\alpha\gamma N^{(1)*}\right), \tag{S7.12}$$

where we have used that $\det\mathbf{C}^* = 0$ at equilibrium and substituted in the expressions for elements of $\mathbf{C}^*$. We note that $N^{(1)*}$ is positive for a feasible equilibrium, so the other

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condition for the Hopf bifurcation is

$$N^{(1)*} > \frac{\beta m + f\gamma - \alpha g}{2\alpha\gamma} = B. \tag{S7.13}$$

Based on our solution to the quadratic eq. S7.4, $N^{(1)*}$ equals B plus some positive quantity, so this always holds. Next, note that the trace is

$$\operatorname{tr} \mathbf{J}^* = r - m - \alpha N^{(2)*} + 2\beta N^{(1)*}. \tag{S7.14}$$

Then the trace becomes greater than zero, one condition for a Hopf bifurcation, when the inequality below goes from false to true:

$$2\beta N^{(1)*} - \alpha N^{(2)*} > m - r. \tag{S7.15}$$

Using the condition for equilibrium, we can rewrite this as:

$$\left(\beta - \frac{f\gamma}{m}\right) N^{(1)*} > \frac{fg}{m} + m, \tag{S7.16}$$

$$(\beta m - \gamma f)\, N^{(1)*} > fg + m^2. \tag{S7.17}$$

Feasibility requires $N^{(1)*} > 0$, and the right-hand side is always positive, so

$$\beta m > \gamma f \tag{S7.18}$$

is a necessary (but not sufficient) condition for a Hopf bifurcation. Dividing by the positive quantity $\beta m - \gamma f$, we find that a Hopf bifurcation occurs when the equilibrium abundance crosses above

$$N_H^{(1)} = \frac{fg + m^2}{\beta m - f\gamma}. \tag{S7.19}$$

S7.3 Boundedness

Let $N = N^{(1)}$ and $Q = N^{(2)}/N^{(1)}$ as before. We will rewrite the system in terms of these transformed variables and, considering Q to be the fast variable, find its quasi-equilibrium Q^*. Rewriting the derivative, we find

$$\frac{dQ}{dt} = \frac{1}{N^{(1)}} \cdot \frac{dN^{(2)}}{dt} - \frac{N^{(2)}}{\left(N^{(1)}\right)^2} \frac{dN^{(1)}}{dt} \tag{S7.20}$$

$$= (\alpha N - f)\, Q^2 - (\beta N + m + r)\, Q + (\gamma N + g) \tag{S7.21}$$

$$= \left(\alpha Q^2 - \beta Q + \gamma\right) N - fQ^2 - (m + r)\, Q + g. \tag{S7.22}$$

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When $N \to \infty$, the terms in N dominate dQ/dt and thus, the quasi-equilibrium Q^* obeys

$$\alpha \left(Q^*\right)^2 - \beta Q^* + \gamma = 0, \tag{S7.23}$$

which has the solutions $Q^* = (\beta \pm \sqrt{\beta^2 - 4\alpha\gamma})/2\alpha$. The discriminant $\beta^2 - 4\alpha\gamma$ determines how many real roots there are. We use the discriminant to consider the effect of parameter values on the long-term behavior of Q; here, we ignore the boundary case where $\beta^2 - 4\alpha\gamma = 0$, where diagnosing long-term behavior requires a more rigorous analysis. If the discriminant is positive ($\beta^2 > 4\alpha\gamma$), both roots of Q^* are real and positive. Since dQ/dt is negative between the two roots, the smaller one is the stable solution; thus,

$$Q^* = \frac{\beta - \sqrt{\beta^2 - 4\alpha\gamma}}{2\alpha}. \tag{S7.24}$$

If the discriminant is negative ($\beta^2 < 4\alpha\gamma$), there are no real roots and dQ/dt is positive for all Q. This suggests that as long as N remains large enough, Q can grow arbitrarily large.

Now consider asymptotic population growth at high N. Writing dN/dt in terms of Q and N, we have

$$\frac{dN}{dt} = (r - \alpha QN + \beta N)\, N + fQN \tag{S7.25}$$

$$= (\beta - \alpha Q)\, N^2 + (r + fQ)\, N, \tag{S7.26}$$

where the terms in N^2 dominate as N becomes large. Thus, population growth at large N is determined by the coefficient $\beta - \alpha Q$. Consider the two cases for the asymptotic behavior of Q derived above:

- If $\beta^2 > 4\alpha\gamma$, Q reaches a quasi-equilibrium Q^*. Substituting the expression for Q^* (eq. S7.24), we see that $\beta - \alpha Q^* = (\beta + \sqrt{\beta^2 - 4\alpha\gamma})/2 > 0$. This means that dN/dt remains positive. Biologically we conclude that the resident settles on some fixed quasi-equilibrium population structure and grows indefinitely.

- If $\beta^2 < 4\alpha\gamma$, Q continues to grow: in terms of the original variables, adults $N^{(2)}$ become increasingly abundant relative to juveniles $N^{(1)}$. This eventually causes the coefficient $\beta - \alpha Q$ to become negative, thus making dN/dt negative as well. Thus, the system is driven back down to lower abundances. Biologically, in this case, as the resident population becomes large, it eventually becomes dominated by adults. Since adults hinder the population's overall growth rate via their competitive effect on juveniles, this has the effect of preventing unbounded population growth.

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Thus, our heuristic analysis suggests that

$$\beta^2 < 4\alpha\gamma \tag{S7.27}$$

is required for population growth to remain bounded. Biologically, it ensures that the facilitative effect β of juveniles on their own replenishment rate is counterbalanced by the population regulation imposed by adults, as determined by the competition coefficient α and the γ parameter controlling facilitation of the juvenile-adult transition.

S7.4 Parameter selection

Finally, we consider how to select parameters such that the system shows a stable limit cycle. The parameter γ is crucial: it regulates the population, preventing unbounded growth, but it must also be small enough that it doesn't damp out oscillations. To see this, note that the necessary condition for the Hopf bifurcation is that

$$\gamma < \frac{\beta m}{f}. \tag{S7.28}$$

Boundedness, however, requires

$$\gamma > \frac{\beta^2}{4\alpha}. \tag{S7.29}$$

Thus, values of γ that allow stable limit cycles exist only when

$$\beta < \frac{4\alpha m}{f}. \tag{S7.30}$$

Having chosen a value of β satisfying this condition, we can then choose a γ value such that $\beta^2/4\alpha < \gamma < \beta m/f$. Finally, we must ensure that the equilibrium juvenile abundance $N^{(1)*}$ (eq. S7.6) is above the threshold $N_H^{(1)}$ (eq. S7.19).

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S8 Stage-structured model with chaotic dynamics

Here, we analyze a multispecies extension of the model in equations 1a–1c of the main text. We consider a model of n interacting species with $n+1$ state variables: the two stages of species 1, $N_1^{(1)}$ and $N_1^{(2)}$, which we will analyze as the invading species, plus $n-1$ non–stage-structured species $N_2, \ldots, N_n$. These follow Lotka–Volterra dynamics,

$$\frac{dN_i}{dt} = N_i \cdot \left(r_i - \sum_j \alpha_{ij} N_j \right), \text{ for } i \geq 2, \tag{S8.1}$$

where, in the residents' competition terms, we take N_1 to represent species 1's total abundance, $N_1 = N_1^{(1)} + N_1^{(2)}$. Here, r_i is the intrinsic growth rate of species i, and α_{ij} is the sensitivity of species i's per-capita growth to competition from species j. Meanwhile, species 1's dynamics obey

$$\frac{dN_1^{(1)}}{dt} = \left(r_1 - \alpha_{11} N_1^{(2)} - \sum_{j=2}^{n} \alpha_{1j} N_j \right) N_1^{(1)} + f_1 N_1^{(2)} \tag{S8.2}$$

$$\frac{dN_1^{(2)}}{dt} = g_1 N_1^{(1)} - m_1 N_1^{(2)}, \tag{S8.3}$$

where r_1 is the juvenile-to-juvenile transition rate (representing juvenile fecundity if positive, or mortality if negative), f_1 is the fecundity of adults, g_1 is the maturation rate of juveniles to adults, and m_1 is the adult mortality rate. As in the stage-structured models in the main text (eqs. 1a–1c and eq. 12), competition affects the invader's juvenile-to-juvenile transition rate (modifying the r_1 term), and the invader's competitive effect on itself (the α_{11} term) is mediated only by adult abundance, $N_1^{(2)}$. Note the different interpretations of α: the competitive effect caused by the invader (the α_{i1} terms) reflects both invader stages because it depends on the total abundance N_1.

Chaotic parameter values We use 5 species (total 6 state variables). Species 1's demographic parameters are as before: $r_1 = -1$, $f_1 = 3$, $g_1 = 1$, $m_1 = 0.5$. We couple this to a chaotic four-species Lotka–Volterra system represented by species 2 through 5, such that the complete set of r_i and α_{ij} coefficients is given by

$$r = \begin{bmatrix} -1 \\ 5 \\ 5 \\ 5 \\ 5 \end{bmatrix}; \quad A = \begin{bmatrix} 5 & 5 & 1 & 3 & 3 \\ 0.6 & 5 & 10 & 6.5 & 0 \\ 0.7 & 0 & 5 & 5 & 5 \\ 0.8 & 10 & 0 & 5 & 0 \\ 0.9 & 5 & 0 & 0 & 5 \end{bmatrix}, \tag{S8.4}$$

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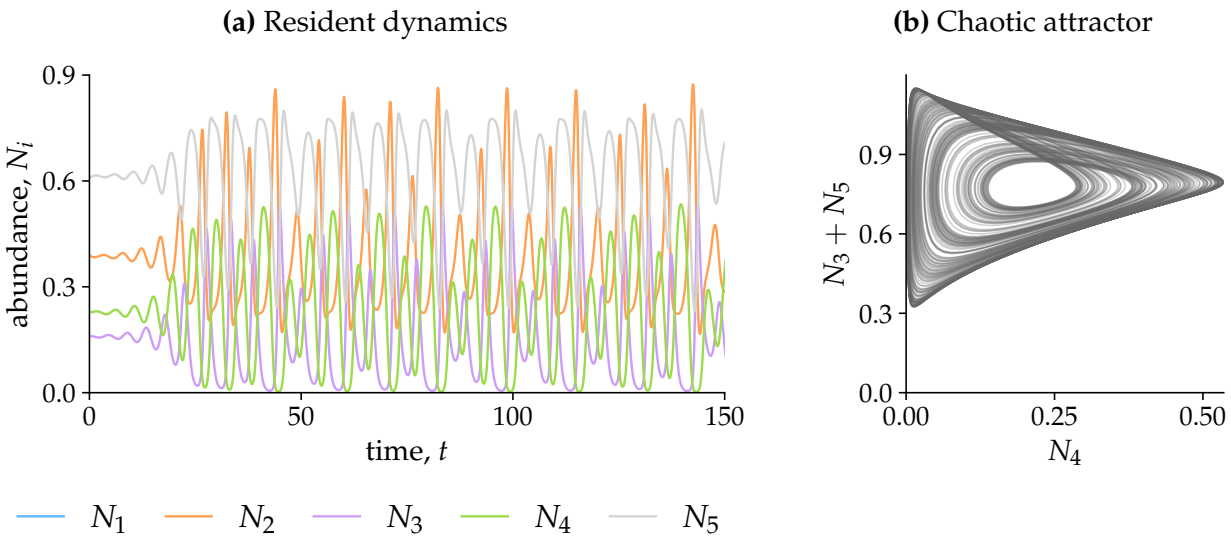


Figure S8.1: Chaotic resident dynamics for our parameter values. **(a)** Time series of species 2 through 5, starting from an initial condition near the unstable fixed point of the resident species' dynamics. **(b)** Phase plane for resident species, based on $t = 25$ to 1000 of the same simulation.

where $(A)_{ij} = \alpha_{ij}$. The Lotka–Volterra parameters for species 2 through 5 (i.e., $r_{2:5}$ and $A_{2:5,2:5}$) are based on the chaotic system discovered by Sprott (2010), scaled by a constant factor. As in the original system (see eq. 6.52 in Sprott, 2010), the non–stage-structured residents exhibit chaotic dynamics. Trajectories starting from near the unstable fixed point give rise to irregular fluctuations (Fig. S8.1a), and a phase plane shows an attractor with the complex interbraided structure typical of chaotic systems (Fig. S8.1b).

Invasion of stage-structured species Just as in the simpler systems in the main text, invasion analysis can be used to study coexistence in this chaotic system. Introducing the stage-structured invader (species 1) to the system, we find that its total abundance increases successfully towards a persistent positive state (Fig. S8.2). Since its dynamics are driven by the residents' chaotic fluctuations, the invader's rate of growth also fluctuates irregularly. The total abundance N_1 (thick blue line) fluctuates irregularly, but follows a broad trend of exponential growth before reaching a final state where it, along with all residents (colored background lines), can persist indefinitely with fluctuating abundance. In Fig. S8.3, we visualize the population structure of species 1 during this invasion trajectory. When the invader remains rare (i.e., until about $t = 600$), the proportion of its adults fluctuates chaotically around a constant average ($N_1^{(2)}/N_1 \approx 0.652$). As the invader becomes more common, its population structure continues to fluctuate, but becomes more adult-dominated on average. Ultimately, the average proportion of adults reaches a plateau ($N_1^{(2)}/N_1 \approx 0.667$) as the invader reaches its final fluctuating abundance in the community (after about $t = 900$).

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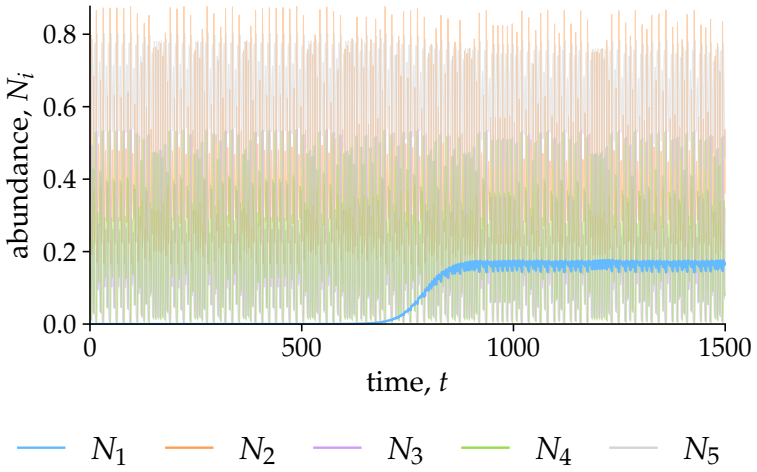


Figure S8.2: Invasion dynamics after introduction of stage-structured species 1. Starting from the final state of the resident simulation (Fig. S8.1), we introduce a small abundance of juveniles ($N_1^{(1)} = 10^{-12}$) at the new time $t = 0$. The invader's total abundance $N_1 = N_1^{(1)} + N_1^{(2)}$ (thick blue line) increases over time, despite the irregular fluctuations in its growth rate driven by the residents' chaotic dynamics (colored background lines).

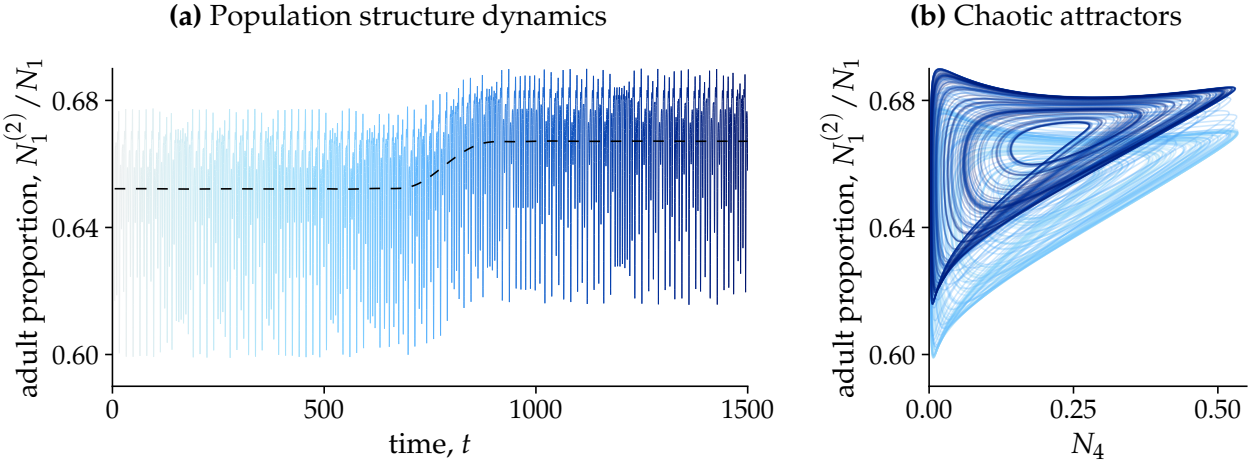


Figure S8.3: Population structure during invasion. As in Main Text Figure 2a, we consider the proportion of adults among the invader's population, $N_1^{(2)}/N_1$. **(a)** Population structure dynamics are shown by plotting the adult proportion over time (thin blue line). To better display long-term dynamics, the first time unit is not visualized ($N_1^{(2)}/N_1$ rapidly increasing from 0). For comparison to the next panel, the color varies from light blue ($t = 0$) to dark blue ($t = 1500$). Dotted line shows the smoothed average over a centered window of 50 time units. **(b)** Phase plane plot of adult proportion vs. the abundance of species 4, N_4. Line color changes over time, matching (a), revealing two distinct attractors: one for resident-driven invasion dynamics (light blue) and one for the entire five-species system (dark blue).

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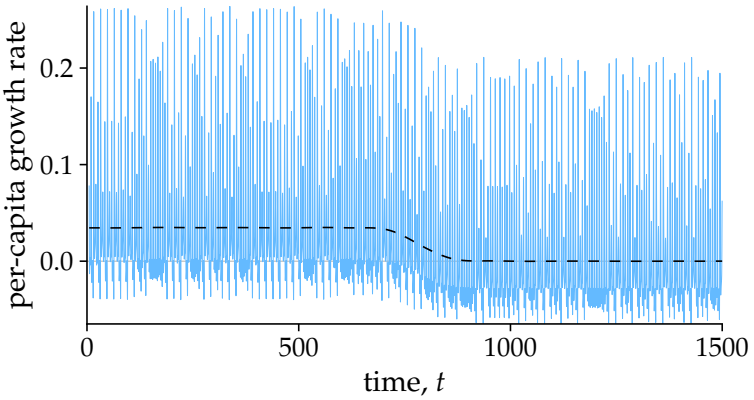


Figure S8.4: Per-capita growth rate of the stage-structured invader (thin blue line) and its smoothed average over a centered window of 50 time units (black dashed line).

Convergent invader dynamics Despite these complex fluctuations, the insights of our framework still apply. While the invader remains rare, the dynamics of its population structure converge on a consistent attractor driven by the chaotic fluctuations of the residents. Indeed, a phase plane reveals that the invader's population structure traces a chaotic attractor during invasion (Fig. S8.3b, light blue portion), before shifting to a different attractor once the invader reaches high abundance (Fig. S8.3b, dark blue). This means that if we keep the invader's abundance low and take a long enough time period, its average properties—including its long-term growth rate—will converge to a consistent value. Indeed, evaluating the invader's per-capita growth rate for our simulation (Fig. S8.4, thin blue line), we can confirm that the average growth rate (black dashed line) converges to a consistent value of IGR $= 0.035$ when the invader remains rare (i.e., until about $t = 600$).

Calculating invasion growth rate using rescaling algorithm As an alternative to simulating the full invasion trajectory, as in Figs. S8.2–S8.4, we can also apply the more efficient numerical algorithms in Appendix S3 to our chaotic system. As noted in the main text, the invasion growth rate of species 1 is equivalent to the Lyapunov exponent of the system around its resident attractor (species 2–5, Fig. S8.1). Accordingly, we can apply a variant of the algorithm for calculating Lyapunov exponents (Fig. S8.5a). Stated simply, we simulate the system as before, but repeatedly rescale the total abundance of the invader N_1 in order to keep it rare. The growth rate GR_i between rescaling events can be calculated using the total abundance before rescaling ($\mathrm{GR}_i = \log\left[N(t_i)/N_0\right]/\Delta t$). To calculate the invasion growth rate, we discard the initial burn-in period where the system has not yet reached its invasion attractor, then take the average of this growth rate over a large number of intervals (e.g., averaging over $t = 5$ to 1500 for the calculation in Fig. S8.5). Comparing the actual invasion dynamics to exponential growth at the rate predicted by our algorithm ($\mathrm{IGR}_1 = 0.035$) confirms that we have

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correctly found the invader's long-term growth rate when it is rare.

Calculating invasion growth rate using Floquet theory Floquet theory rigorously holds only for periodic systems. However, by approximating our chaotic invasion system as a periodic system with a very long period ($T \to \infty$), we can apply the Floquet theory calculation as a numerical algorithm for calculating the invasion growth rate in our chaotic system (Farrell & Ioannou, 1999; Sharma & Sinha, 2018). Note that when the invader is rare, the invader's abundances $\vec{N} = [N_1^{(1)}\ N_1^{(2)}]^\top$ grow according to the time-varying linear system

$$\frac{d\vec{N}}{dt} = \mathbf{J}(t)\,\vec{N}, \tag{S8.5}$$

where $\mathbf{J}$ is the time-dependent Jacobian for species 1's two stages,

$$\mathbf{J}(t) = \begin{bmatrix} r_1 - \sum_{j>1} \alpha_{1j} N_j(t) & f_1 \\ g_1 & -m_1 \end{bmatrix}, \tag{S8.6}$$

and the resident abundances $N_j(t)$ fluctuate chaotically over time. Applying the general method for calculating the monodromy matrix $\mathbf{M}$ (main text Box 3, eq. B3.5), we simulate the linear system in eq. S8.5 starting from the standard basis vectors $\vec{e}_1 = [1\ 0]^\top$ and $\vec{e}_2 = [0\ 1]^\top$ for T time units, where T is the large period used to approximate the chaotic system. For $T = 1500$ (matching the previous simulation), this gives

$$\mathbf{M}_T = \begin{bmatrix} | & | \\ \vec{N}_{\vec{e}_1}(T) & \vec{N}_{\vec{e}_2}(T) \\ | & | \end{bmatrix} = \begin{bmatrix} 0.37 & 0.67 \\ 2.15 & 3.87 \end{bmatrix} \times 10^{22}, \tag{S8.7}$$

with the leading eigenvalue (i.e., Floquet multiplier) $\rho = 4.24 \times 10^{22}$. This corresponds to a Floquet exponent of $(\log \rho)\,/T = 0.035$, matching the previous calculations. Comparing the results of this Floquet method to those from the previous algorithm for different simulation times (Fig. S8.6) highlights that the results converge to the same true invasion growth rate. Because the abundances do not need to be rescaled, this method also runs somewhat faster ($\sim 2.5\times$) than the previous algorithm.

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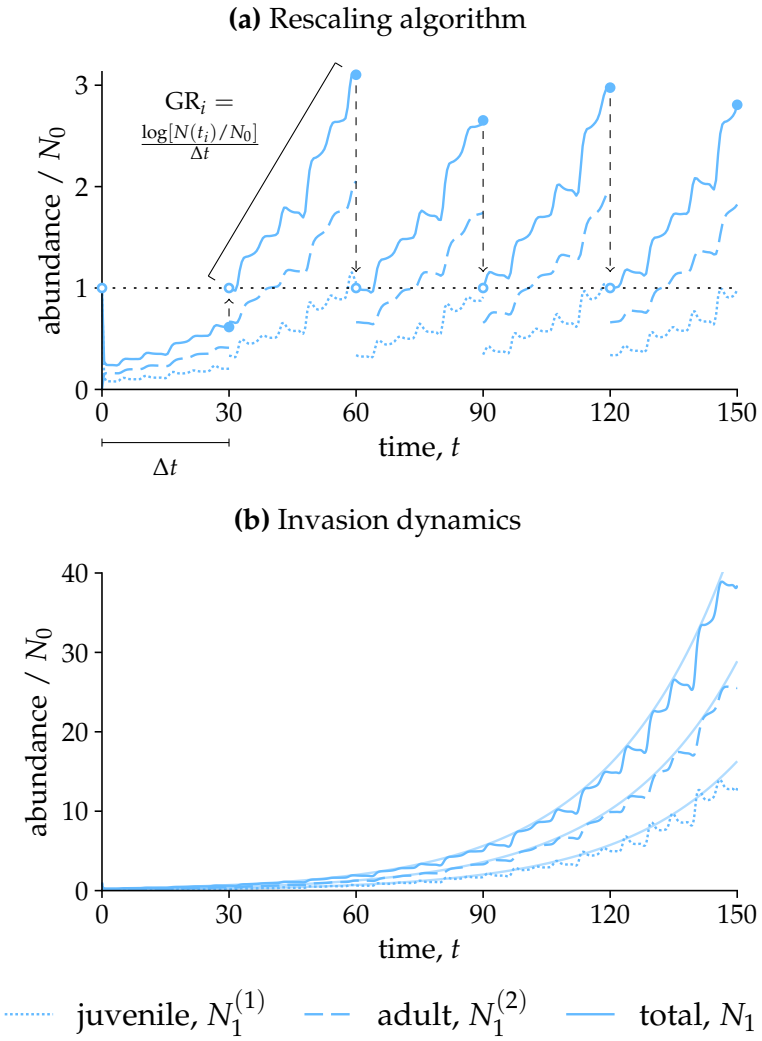


Figure S8.5: Finding the invasion growth rate in the chaotic model. **(a)** Applying the rescaling algorithm to simulate the invasion growth rate. The invader is introduced at a small target abundance $N_0 = 10^{-12}$ (dotted black horizontal line) and simulated for intervals of Δt (here, for visualization purposes, 30 time units). After each interval i, the populations are rescaled proportionally, such that the total abundance (solid blue line) is scaled from $N(t_i)$ back to N_0 (dashed arrows). The exponential growth rate GR$_i$ is calculated for each interval, and the final IGR is obtained by averaging the interval growth rates over a large number of intervals after some burn-in period (Appendix S3). **(b)** Comparison of model dynamics (solid, dashed, and dotted blue lines) to exponential growth (faint blue curves) at the invasion growth rate IGR$_1 = 0.035$ obtained using the rescaling algorithm. For greater accuracy, the invasion growth rate calculation uses a smaller interval ($\Delta t = 0.1$) and a much larger time horizon (averaging from $t = 5$ to 1500) than depicted in panel (a).

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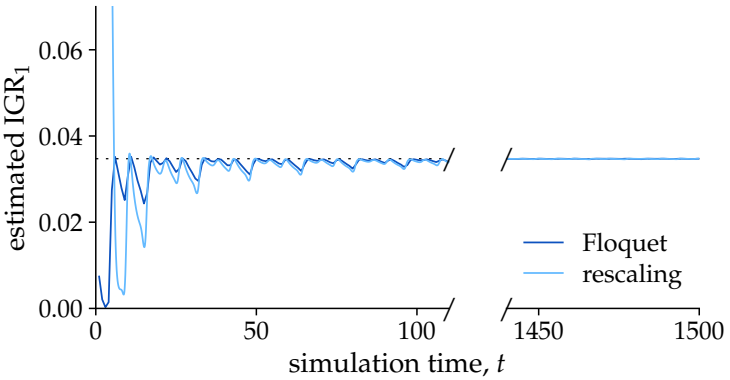


Figure S8.6: Comparison of methods for calculating invasion growth rate. The x-axis shows the length of simulation used to calculate invasion growth rate. The y-axis shows the estimated invasion growth rate: for the rescaling algorithm (light blue), this is the average growth rate between times 5 and t; for the Floquet method, this is $(\log \rho)\,/\,t$ for a monodromy matrix calculated for period t. Both methods converge to the same value of IGR_1 (dotted line).

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S9 Floquet theory

Floquet theory considers solutions to the periodic system of differential equations

$$\frac{d\vec{N}}{dt} = \mathbf{C}(t)\,\vec{N}, \tag{S9.1}$$

where $\mathbf{C}$ is a periodically varying matrix such that $\mathbf{C}(t+T) = \mathbf{C}(t)$ for period T. Here, we provide a summary of key results in Floquet theory. Rather than a full set of formal proofs, we provide intuitive justifications for key derivations and refer readers to standard texts such as Chicone (1999) for a more rigorous treatment.

S9.1 Principal fundamental matrix solution

For any system of the form in eq. S9.1, there exists a fundamental matrix solution $\mathbf{\Phi}(t)$ such that $\mathbf{\Phi}(0) = I$, known as the *principal fundamental matrix solution*. To see this, let $\mathbf{\Psi}(t)$ be any fundamental matrix solution. Then,

$$\mathbf{\Phi}(t) = \mathbf{\Psi}(t)\,\mathbf{\Psi}^{-1}(0). \tag{S9.2}$$

Indeed, this is a linear combination of columns of $\mathbf{\Psi}$ and is therefore also a solution. Moreover, when $t = 0$, the matrices cancel to give the identity matrix I, ensuring that $\mathbf{\Phi}(0) = I$ as required.

Note that the principal fundamental matrix solution has the convenient property that any solution to the system can be written

$$\vec{N}(t) = \mathbf{\Phi}(t)\,\vec{N}(0), \tag{S9.3}$$

where $\vec{N}(0)$ is the initial condition. This also gives a convenient way to numerically calculate $\mathbf{\Phi}$. Let $\vec{e}_i$ be the i-th standard basis vector; i.e., a vector with one at position i and zeros elsewhere. Then the product $\mathbf{\Phi}(t)$ with $\vec{e}_i$ is simply its i-th column; on the other hand, it also represents the solution at time t of a system starting with $\vec{N}(0) = \vec{e}_i$. Thus, we can assemble $\mathbf{\Phi}(t)$ from a series of columns representing different solutions to eq. S9.1 at time t:

$$\mathbf{\Phi}(t) = \begin{bmatrix} | & | & & | \\ \vec{N}_{\vec{e}_1}(t) & \vec{N}_{\vec{e}_2}(t) & \cdots & \vec{N}_{\vec{e}_m}(t) \\ | & | & & | \end{bmatrix}, \tag{S9.4}$$

where the i-th column $\vec{N}_{\vec{e}_i}(t)$ uses the i-th standard basis vector $\vec{e}_i$ as its initial population vector.

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S9.2 The monodromy matrix M

Floquet proved that for a principal fundamental matrix solution $\mathbf{\Phi}(t)$,

$$\mathbf{\Phi}(t+T) = \mathbf{\Phi}(t)\,\mathbf{M}, \tag{S9.5}$$

for all t, where $\mathbf{M}$ is the *monodromy matrix*, defined as

$$\mathbf{M} = \mathbf{\Phi}^{-1}(0)\,\mathbf{\Phi}(T). \tag{S9.6}$$

To see this, let $\mathbf{\Psi}(t) = \mathbf{\Phi}(t+T)$. This time-shifted function is also a fundamental matrix solution of the system due to the fact that $\mathbf{C}(t+T) = \mathbf{C}(t)$, which we can verify by showing that its derivative satisfies the original system (eq. S9.1). Thus, there should exist some linear combination of columns of $\mathbf{\Phi}$ that gives us $\mathbf{\Psi}$.

We show that the monodromy matrix $\mathbf{M}$ defined above fulfills this condition. Consider the product of the original solution with $\mathbf{M}$ on the right: rewriting in terms of $\mathbf{\Psi}$, this is $\mathbf{\Phi}(t)\,\mathbf{M} = \mathbf{\Phi}(t)\,\mathbf{\Phi}^{-1}(0)\,\mathbf{\Psi}(0)$. Due to the fact that the system is linear, $\mathbf{\Phi}(t)\,\mathbf{M}$ (where each column is a linear combination of the columns of $\mathbf{\Phi}$) is also a solution of eq. S9.1. Moreover, its value matches that of $\mathbf{\Psi}$ at $t = 0$ due to cancellation of the inverse in

$$\mathbf{\Phi}(0)\,\mathbf{M} = \mathbf{\Phi}(0)\,\mathbf{\Phi}^{-1}(0)\,\mathbf{\Psi}(0) = \mathbf{\Psi}(0), \tag{S9.7}$$

i.e., the solutions share an initial value. The solutions of a linear system are unique, meaning that $\mathbf{\Phi}(t)\,\mathbf{M} = \mathbf{\Psi}(t) = \mathbf{\Phi}(t+T)$ for all t, as required.

Note that eq. S9.5 involves the product of the principal fundamental matrix solution with $\mathbf{M}$ on the right. We now derive the result in the main text, which concerns $\mathbf{M}$ on the left multiplying an individual vector solution $\vec{N}$. Note that the monodromy matrix is specific to a particular choice of fundamental matrix solution. Accordingly, let $\mathbf{\Psi}(t)$ be any fundamental matrix solution and let $\mathbf{M}'$ represent its monodromy matrix. Then, $\vec{N}(t) = \mathbf{\Psi}(t)\,\mathbf{\Psi}^{-1}(t_0)\,\vec{N}(t_0)$ since the right-hand side equals $\vec{N}(t_0)$ at $t = t_0$. Then, we can write

$$\begin{aligned}
\vec{N}(t_0 + kT) &= \mathbf{\Psi}(t_0 + kT)\,\mathbf{\Psi}^{-1}(t_0)\,\vec{N}(t_0) && \text{(S9.8)}\\
&= \mathbf{\Psi}(t_0)\,\mathbf{M}'\cdots\mathbf{M}'\mathbf{\Psi}^{-1}(t_0)\,\vec{N}(t_0) && \text{(S9.9)}\\
&= (\mathbf{M}_{t_0})^k\,\vec{N}(t_0), && \text{(S9.10)}
\end{aligned}$$

where

$$\mathbf{M}_{t_0} = \mathbf{\Psi}(t_0)\,\mathbf{M}'\mathbf{\Psi}^{-1}(t_0). \tag{S9.11}$$

Unlike in eq. S9.5, the matrix that advances a single vector by one period depends on phase. Nonetheless, $\mathbf{M}_{t_0}$ still corresponds to a specific monodromy matrix; in particular,

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it belongs to the specific fundamental matrix solution with $\mathbf{\Psi}(t_0) = I$. Thus, for the case $t_0 = 0$ considered in the main text, $\mathbf{M}$ is precisely the monodromy matrix of the *principal* fundamental matrix solution, matching the suggested procedure for its column-wise calculation. More generally, though the actual value of $\mathbf{M}$ depends on the choice of fundamental matrix solution, its eigenvalues do not change, meaning the invasion and stability properties are well defined.

S9.3 The Floquet exponent matrix B and the Floquet normal form

It is possible to show that, because $\mathbf{M}$ (the monodromy matrix for $t_0 = 0$) is an invertible matrix, there exists some matrix $\mathbf{B}$ such that

$$e^{T\mathbf{B}} = \mathbf{M}; \tag{S9.12}$$

though the formal proof is beyond the scope of this appendix, interested readers may consult Chicone (1999, Chapter 2). Just as $\mathbf{M}$ captures the growth of the principal fundamental matrix solution over entire periods (i.e., in discrete time), $\mathbf{B}$ captures its overall growth in continuous time. However, we need another ingredient to capture the system's variation within each period. To do so, we define a periodically varying matrix $\mathbf{P}(t)$ as

$$\mathbf{P}(t) = \mathbf{\Phi}(t) e^{-t\mathbf{B}}. \tag{S9.13}$$

This function is also periodic with period T because $\mathbf{P}(t+T) = \mathbf{\Phi}(t+T) e^{-T\mathbf{B}} e^{-t\mathbf{B}} = \mathbf{\Phi}(t) \mathbf{M}\mathbf{M}^{-1} e^{-t\mathbf{B}} = \mathbf{\Phi}(t) e^{-t\mathbf{B}} = \mathbf{P}(t)$. Furthermore, we can rearrange the equation to write

$$\mathbf{\Phi}(t) = \mathbf{P}(t) e^{t\mathbf{B}}, \tag{S9.14}$$

known as a *Floquet normal form* for the fundamental matrix $\mathbf{\Phi}(t)$. Here, $e^{t\mathbf{B}}$ is a matrix solution of the linear ordinary differential equation system $d\mathbf{X}/dt = \mathbf{B}\mathbf{X}$ where $\mathbf{X}(0) = \mathbf{I}$. Accordingly, we can view $\mathbf{P}(t)$ as a periodically varying variable transformation mapping the underlying linear system described by $\mathbf{B}$ to the original periodic Floquet system described by eq. S9.1.

S9.4 Eigenvalue analysis: the Floquet multipliers ρ_i and Floquet exponents λ_i

Just as in non-periodic systems, eigenvalue analysis can tell us important information about the long-term fate of the system. Because $\mathbf{\Phi}(kT) = \mathbf{M}^k$ for any integer k, we can write the solution after a whole number k of periods using the monodromy matrix $\mathbf{M}$

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and the initial condition $\vec{N}(0)$ as

$$\vec{N}(kT) = \mathbf{M}^k \vec{N}(0). \tag{S9.15}$$

Letting the eigenvalues ρ_i and corresponding eigenvectors $\vec{w}_i$ be solutions of $\mathbf{M}\vec{w}_i = \rho_i \vec{w}_i$, we can also write this whole-period solution as a sum of contributions from different eigenvectors:

$$\vec{N}(kT) = \sum_i a_i \rho_i^k \vec{w}_i. \tag{S9.16}$$

Thus, the eigenvalues and eigenvectors of $\mathbf{M}$ tell us the factor by which contributions to the solution can grow or shrink over one whole period; accordingly, the values of ρ_i are termed Floquet multipliers of the system.

Meanwhile, in continuous time, the eigenvalues and eigenvectors of $\mathbf{B}$ play an analogous role after we have accounted for the transformation $\mathbf{P}(t)$. Using eq. S9.14, we can write

$$\vec{N}(t) = \mathbf{P}(t)\, e^{t\mathbf{B}} \vec{N}(0), \tag{S9.17}$$

and analogously, we can write the $e^{t\mathbf{B}}\vec{N}(0)$ term as a sum of contributions from eigenvalues λ_i and eigenvectors $\vec{w}_i$ of $\mathbf{B}$:

$$\vec{N}(t) = \mathbf{P}(t) \sum_i k_i e^{\lambda_i t} \vec{w}_i. \tag{S9.18}$$

Since $e^{\lambda_i t}$ is a scalar, we can let $\vec{p}_i(t) = \mathbf{P}(t)\,\vec{w}_i$ represent the periodically varying contribution associated with each eigenvector of the underlying linear system and rewrite the system as

$$\vec{N}(t) = \sum_i k_i e^{\lambda_i t} \vec{p}_i(t). \tag{S9.19}$$

Here, the eigenvalues λ_i represent the continuous-time exponential growth rates of the system and are termed the Floquet exponents of the system.

Finally, note that it is possible to map between the discrete- and continuous-time representations of the system. Generally, the eigenvalues of a matrix $\mathbf{A}$ correspond to the eigenvalues of its exponential $e^{\mathbf{A}}$: if λ_i is an eigenvalue of $\mathbf{A}$, e^{λ_i} is an eigenvalue of $e^{\mathbf{A}}$ (Chicone, 1999, Chapter 2); furthermore, we can prove that the same eigenvector $\vec{w}_i$ is associated with both values. The ith eigenvalue λ_i of $\mathbf{B}$ corresponds to an eigenvalue $T\lambda_i$ of $T\mathbf{B}$ and, since $\mathbf{M} = e^{T\mathbf{B}}$, the ith eigenvalue ρ_i of $\mathbf{M}$ is equal to $e^{T\lambda_i}$. This allows us to relate the two sets of eigenvalues:

$$\rho_i = e^{T\lambda_i}. \tag{S9.20}$$

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S9.5 Long-term population dynamics

To apply this theory to population dynamics, we consider the special case when the original periodic system is positive: all solutions starting at positive abundances remain positive for all time. As discussed in Appendix S1, this is a natural requirement for a structured population model to be biologically realistic. Additionally, we will assume that after some integer number of periods k, a system starting from individuals only in one state i will eventually contain a nonzero quantity of state j.

Because multiplying by the monodromy matrix $\mathbf{M}$ advances the system by one period, $\mathbf{M}$ must also satisfy the conditions to maintain discrete-time positivity. As discussed for discrete-time matrices in Appendix S1, Box S1.B1, this means that all entries of $\mathbf{M}$ must be nonnegative. Meanwhile, our assumption that all states can be reached from all other states after some integer number of periods means that $\mathbf{M}$ must also be irreducible. Because $\mathbf{M}$ is nonnegative and irreducible, the Perron–Frobenius theorem (Appendix S1, Box S1.B2) guarantees that it has a unique, simple leading eigenvalue ρ_1 and associated positive eigenvector $\vec{w}_1$ which is uniquely defined up to a constant.

We can also map these properties onto the continuous-time description of the system given by $\mathbf{B}$. Because $e^{T\mathbf{B}} = \mathbf{M}$, every eigenvector of $\mathbf{B}$ is also an eigenvector of $\mathbf{M}$ and their eigenvalues ρ_i and λ_i are related by $\rho_i = e^{T\lambda_i}$. It is also possible to prove that $\mathbf{B}$ has a simple real leading eigenvalue

$$\lambda_1 = \frac{\log \rho_1}{T}, \tag{S9.21}$$

associated with the same unique positive eigenvector $\vec{w}_1$ from above. This gives the key result of the main text, that population dynamics can be decomposed as

$$\vec{N}(t) = k_1 e^{\lambda_1 t}\vec{p}_1(t) + k_2 e^{\lambda_2 t}\vec{p}_2(t) + \cdots \tag{S9.22}$$

$$= k_1 e^{\lambda_1 t}\vec{p}_1(t) + o\left(e^{\lambda_1 t}\right), \tag{S9.23}$$

where λ_2 has a strictly smaller real part than λ_1. Here, the little-o notation $o(e^{\lambda_1 t})$ indicates that the transient terms grow more slowly than $e^{\lambda_1 t}$.

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S10 Discrete-time perennial plant model

In this Appendix, we study competition between stage-structured perennial plant populations in discrete time. We use a model structure commonly applied to studies of plants experiencing discrete growth seasons. In particular, we consider a simplified version of the perennial model of Song *et al.* (2021) and Uricchio *et al.* (2019), without density-dependent seedling-to-adult recruitment. Closely related models have been applied in other studies modeling annual–perennial plant competition (e.g., Larios *et al.*, 2017; Mordecai *et al.*, 2015; Werner *et al.*, 2025), and a model with the same structure was also analyzed in a general theoretical study by Fujiwara *et al.* (2011).

Each species i has three stages: seeds N_i, juveniles J_i, and adults A_i. Each year, a proportion γ_i of seeds germinates and, of the remaining portion (the seed bank), a fraction ℓ_i loses viability. A proportion b_i of germinants successfully becomes juveniles in the current year. A proportion h_i of juveniles successfully becomes adults in the next year, and the remainder is lost at the end of the year. Finally, a fraction m_i of adults dies each year, and all adults in a given year produce seeds (in the next year) according to some density-dependent fecundity f_i, for instance the Beverton–Holt functional response (used in, e.g., Fujiwara *et al.*, 2011; Song *et al.*, 2021; Uricchio *et al.*, 2019):

$$f_i = \frac{r_i}{1 + \sum_j \alpha_{ij} A_j}. \tag{S10.1}$$

As a full analysis of the model is beyond the scope of this simplified case study, we write this functional response generally as $f_i\left(A_1, \ldots, A_n\right)$, where n is the number of species in the system.

Note that we census the seed population N_i at the beginning of a year, before any germination events, and juveniles J_i and adults A_i at the end of a year. With this in mind, the dynamics of seeds are given by

$$N_i\left(t+1\right) = \left(1-\gamma_i\right)\left(1-\ell_i\right) \cdot N_i\left(t\right) + f_i\left(A_1, \ldots, A_n\right) \cdot A_i\left(t\right). \tag{S10.2}$$

Juvenile abundances are proportional to the seed population in the same year,

$$J_i\left(t+1\right) = \gamma_i b_i N_i\left(t+1\right), \tag{S10.3}$$

and adult abundances are given by

$$A_i\left(t+1\right) = h_i \cdot J_i\left(t\right) + \left(1-m_i\right) \cdot A_i\left(t\right). \tag{S10.4}$$

Note we do not need to track juveniles as a separate state variable, since their abundance can be written in terms of the seed population. Thus the entire system can

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be written as a recurrence relation giving seed and adult abundances in year $t+1$ as a function of those in the current year t,

$$N_i(t+1) = (1-\gamma_i)(1-\ell_i) \cdot N_i(t) + f_i(A_1, \ldots, A_n) \cdot A_i(t), \tag{S10.5}$$

$$A_i(t+1) = \gamma_i b_i h_i \cdot N_i(t) + (1-m_i) \cdot A_i(t). \tag{S10.6}$$

Define composite parameters: let $p_i = 1-(1-\gamma_i)(1-\ell_i) = \gamma_i + (1-\gamma_i)\ell_i$ represent the fraction of seeds lost each year (either due to successful germination or loss from the seed bank), and let $g_i = \gamma_i b_i h_i$ represent the overall recruitment rate from seeds to adults in the next year (combining germination, seedling survival, and the juvenile–adult transition). Then

$$N_i(t+1) = (1-p_i) \cdot N_i(t) + f_i(A_1, \ldots, A_n) \cdot A_i(t), \tag{S10.7}$$

$$A_i(t+1) = g_i \cdot N_i(t) + (1-m_i) \cdot A_i(t). \tag{S10.8}$$

Now we derive the long-term growth rate of a species. Note f_i is the only demographic transition rate which is density-dependent; we will omit the function arguments for brevity. Defining $\vec{y}(t) = [N_i(t) \;\; A_i(t)]^\top$ as a vector giving the state of the focal species i's population at time t, we can write the dynamics in matrix form as

$$\vec{y}(t+1) = \mathbf{C}\vec{y}(t); \quad \mathbf{C} = \begin{bmatrix} 1-p_i & f_i \\ g_i & 1-m_i \end{bmatrix}, \tag{S10.9}$$

where $\mathbf{C}$ depends on adult densities through its upper-right f_i entry. Note that the matrix $\mathbf{C}$ describes discrete-time transitions (Appendix S1, Section S1.2); though this matrix was denoted $\mathbf{A}$ in Appendix S1, we use $\mathbf{C}$ here to avoid confusion with the adult abundance variable A_i. Assuming the value of f_i is fixed at its value during invasion, the long-term growth factor per time step of species i is given by the leading eigenvalue of $\mathbf{C}$, and we can derive the condition for invasibility in several ways.

S10.1 Method 1.2: Strategic linear combination

Define the new variable

$$Y_i(t) = N_i(t) + \frac{p_i}{g_i} \cdot A_i(t), \tag{S10.10}$$

and assume that under long-term population growth, $A_i(t)$ is some proportion of $Y_i(t)$, i.e.,

$$A_i(t) = k \cdot Y_i(t), \tag{S10.11}$$

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where k is positive. Then we can rewrite the recurrence relation in terms of only Y_i and A_i:

$$Y_i(t+1) = N_i(t) + \left[f_i + \frac{p_i}{g_i}(1-m_i)\right] A_i(t) \tag{S10.12}$$

$$= N_i(t) + \frac{p_i}{g_i} \cdot A_i(t) + \left(f_i - \frac{p_i m_i}{g_i}\right) A_i(t) \tag{S10.13}$$

$$= Y_i(t) + \left(f_i - \frac{p_i m_i}{g_i}\right) A_i(t). \tag{S10.14}$$

Furthermore, using eq. S10.11, we can write

$$\frac{Y_i(t+1)}{Y_i(t)} = 1 + k\left(f_i - \frac{p_i m_i}{g_i}\right). \tag{S10.15}$$

Since we are in a discrete-time system, positive population growth requires that this ratio is greater than unity. Since $k > 0$, we require that $f_i - p_i m_i / g_i > 0$, giving the simple persistence condition

$$f_i g_i > p_i m_i, \tag{S10.16}$$

i.e., the positive effects of fecundity f_i and recruitment g_i on population growth must be great enough to account for the negative effects of seed loss p_i and adult mortality m_i.

S10.2 Method 2.1: Direct solution for eigenvalues

To find this specific value, we compute the characteristic polynomial of $\mathbf{C}$:

$$\det \begin{bmatrix} 1-p_i-\lambda & f_i \\ g_i & 1-m_i-\lambda \end{bmatrix} = 0 \tag{S10.17}$$

$$(1-p_i-\lambda)(1-m_i-\lambda) - g_i f_i = 0 \tag{S10.18}$$

$$\lambda^2 - [(1-p_i)+(1-m_i)]\lambda + (1-p_i)(1-m_i) - g_i f_i = 0. \tag{S10.19}$$

Then the eigenvalue with the greatest real part is given by

$$\lambda = 1 - \frac{p_i + m_i}{2} + \sqrt{\left(\frac{p_i - m_i}{2}\right)^2 + g_i f_i}, \tag{S10.20}$$

where the value under the square root is positive (due to g_i and f_i necessarily being positive). This matches the expression given in Fujiwara *et al.* (2011, see eq. 4 there).

Note that, as in the continuous-time case, λ determines the long-term growth rate of a species. However, its interpretation is different from that of the continuous-time invasion growth rate IGR: persistence requires $\lambda > 1$, not $\lambda > 0$ as in continuous time. One option is to define an analogous IGR in terms of λ such that IGR > 0 is the

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persistence condition: e.g., IGR $= \log \lambda$ (Spaak & De Laender, 2020) or IGR $= \lambda - 1$. Alternatively, we can directly solve for the conditions under which $\lambda > 1$. For this to occur, we require that

$$\sqrt{\left(\frac{p_i - m_i}{2}\right)^2 + g_i f_i} > \frac{p_i + m_i}{2} \tag{S10.21}$$

$$g_i f_i > \left(\frac{p_i + m_i}{2}\right)^2 - \left(\frac{p_i - m_i}{2}\right)^2, \tag{S10.22}$$

which we can simplify using the difference of squares to $g_i f_i > p_i m_i$, identical to the simple condition (eq. S10.16) derived above.

S10.3 Method 2.2: Stability conditions

In the previous calculation, we looked for the conditions under which the maximum eigenvalue of $\mathbf{C}$ was greater than unity:

$$\max(\lambda_1, \ldots, \lambda_n) > 1. \tag{S10.23}$$

In the case where $n = 2$, this was feasible by directly solving for λ and rearranging the quadratic root in the solution. However, when $n > 2$, solving the characteristic polynomial may be difficult or impossible; moreover, eq. S10.23 does not match the conditions which the Routh–Hurwitz criteria diagnose (i.e., whether the leading eigenvalue is positive or negative). As described in Appendix S1, section S1.2, we can shift the matrix by the identity to obtain a problem where the Routh–Hurwitz criteria can be applied. In brief, subtracting the identity matrix,

$$\mathbf{C}' = \mathbf{C} - \mathbf{I}, \tag{S10.24}$$

shifts each eigenvalue to $\lambda_i' = \lambda_i - 1$. Thus, the discrete-time growth threshold $\lambda > 1$ becomes $\lambda' > 0$, allowing us to apply the Routh–Hurwitz criteria to $\mathbf{C}'$.

Two-dimensional example Directly using the system above (eq. S10.9), we can derive $\mathbf{C}'$ by subtracting one from the diagonal elements, giving

$$\mathbf{C}' = \begin{bmatrix} -p_i & f_i \\ g_i & -m_i \end{bmatrix}. \tag{S10.25}$$

The Routh–Hurwitz conditions for a 2×2 matrix state that all eigenvalues have negative real parts if and only if $\det \mathbf{C}' > 0$ and $\operatorname{tr} \mathbf{C}' < 0$. For nonnegative population growth (one or more eigenvalues with nonnegative real parts), at least one of these conditions

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must not hold. Note that $\operatorname{tr} \mathbf{C}' = -p_i - m_i$ is always negative due to seed loss p_i and mortality m_i being non-zero proportions. Then nonnegative population growth requires that the determinant condition not hold; i.e., $\det \mathbf{C}' \leq 0$; moreover, we can infer that positive population growth requires $\det \mathbf{C}' < 0$. Thus, the persistence condition is

$$\det \mathbf{C}' = p_i m_i - g_i f_i < 0, \tag{S10.26}$$

matching eq. S10.16.

Three-dimensional example As an illustration of the advantage of using the Routh–Hurwitz conditions, say we had not simplified the system by removing the variable for juvenile population. In this case, the juvenile population (censused after germination) is determined by multiplying the seed abundance from the same year (censused before germination) by germination and juvenile recruitment $\gamma_i b_i$ (eq. S10.3). This gives the full three-stage system

$$N_i(t+1) = (1-p_i) \cdot N_i(t) + f_i \cdot A_i(t) \tag{S10.27}$$
$$J_i(t+1) = (1-p_i)\,\gamma_i b_i \cdot N_i(t) + f_i \gamma_i b_i \cdot A_i(t) \tag{S10.28}$$
$$A_i(t+1) = h_i \cdot J_i(t) + (1-m_i) \cdot A_i(t), \tag{S10.29}$$

where f_i represents the density-dependent fecundity function $f_i(A_1, \ldots, A_n)$. Furthermore, we modify this system by assuming that juveniles can persist between years, such that a proportion $1 - q_i$ is retained each year into the next:

$$N_i(t+1) = (1-p_i) \cdot N_i(t) + f_i \cdot A_i(t) \tag{S10.30}$$
$$J_i(t+1) = (1-p_i)\,\gamma_i b_i \cdot N_i(t) + (1-q_i) \cdot J_i(t) + f_i \gamma_i b_i \cdot A_i(t) \tag{S10.31}$$
$$A_i(t+1) = h_i \cdot J_i(t) + (1-m_i) \cdot A_i(t). \tag{S10.32}$$

Letting $\vec{y}(t) = [N_i(t) \;\; J_i(t) \;\; A_i(t)]^\top$ with an extra juvenile stage, the system is now $\vec{y}(t+1) = \mathbf{C}\vec{y}(t)$ with

$$\mathbf{C} = \begin{bmatrix} 1-p_i & 0 & f_i \\ \gamma_i b_i (1-p_i) & 1-q_i & \gamma_i b_i f_i \\ 0 & h_i & 1-m_i \end{bmatrix}, \tag{S10.33}$$

and the shifted matrix

$$\mathbf{C}' = \begin{bmatrix} -p_i & 0 & f_i \\ \gamma_i b_i (1-p_i) & -q_i & \gamma_i b_i f_i \\ 0 & h_i & -m_i \end{bmatrix}. \tag{S10.34}$$

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The characteristic polynomial is now

$$-\left(\lambda'+p_i\right)\left(\lambda'+q_i\right)\left(\lambda'+m_i\right)+\left(1-p_i\right)\gamma_i b_i h_i f_i+\left(\lambda'+p_i\right)\gamma_i b_i h_i f_i=0 \tag{S10.35}$$

$$-\left(\lambda'+p_i\right)\left(\lambda'+q_i\right)\left(\lambda'+m_i\right)+\left(\lambda'+1\right)g_i f_i=0. \tag{S10.36}$$

Expanding and taking the negative of both sides to obtain a positive leading coefficient of one,

$$\lambda'^3+\underbrace{\left(p_i+q_i+m_i\right)}_{a_1}\lambda'^2+\underbrace{\left(p_iq_i+q_im_i+p_im_i-g_if_i\right)}_{a_2}\lambda'+\underbrace{p_iq_im_i-g_if_i}_{a_3}=0. \tag{S10.37}$$

The Routh–Hurwitz conditions state that all eigenvalues have negative real parts if and only if $a_1>0$, $a_3>0$, and $a_1a_2>a_3$. Note that $a_1>0$ holds automatically because all three parameters (loss from seed, juvenile, and adult stages) are positive. Then, $a_3>0$ is necessary and sufficient for all eigenvalues to have negative real parts. To see this, rearrange the last condition to $a_1a_2-a_3>0$ and substitute:

$$\left(p_i+q_i+m_i\right)\left(p_iq_i+q_im_i+p_im_i-g_if_i\right)-p_iq_im_i+g_if_i>0 \tag{S10.38}$$

$$\left(q_i+m_i\right)\underbrace{\left(p_iq_i+q_im_i+p_im_i-g_if_i\right)}_{a_2}+p_i^2\left(q_i+m_i\right)+\left(1-p_i\right)g_if_i>0, \tag{S10.39}$$

where the second term is necessarily positive and the last term is nonnegative (because $p_i\leq 1$). Note that $a_3<a_2$ because

$$a_3=p_iq_im_i-g_if_i\leq p_iq_i-g_if_i<p_iq_i+q_im_i+p_im_i-g_if_i=a_2, \tag{S10.40}$$

and thus, as long as $a_3\geq 0$, the entire left-hand side is positive. Furthermore, note that in the boundary case $a_3=0$, the leading eigenvalue is exactly zero.

Thus, persistence $\lambda'>0$ occurs if and only if $a_3<0$, giving the persistence condition

$$g_if_i>p_iq_im_i, \tag{S10.41}$$

which has an interpretation paralleling the previous conclusion: recruitment g_i and fecundity f_i must outweigh losses in the three stages. When $q_i=1$ (complete loss of juveniles), we recover the simpler two-stage model and its persistence condition (eq. S10.16). This simple criterion also generalizes to other specialized reproductive strategies, e.g., semelparous (juvenile persistence $q_i<1$ and complete adult mortality $m_i=1$) or biennial life history (no juvenile persistence $q_i=1$ and complete adult mortality $m_i=1$).

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References

Caswell, H. (2001) *Matrix Population Models: Construction, Analysis, and Interpretation*. Sinauer Associates, Sunderland, Mass, 2nd ed.

Chicone, C. (1999) *Ordinary Differential Equations with Applications*. Number 34 in Texts in Applied Mathematics. Springer New York, New York, NY.

Farrell, B.F. & Ioannou, P.J. (1999) Perturbation Growth and Structure in Time-Dependent Flows. *Journal of the Atmospheric Sciences*, **56**, 3622–3639. https://doi.org/10.1175/1520-0469(1999)056<3622:PGASIT>2.0.CO;2.

Frobenius, F.G. (1912) Über Matrizen aus nicht negativen Elementen. *Sitzungsberichte der Königlich Preussischen Akademie der Wissenschaften zu Berlin*, **1912**, 456–477.

Fujiwara, M., Pfeiffer, G., Boggess, M., Day, S. & Walton, J. (2011) Coexistence of competing stage-structured populations. *Scientific Reports*, **1**, 107. https://doi.org/10.1038/srep00107.

Hurwitz, A. (1895) Ueber die Bedingungen, unter welchen eine Gleichung nur Wurzeln mit negativen reellen Theilen besitzt. *Mathematische Annalen*, **46**, 273–284. https://doi.org/10.1007/BF01446812.

Iritani, R. & Day, T. (2026) Evolutionary invasion analysis for structured populations: a synthesis. https://doi.org/10.64898/2026.03.23.713828.

Ke, P.-J. & Levine, J.M. (2021) The temporal dimension of plant-soil microbe interactions: mechanisms promoting feedback between generations. *The American Naturalist*, **198**, E80–E94. https://doi.org/10.1086/715577.

Larios, L., Hallett, L.M. & Suding, K.N. (2017) Where and how to restore in a changing world: a demographic-based assessment of resilience. *Journal of Applied Ecology*, **54**, 1040–1050. https://doi.org/10.1111/1365-2664.12946.

Lefkovitch, L.P. (1965) The Study of Population Growth in Organisms Grouped by Stages. *Biometrics*, **21**, 1. https://doi.org/10.2307/2528348.

Leslie, P.H. (1945) On the use of matrices in certain population mathematics. *Biometrika*, **33**, 183–212. https://doi.org/10.1093/biomet/33.3.183.

Liénard, A. & Chipart, M.H. (1914) Sur le signe de la partie réelle des racines d'une équation algébrique. *Journal de Mathématiques Pures et Appliquées*, **10**, 291–346.

Metzler, L.A. (1945) Stability of Multiple Markets: The Hicks Conditions. *Econometrica*, **13**, 277–292. https://doi.org/10.2307/1906922.

Mordecai, E.A., Molinari, N.A., Stahlheber, K.A., Gross, K. & D'Antonio, C. (2015) Controls over native perennial grass exclusion and persistence in California grasslands invaded by annuals. *Ecology*, **96**, 2643–2652. https://doi.org/10.1890/14-2023.1.

Perron, O. (1907) Zur Theorie der Matrices. *Mathematische Annalen*, **64**, 248–263. https://doi.org/10.1007/BF01449896.

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Routh, E.J. (1877) *A Treatise on the Stability of a Given State of Motion, Particularly Steady Motion*. Macmillan and Co., London.

Sharma, A. & Sinha, S.C. (2018) An Approximate Analysis of Quasi-Periodic Systems Via Floquét Theory. *Journal of Computational and Nonlinear Dynamics*, **13**, 021008. https://doi.org/10.1115/1.4037797.

Song, C., Uricchio, L.H., Mordecai, E.A. & Saavedra, S. (2021) Understanding the emergence of contingent and deterministic exclusion in multispecies communities. *Ecology Letters*, **24**, 2155–2168. https://doi.org/10.1111/ele.13846.

Spaak, J.W. & De Laender, F. (2020) Intuitive and broadly applicable definitions of niche and fitness differences. *Ecology Letters*, **23**, 1117–1128. https://doi.org/10.1111/ele.13511.

Sprott, J.C. (2010) *Elegant Chaos: Algebraically Simple Chaotic Flows*. World Scientific, Singapore.

Uricchio, L.H., Daws, S.C., Spear, E.R. & Mordecai, E.A. (2019) Priority effects and nonhierarchical competition shape species composition in a complex grassland community. *The American Naturalist*, **193**, 213–226. https://doi.org/10.1086/701434.

Ważewski, T. (1950) Systèmes des équations et des inégalités différentielles ordinaires aux deuxièmes membres monotones et leurs applications. *Annales de la Société Polonaise de Mathématique*, **23**, 112–166.

Werner, C.M., Hallett, L.M. & Shoemaker, L.G. (2025) Fluctuation-Dependent Coexistence of Stage-Structured Species. *The American Naturalist*, **205**, 327–341. https://doi.org/10.1086/733382.